

22050 Continuous-Time Signals and Linear Systems

Kaj-Åge Henneberg

L13

Butterworth filter design – part 2

Instrumentation amplifier

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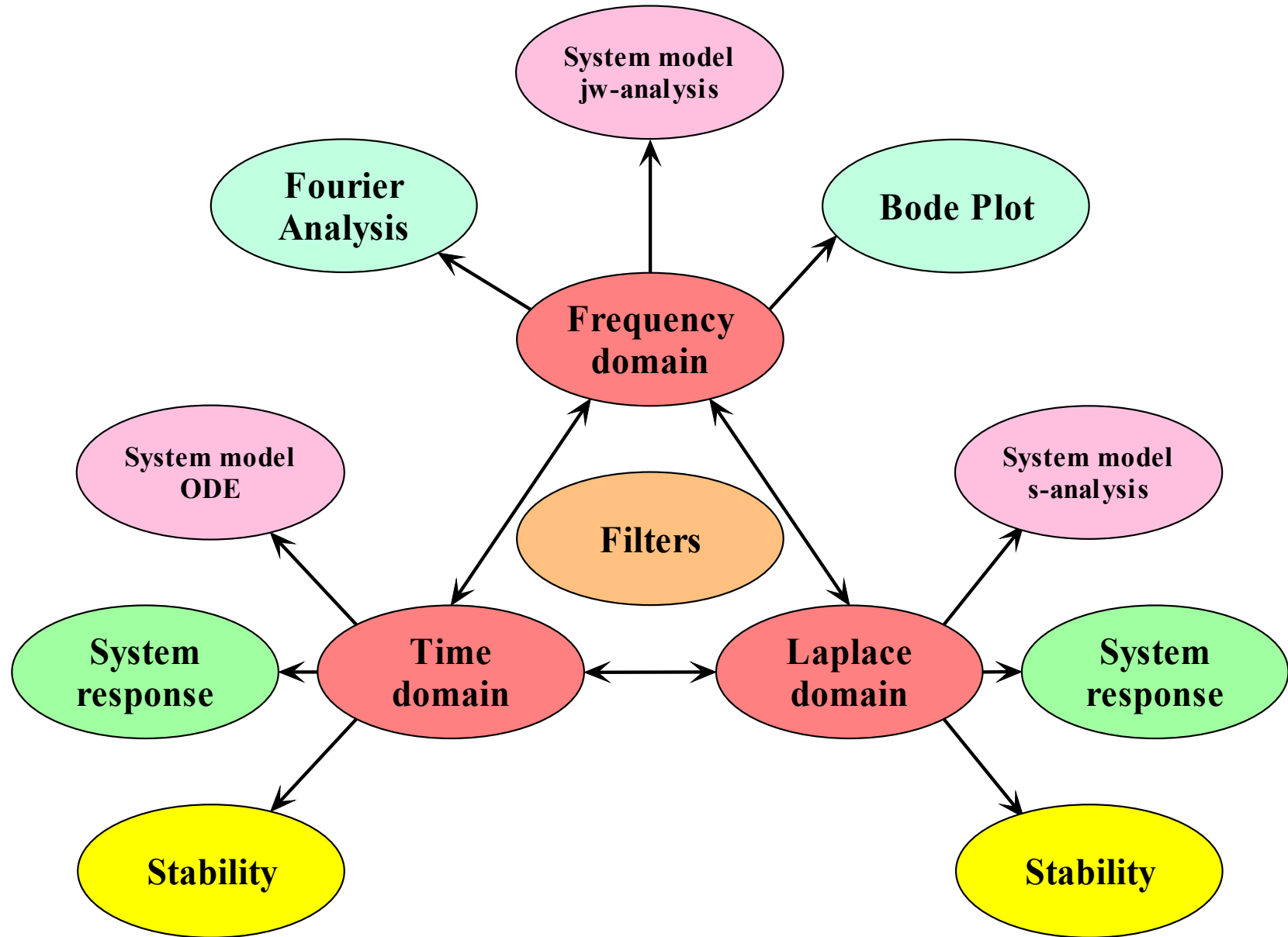
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- Frequency transformation from low-pass to high-pass – Lathi 7.7
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- Instrumentation amplifier (in-amp) – Not in Lathi
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From low-pass to high-pass

Filter specifications

Video

low-pass to high-pass Transformation

Transformation of a prototype low-pass filter into the final high-pass is accomplished by mapping of the frequency axis:

$$\hat{S}_{LP} \leftarrow \frac{\omega_p}{S_{HP}}$$

$$j\hat{\omega}_{LP} \leftarrow \frac{\omega_p}{j\omega_{HP}} = -j \frac{\omega_p}{\omega_{HP}}$$

$$\hat{\omega}_{LP} \leftarrow -\frac{\omega_p}{\omega_{HP}}$$

$$\hat{\omega} \leftarrow -\frac{k}{\omega}$$

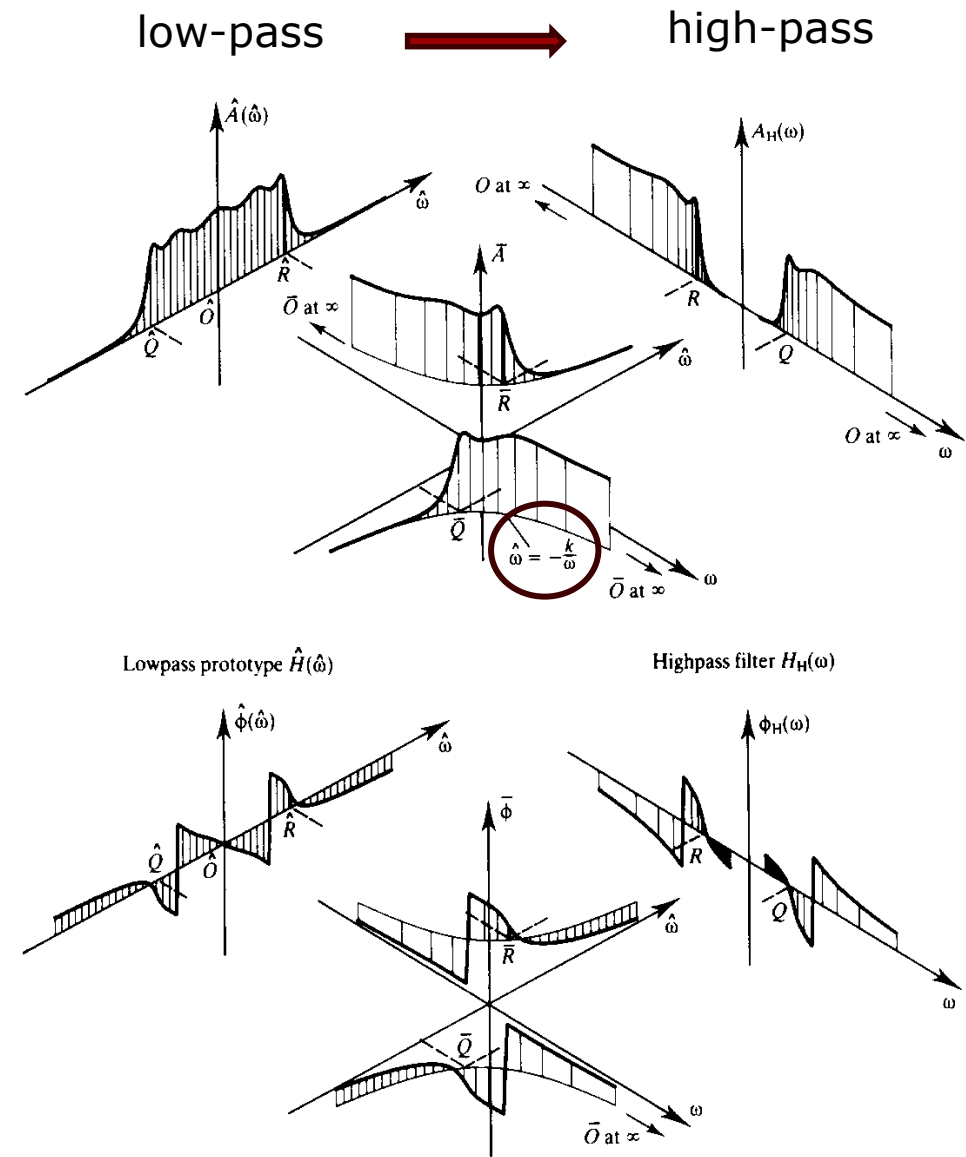


Figure 10.34 Lowpass-to-highpass frequency mapping: $\hat{\omega} = -k/\omega$.

P. Kraniuskas, Transforms in Signals and Systems, Addison Wesley.

Using frequency transformation, we can rewrite a low-pass filter characteristic into one for a high-pass filter.

Consequently, if we start out with a frequency-normalized low-pass filter, we can transform it into a frequency-normalized high-pass filter.

For this reason, we will rarely see filter coefficient tables for high-pass, band-pass and band-stop filters.

Frequency-normalized high-pass filter:

$$H_{LP}(s) = \frac{b_0}{s^2 + a_1 s + a_0} = \frac{\frac{b_0}{a_0}}{\left(\frac{s}{\sqrt{a_0}}\right)^2 + \frac{a_1}{\sqrt{a_0}} \frac{s}{\sqrt{a_0}} + 1}$$

$$H_{LP}(\hat{s}_{LP}) = \frac{\frac{b_0}{a_0}}{\hat{s}_{LP}^2 + \frac{a_1}{\sqrt{a_0}} \hat{s}_{LP} + 1} \quad \hat{s}_{LP} \stackrel{\text{def}}{=} \frac{s_{LP}}{\sqrt{a_0}}$$

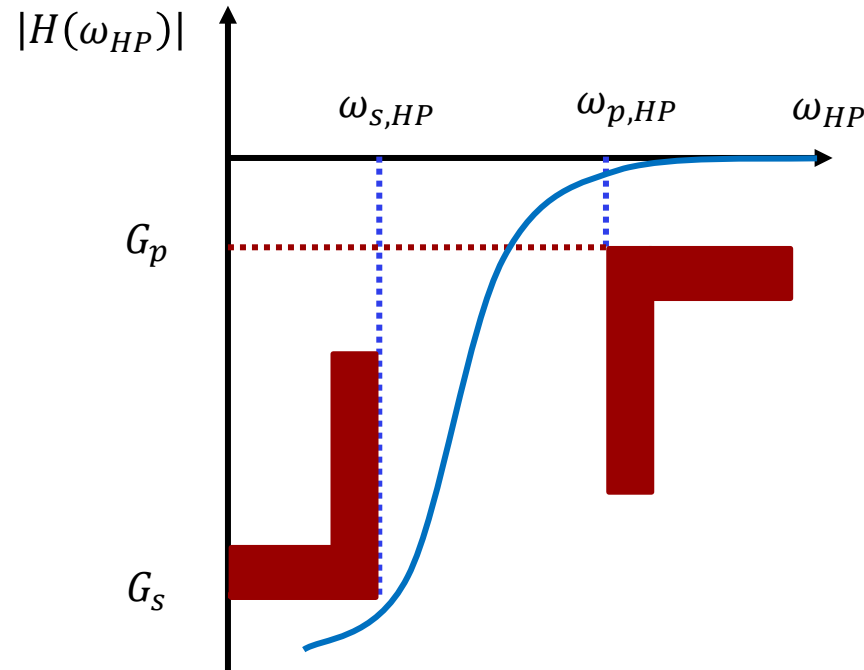
$$\frac{1}{\hat{s}_{HP}} \rightarrow \hat{s}_{LP}$$

$$H_{HP}(\hat{s}_{HP}) = \frac{\frac{b_0}{a_0}}{\left(\frac{1}{\hat{s}_{HP}}\right)^2 + \frac{a_1}{\sqrt{a_0}} \frac{1}{\hat{s}_{HP}} + 1} \quad \hat{s}_{HP} \stackrel{\text{def}}{=} \frac{s_{HP}}{\omega_p}$$

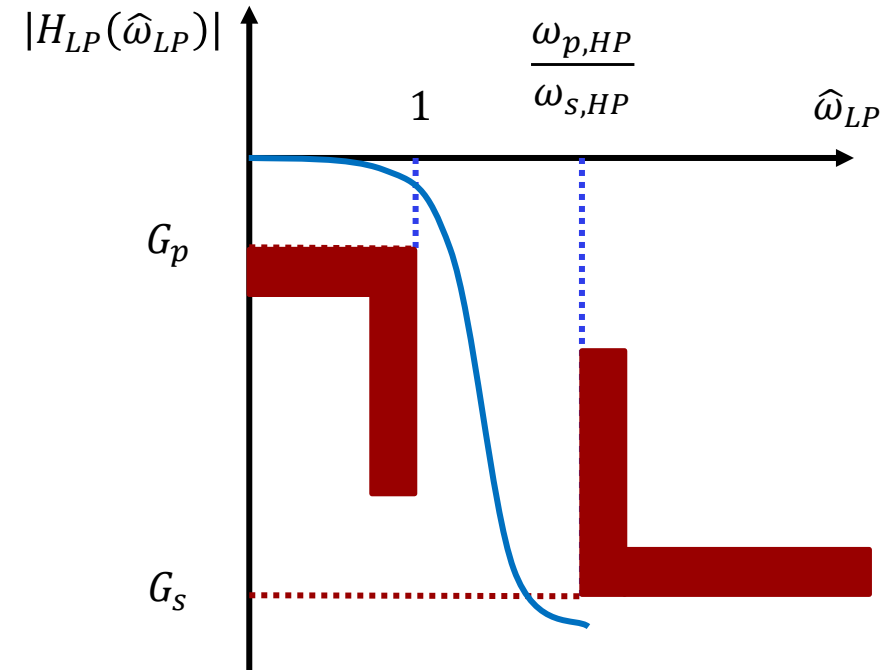
$$H_{HP}(\hat{s}_{HP}) = \frac{b_2 \hat{s}_{HP}^2}{\hat{s}_{HP}^2 + \frac{a_1}{\sqrt{a_0}} \hat{s}_{HP} + 1} \quad b_2 = \frac{b_0}{a_0}$$

Butterworth High-pass Filter Specifications

Filter performance specifications for a high-pass filter are formulated for the high-pass filter and then transformed to the frequency-normalized low-pass filter, so that we can determine the required filter order.



$$\hat{\omega}_{LP} \leftarrow -\frac{\omega_{p,HP}}{\omega_{HP}}$$



$$\hat{\omega}_{p,LP} \leftarrow -\frac{\omega_{p,HP}}{\omega_{p,HP}} = -1 \quad \hat{\omega}_{s,LP} \leftarrow -\frac{\omega_{p,HP}}{\omega_{s,HP}}$$

$$n = \frac{1}{2\log_{10}(\hat{\omega}_{s,LP}/\hat{\omega}_{p,LP})} \log_{10} \left(\frac{10^{-\frac{G_{S,dB}}{10}} - 1}{10^{-\frac{G_{p,dB}}{10}} - 1} \right) = \frac{1}{2\log_{10}(\omega_{p,HP}/\omega_{s,HP})} \log_{10} \left(\frac{10^{-\frac{G_{S,dB}}{10}} - 1}{10^{-\frac{G_{p,dB}}{10}} - 1} \right)$$

Example: high-pass Butterworth Filter

Find the transfer function order for the filter satisfying the following specifications:

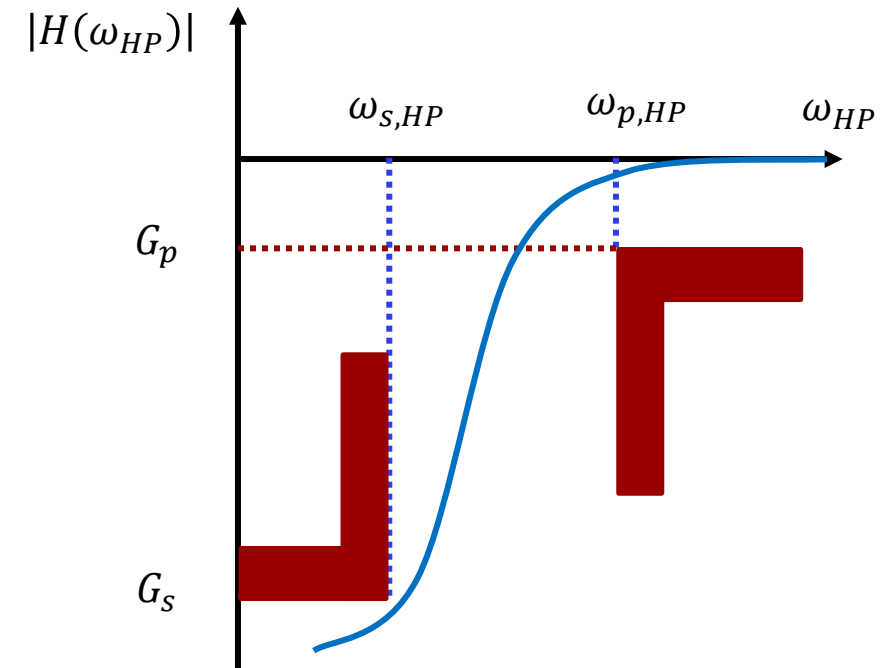
$$\begin{aligned} G_s &\leq -26 \text{ dB}, & \omega_s &= 2\pi \cdot 0.01 \text{ rad/s} \\ G_p &= -3 \text{ dB}, & \omega_p &= 2\pi \cdot 0.05 \text{ rad/s} \end{aligned}$$

$$\hat{\omega}_{p,LP} \leftarrow -\frac{\omega_{p,HP}}{\omega_{p,HP}} = -\frac{0.05}{0.05} = -1$$

$$\hat{\omega}_{s,LP} \leftarrow -\frac{\omega_{p,HP}}{\omega_{s,HP}} = -\frac{0.05}{0.01} = -5$$

$$\begin{aligned} n &= \frac{1}{2 \log_{10}(\hat{\omega}_{s,LP} / \hat{\omega}_{p,LP})} \log_{10} \left(\frac{10^{-\frac{G_s, dB}{10}} - 1}{10^{-\frac{G_p, dB}{10}} - 1} \right) \\ &= \frac{1}{2 \log_{10}(\omega_{p,HP} / \omega_{s,HP})} \log_{10} \left(\frac{10^{-\frac{G_s, dB}{10}} - 1}{10^{-\frac{G_p, dB}{10}} - 1} \right) \end{aligned}$$

$$n = 1.861 \rightarrow 2$$

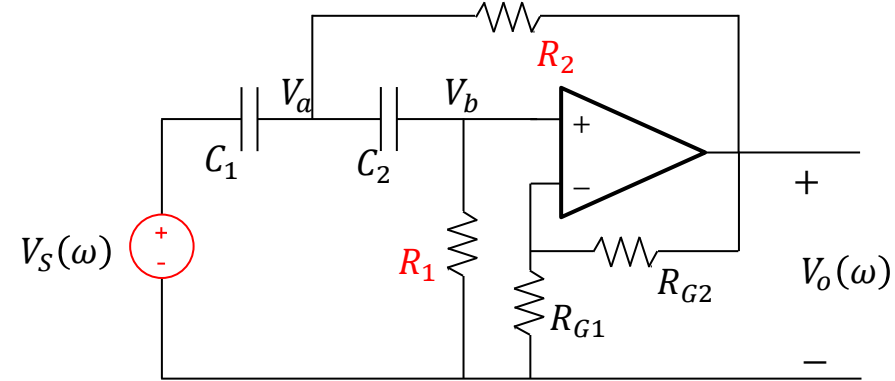


From low-pass to high-pass Filter circuit

Video

2nd order Sallen-Key high-pass filter

To determine the component values, we must match the circuit equation with the performance equation.



$$H_{HP}(j\omega) = \frac{K(j\omega)^2}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_1} + \frac{1}{R_1 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

The performance equation for the **frequency-normalized** Butterworth high-pass filter:

$$H_{HP}(\hat{\omega}_{HP}) = \frac{b_2(j\hat{\omega}_{HP})^2}{(j\hat{\omega}_{HP})^2 + \frac{a_1}{\sqrt{a_0}}(j\hat{\omega}_{HP}) + 1}$$

Sensitivity analysis on coefficients.

$$\frac{1}{Q} \stackrel{\text{def}}{=} 2\zeta \Rightarrow$$

$$Q \stackrel{\text{def}}{=} \frac{1}{2\zeta} = \frac{1}{2} \frac{2\sqrt{a_0}}{a_1} = \frac{\sqrt{a_0}}{a_1}$$

We will set the gain to unity, $K = 1$.

$$H_{HP}(j\omega) = \frac{K(j\omega)^2}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_1} + \frac{1}{R_1 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\omega_n = \sqrt{\frac{1}{R_1 R_2 C_1 C_2}}$$

$$S_x^y \stackrel{\text{def}}{=} \frac{x}{y} \frac{\partial y}{\partial x}$$

$$Q_{HP} = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(C_1 + C_2)R_2 + R_1 C_2(1-K)}$$

$$S_x^{\omega_n} = -\frac{1}{2}$$

$$Q_{HP} = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(C_1 + C_2)R_2} = \sqrt{\frac{R_1}{R_2}} \cdot \frac{\sqrt{C_1 C_2}}{C_1 + C_2}$$

$$S_{R_1}^Q = \frac{1}{2} \quad S_{R_2}^Q = -\frac{1}{2}$$

$$S_{C_1}^Q = \frac{C_2 - C_1}{2(C_1 + C_2)}$$

$$S_{C_2}^Q = \frac{C_1 - C_2}{2(C_1 + C_2)}$$

Observation:

For a Sallen-Key high-pass filters we obtain low sensitivity for Q if we match the capacitors.

By matching coefficients in the above two equations, we obtain two equations:

$$H_{HP}(j\hat{\omega}) = \frac{K(j\omega)^2}{(j\hat{\omega})^2 + j\hat{\omega} \left(\frac{1}{\hat{R}_1\hat{C}_1} + \frac{1}{\hat{R}_1\hat{C}_2} + \frac{(1-K)}{\hat{R}_2\hat{C}_1} \right) + \frac{1}{\hat{R}_1\hat{R}_2\hat{C}_1\hat{C}_2}}$$

$$H_{HP}(j\hat{\omega}) = \frac{K(j\hat{\omega})^2}{(j\hat{\omega})^2 + a_1(j\hat{\omega}) + a_0}$$

This time we match capacitors.

$$K \stackrel{\text{def}}{=} 1, \hat{C}_1 \stackrel{\text{def}}{=} \hat{C}_2 \stackrel{\text{def}}{=} \hat{C}$$

$$\frac{2}{\hat{R}_1\hat{C}} = a_1 \qquad \hat{R}_1\hat{C} = \frac{2}{a_1}$$

$$\hat{R}_1\hat{R}_2\hat{C}^2 = \frac{1}{a_0} \qquad \hat{R}_2\hat{C} \frac{2}{a_1} = \frac{1}{a_0} \qquad \hat{R}_2\hat{C} = \frac{a_1}{2a_0}$$

We eliminate the capacitor and obtain a ratio for the resistors:

$$\frac{\hat{R}_1}{\hat{R}_2} = \frac{2}{a_1} \frac{2a_0}{a_1} = \frac{4a_0}{a_1^2} = \frac{4 \cdot 1}{(\sqrt{2})^2} = 2 \approx \frac{68}{33}$$

We have several options.

We can use a look-up table for time constants and in one and the same column (same C value) find the two best time constants.

But there might be more than one useful column (C value).

If we decide to stick to a ratio of ~ 2 for the resistors, we can obtain a suggestion for the best capacitor value:

We see that the product of the two time constants is 1.

This is best obtained for $C = 1.2\text{F}$, $R_1 = 1.2\Omega$, $R_2 = 0.56\Omega$

$$\hat{\tau}_1 = \hat{R}_1 \hat{C} = \frac{2}{a_1} = 1.414 \Rightarrow \hat{C} = \frac{1.414\text{s}}{0.68\Omega} = 2.079\text{F}$$

$$\hat{\tau}_2 = \hat{R}_2 \hat{C} = \frac{a_1}{2a_0} = 0.707 \Rightarrow \hat{C} = \frac{0.707\text{s}}{0.33\Omega} = 2.142\text{F}$$

Good options are 1.2, 1.5, and 2.2F

$$\hat{R}_1 \hat{R}_2 \hat{C}^2 = \frac{1}{a_0} \Rightarrow \hat{C}^2 = \frac{1}{a_0 \hat{R}_1 \hat{R}_2}$$

$$\hat{C} = \frac{1}{\sqrt{a_0 \hat{R}_1 \hat{R}_2}} = \frac{1}{\sqrt{0.68 \times 0.33}} = 2.11 \approx 2.2$$

$$\hat{R}_1 \hat{R}_2 \hat{C}^2 = \frac{1}{a_0} = 1 \Rightarrow \tau_1 \tau_2 = 1$$

2nd order Sallen-Key high-pass filter - Butterworth

$$\hat{t}_1 = \hat{R}_1 \hat{C} = \frac{2}{a_1} = \frac{2}{\sqrt{2}} = \sqrt{2} = 1.414$$

$$\hat{t}_2 = \hat{R}_2 \hat{C} = \frac{a_1}{2a_0} = \frac{\sqrt{2}}{2} = 0.707$$



R1*C1	0,1	0,12	0,15	0,18	0,22	0,27	0,33	0,39	0,47	0,56	0,68	0,82	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10
0,1	0,010	0,012	0,015	0,018	0,022	0,027	0,033	0,039	0,047	0,056	0,068	0,082	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000
0,12	0,012	0,014	0,018	0,022	0,026	0,032	0,040	0,047	0,056	0,067	0,082	0,098	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200
0,15	0,015	0,018	0,023	0,027	0,033	0,041	0,050	0,059	0,071	0,084	0,102	0,123	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500
0,18	0,018	0,022	0,027	0,032	0,040	0,049	0,059	0,070	0,085	0,101	0,122	0,148	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800
0,22	0,022	0,026	0,033	0,040	0,048	0,059	0,073	0,086	0,103	0,123	0,150	0,180	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200
0,27	0,027	0,032	0,041	0,049	0,059	0,073	0,089	0,105	0,127	0,151	0,184	0,221	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700
0,33	0,033	0,040	0,050	0,059	0,073	0,089	0,109	0,129	0,155	0,185	0,224	0,271	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300
0,39	0,039	0,047	0,059	0,070	0,086	0,105	0,129	0,152	0,183	0,218	0,265	0,320	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900
0,47	0,047	0,056	0,071	0,085	0,103	0,127	0,155	0,183	0,221	0,263	0,320	0,385	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700
0,56	0,056	0,067	0,084	0,101	0,123	0,151	0,185	0,218	0,263	0,314	0,381	0,459	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600
0,68	0,068	0,082	0,102	0,122	0,150	0,184	0,224	0,265	0,320	0,381	0,462	0,558	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800
0,82	0,082	0,098	0,123	0,148	0,180	0,221	0,271	0,320	0,385	0,459	0,558	0,672	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200
1	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000
1,2	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200	1,440	1,800	2,160	2,640	3,240	3,960	4,680	5,640	6,720	8,160	9,840	12,000
1,5	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500	1,800	2,250	2,700	3,300	4,050	4,950	5,850	7,050	8,400	10,200	12,300	15,000
1,8	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800	2,160	2,700	3,240	3,960	4,860	5,940	7,020	8,460	10,080	12,240	14,760	18,000
2,2	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200	2,640	3,300	3,960	4,840	5,940	7,260	8,580	10,340	12,320	14,960	18,040	22,000
2,7	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700	3,240	4,050	4,860	5,940	7,290	8,910	10,530	12,690	15,120	18,360	22,140	27,000
3,3	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300	3,960	4,950	5,940	7,260	8,910	10,890	12,870	15,510	18,480	22,440	27,060	33,000
3,9	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900	4,680	5,850	7,020	8,580	10,530	12,870	15,210	18,330	21,840	26,520	31,980	39,000
4,7	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700	5,640	7,050	8,460	10,340	12,690	15,510	18,330	22,090	26,320	31,960	38,540	47,000
5,6	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600	6,720	8,400	10,080	12,320	15,120	18,480	21,840	26,320	31,360	38,080	45,920	56,000
6,8	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800	8,160	10,200	12,240	14,960	18,360	22,440	26,520	31,960	38,080	46,240	55,760	68,000
8,2	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200	9,840	12,300	14,760	18,040	22,140	27,060	31,980	38,540	45,920	55,760	67,240	82,000
10	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000	12,000	15,000	18,000	22,000	27,000	33,000	39,000	47,000	56,000	68,000	82,000	100,00

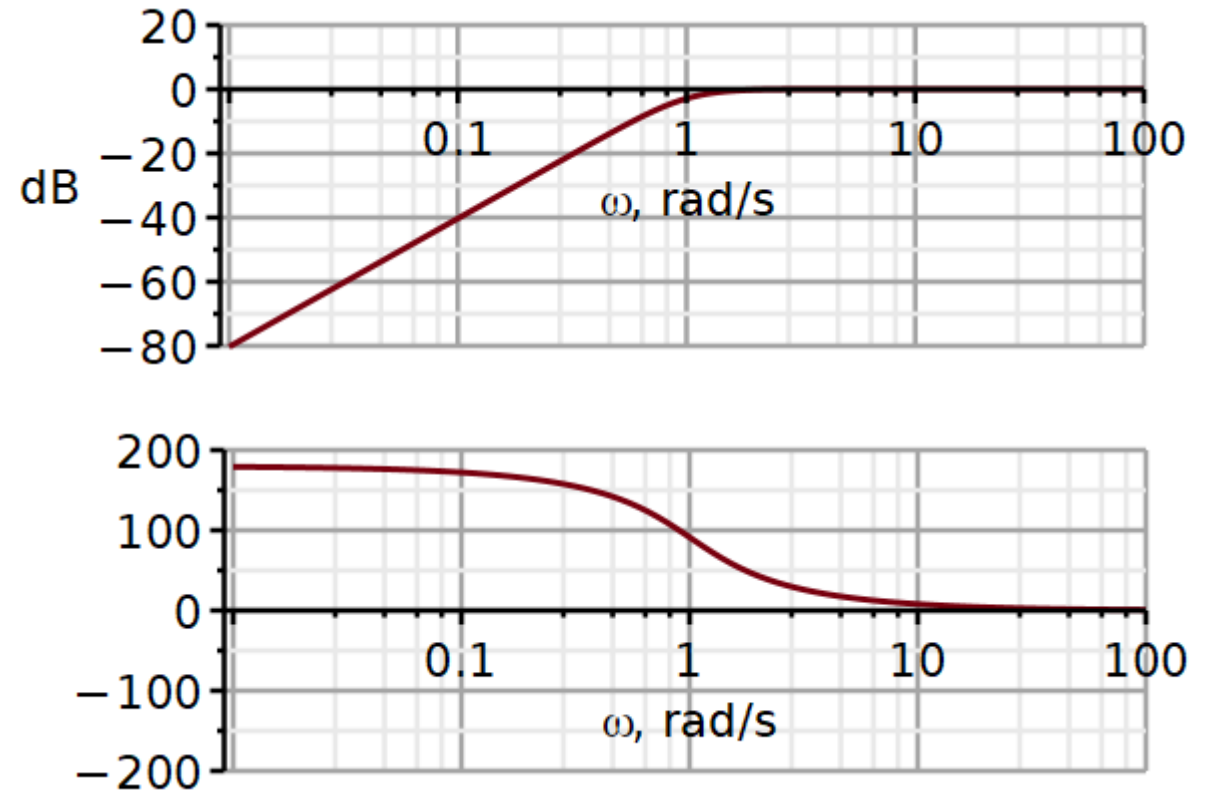
$$\tau_1 \tau_2 = 0.968 \ 1.0575 \ 1.086$$

In the RC look-up table on the previous slide we see that there are several options. Here we have chosen this combination:

$$\hat{R}_1 \approx 1.2\Omega$$

$$\hat{R}_2 \approx 0.56\Omega$$

$$\hat{C} \approx 1.2F$$



The low-frequency asymptote has a slope of +40 dB/decade, the cut-off is at 1 rad/s and the phase is 90 degrees at the cut-off. So, it looks promising.

The frequency characteristic for the frequency-normalized high-pass filter. The hats indicate that the components have the value for a cut-off frequency of 1 rad/s.

We can multiply the normalized frequency variable $\hat{\omega}$ by a factor K_F if we also divide all capacitors by the same factor.

$$\omega \stackrel{\text{def}}{=} K_F \hat{\omega}$$

$$H_{HP}(j\hat{\omega}) = \frac{K(j\hat{\omega})^2}{(j\hat{\omega})^2 + j\hat{\omega} \left(\frac{1}{\hat{R}_1 \hat{C}_1} + \frac{1}{\hat{R}_1 \hat{C}_2} + \frac{(1-K)}{\hat{R}_2 \hat{C}_1} \right) + \frac{1}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}$$

$$H_{HP}(j\hat{\omega}) = \frac{K(j\hat{\omega})^2 \hat{C}_1 \hat{C}_2}{(j\hat{\omega})^2 \hat{C}_1 \hat{C}_2 + j\hat{\omega} \left(\frac{\hat{C}_2}{\hat{R}_1} + \frac{\hat{C}_1}{\hat{R}_1} + \frac{\hat{C}_2(1-K)}{\hat{R}_2} \right) + \frac{1}{\hat{R}_1 \hat{R}_2}}$$

$$H_{HP}(j\hat{\omega}) = \frac{K(jK_F \hat{\omega})^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F}}{(jK_F \hat{\omega})^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F} + jK_F \hat{\omega} \left(\frac{\hat{C}_2}{K_F \hat{R}_1} + \frac{\hat{C}_1}{K_F \hat{R}_1} + \frac{\hat{C}_2(1-K)}{K_F \hat{R}_2} \right) + \frac{1}{\hat{R}_1 \hat{R}_2}}$$

$$H_{HP}(j\omega) = \frac{K(j\omega)^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F}}{(j\omega)^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F} + j\omega \left(\frac{\hat{C}_2}{K_F \hat{R}_1} + \frac{\hat{C}_1}{K_F \hat{R}_1} + \frac{\hat{C}_2(1-K)}{K_F \hat{R}_2} \right) + \frac{1}{\hat{R}_1 \hat{R}_2}}$$

By frequency scaling, we scale the normalized frequency axis while also downscaling all capacitors.

The shape of the magnitude and phase curves are determined by the resistance values and the values of the capacitor impedances. At the normalized cut-off frequency $\hat{\omega}_c$, the resistors and the capacitor impedances have certain values.

The frequency scaling changes the frequency scale and the capacitor values without changing the capacitor impedance. So, we now have the same capacitor impedance, but at a new cut-off frequency ω_c .

$$H_{HP}(j\omega) = \frac{K(j\omega)^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F}}{(j\omega)^2 \frac{\hat{C}_1}{K_F} \frac{\hat{C}_2}{K_F} + j\omega \left(\frac{\hat{C}_2}{K_F \hat{R}_1} + \frac{\hat{C}_1}{K_F \hat{R}_1} + \frac{\hat{C}_2(1-K)}{K_F \hat{R}_2} \right) + \frac{1}{\hat{R}_1 \hat{R}_2}}$$

$$H_{HP}(j\omega) = \frac{K(j\omega)^2 C'_1 C'_2}{(j\omega)^2 C'_1 C'_2 + j\omega \left(\frac{C'_2}{R'_1} + \frac{C'_2}{R'_1} + \frac{C'_2(1-K)}{R'_2} \right) + \frac{1}{R'_1 R'_2}}$$

$$R' = \hat{R} \quad C' \stackrel{\text{def}}{=} \frac{\hat{C}}{K_F}$$

$$Z_C = \frac{1}{j\hat{\omega}_c \hat{C}} = \frac{1}{j(K_F \hat{\omega}_c) \left(\frac{\hat{C}}{K_F} \right)} = \frac{1}{j\omega_c C'}$$

Frequency scaling

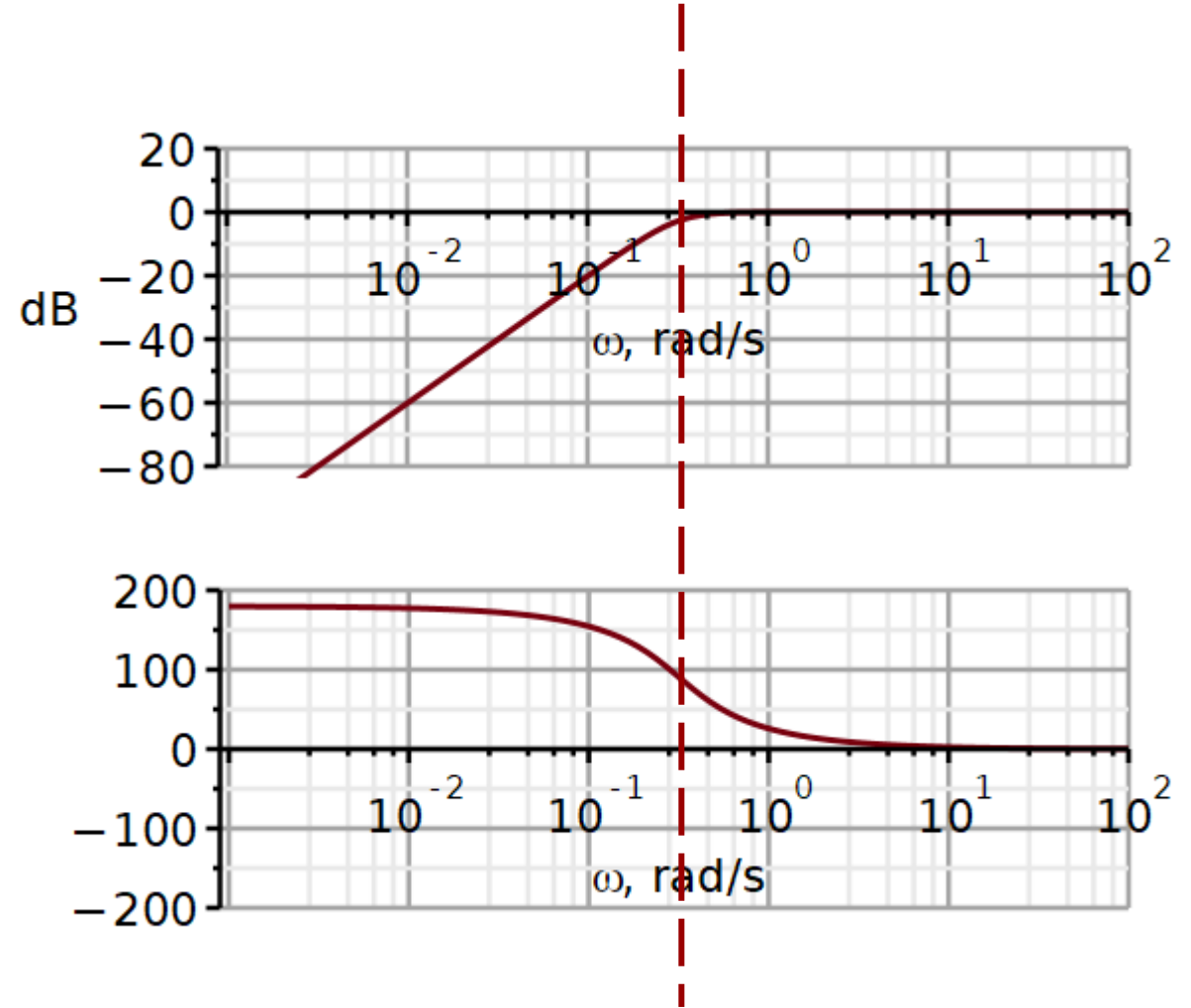
$$K_F = 2\pi \cdot 0.05 = 0.314159$$

$$C'_1 = \frac{\hat{C}_1}{K_F} = \frac{1.2\text{F}}{0.314159} = 3.82\text{F} \approx 3.9\text{ F}$$

$$C'_2 = \frac{\hat{C}_1}{K_F} = 3.9\text{ F}$$

$$R'_1 = \hat{R}_1 = 1.2\Omega$$

$$R'_2 = \hat{R}_2 = 0.56\Omega$$



2nd order Sallen-Key high pass filter - Butterworth

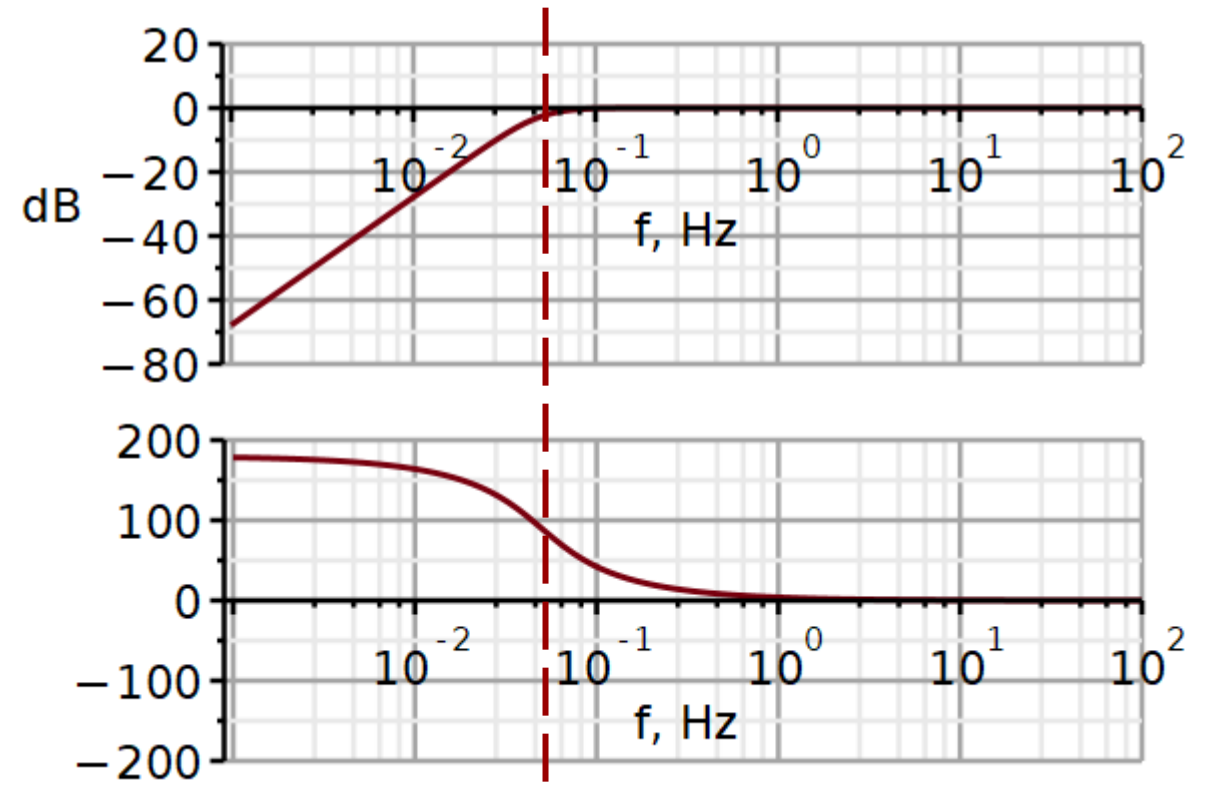
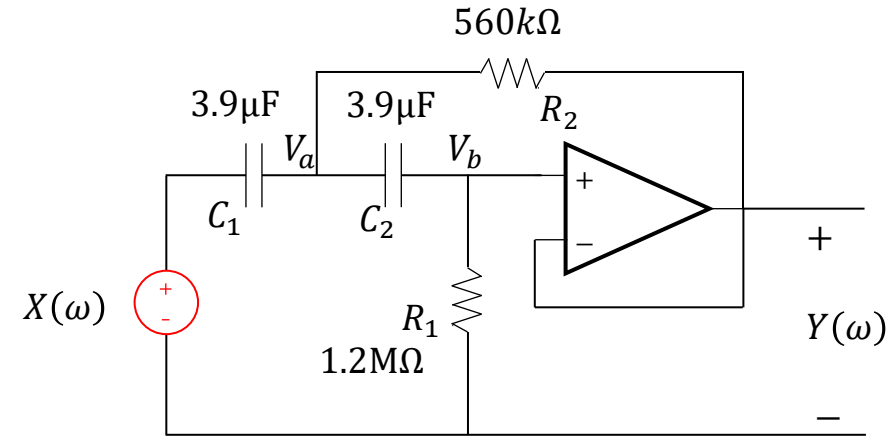
Impedance scaling:

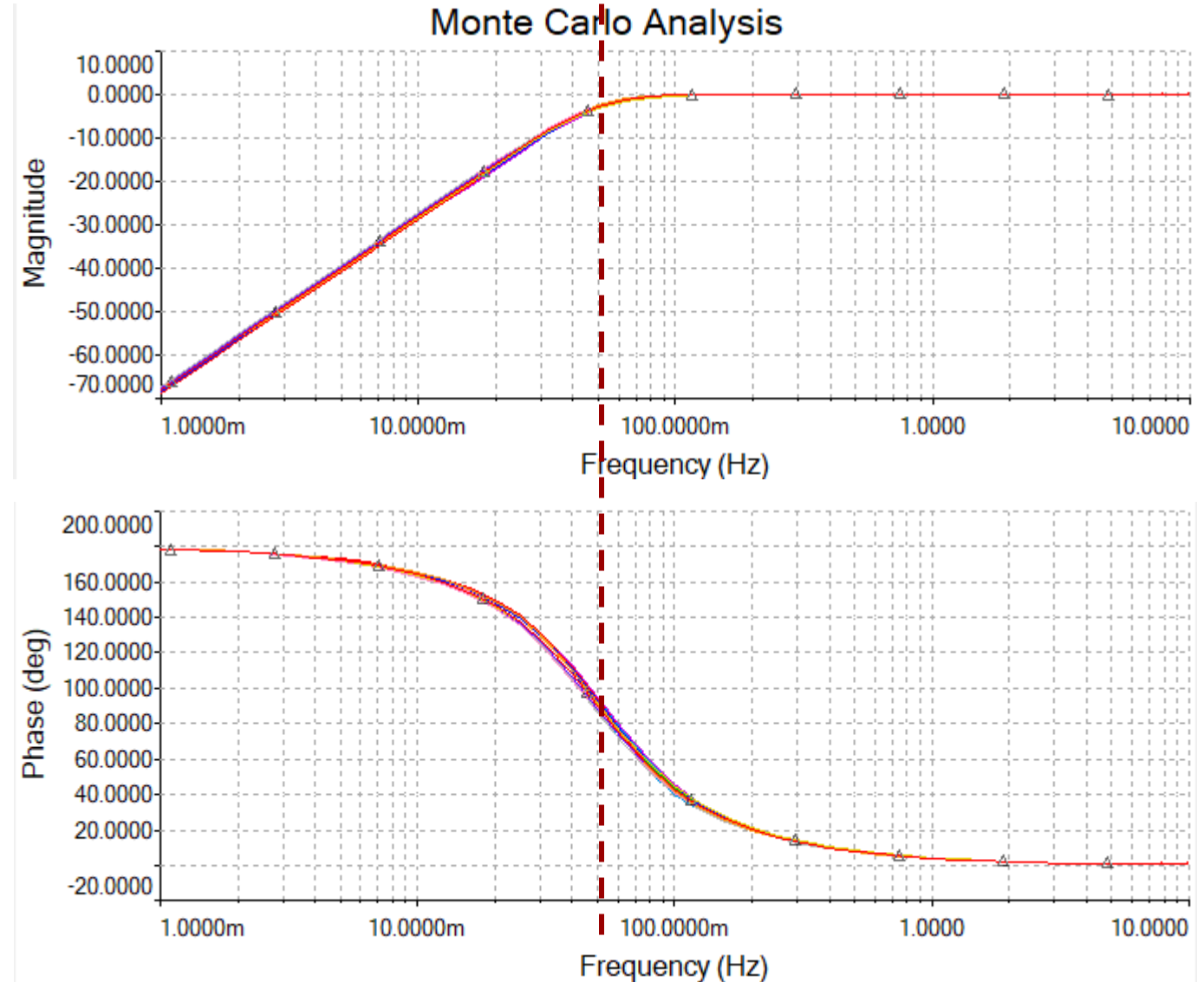
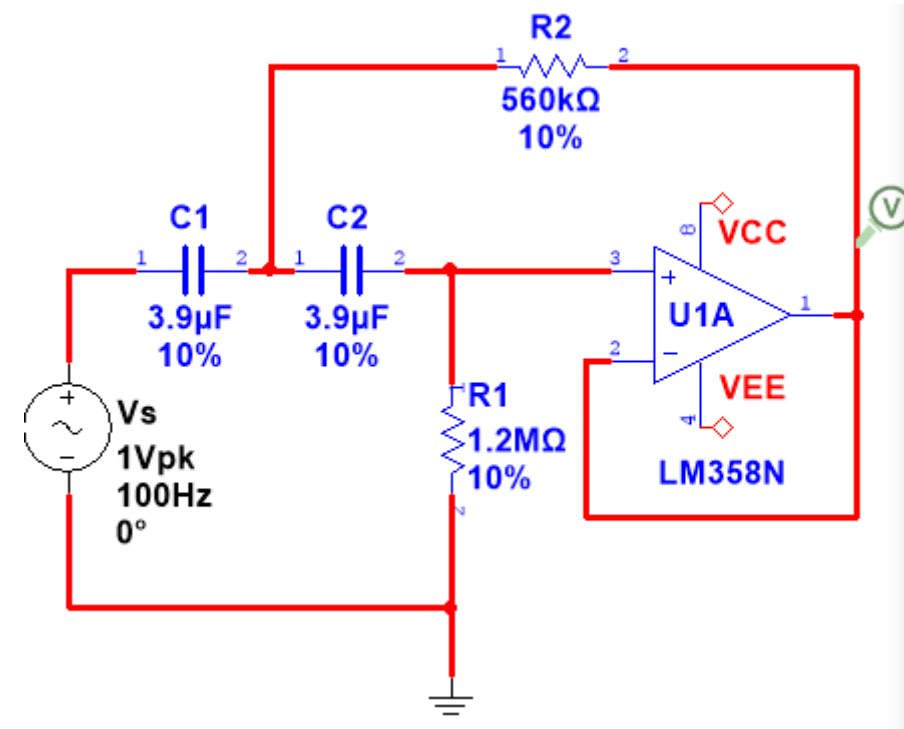
$$C_1 = C_2 = 3.9\mu\text{F} \quad \leftarrow \text{desired capacitor value}$$

$$K_z = \frac{C'_1}{3.9 \cdot 10^{-6}\text{F}} = \frac{3.9\text{F}}{3.9 \cdot 10^{-6}\text{F}} = 1 \times 10^6$$

$$R_1 = K_z \cdot R'_1 = K_z \cdot 1.2\Omega = 1.2\text{M}\Omega$$

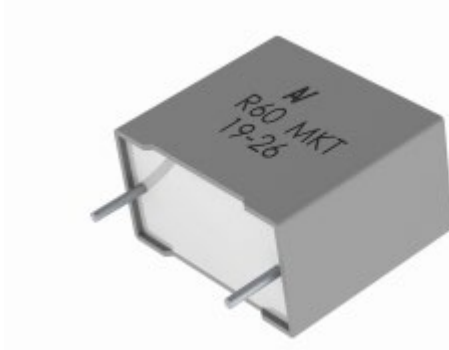
$$R_2 = K_z \cdot R'_2 = K_z \cdot 0.56\Omega = 560\text{k}\Omega$$





Capacitors for filter circuits can be quite large.

This one will not fit in a Smartwatch.



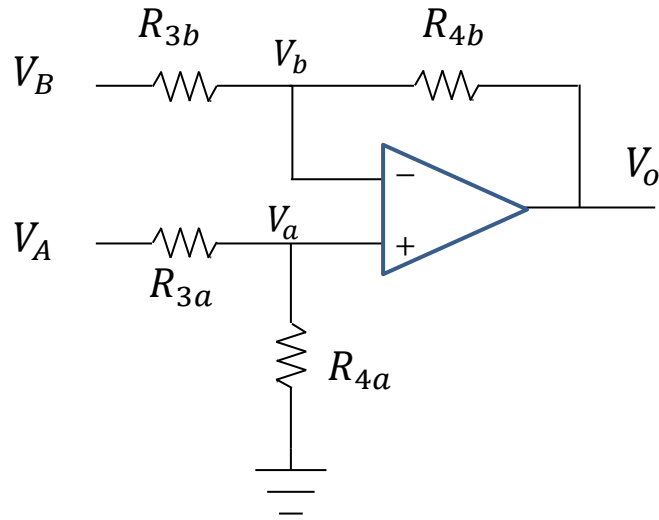
<https://docs.rs-online.com/74f3/0900766b8171c883.pdf>

VDC	VAC	Capacitance Value (μF)	Dimensions in mm			Lead Spacing
			B	H	L	
63	40	0.68	5.0	11.0	18.0	15.0
63	40	1.0	5.0	11.0	18.0	15.0
63	40	1.5	5.0	11.0	18.0	15.0
63	40	2.2	6.0	12.0	18.0	15.0
63	40	3.3	7.5	13.5	18.0	15.0
63	40	3.3	9.0	12.5	18.0	15.0
63	40	4.7	8.5	14.5	18.0	15.0
63	40	4.7	13.0	12.0	18.0	15.0
63	40	6.8	10.0	16.0	18.0	15.0
63	40	3.3	6.0	15.0	26.5	22.5
63	40	4.7	7.0	16.0	26.5	22.5
63	40	6.8	7.0	16.0	26.5	22.5
63	40	10.0	8.5	17.0	26.5	22.5
63	40	15.0	11.0	20.0	26.5	22.5
63	40	4.7	13.0	12.0	32.0	27.5
63	40	6.8	13.0	12.0	32.0	27.5
63	40	10.0	9.0	17.0	32.0	27.5
63	40	15.0	11.0	20.0	32.0	27.5
63	40	15.0	24.0	15.0	32.0	27.5
63	40	22.0	13.0	22.0	32.0	27.5
63	40	22.0	24.0	15.0	32.0	27.5
63	40	22.0	11.0	20.0	32.0	27.5
63	40	33.0	18.0	33.0	32.0	27.5
63	40	47.0	18.0	33.0	32.0	27.5
63	40	47.0	22.0	37.0	32.0	27.5
63	40	68.0	22.0	37.0	32.0	27.5
63	40	22.0	11.0	22.0	41.5	37.5
63	40	22.0	24.0	15.0	41.5	37.5
63	40	33.0	16.0	28.5	41.5	37.5
63	40	33.0	24.0	15.0	41.5	37.5
63	40	47.0	16.0	28.5	41.5	37.5
63	40	47.0	24.0	19.0	41.5	37.5
63	40	68.0	19.0	32.0	41.5	37.5
63	40	100.0	20.0	40.0	41.5	37.5
63	40	100.0	24.0	44.0	41.5	37.5
100	63	0.33	5.0	11.0	18.0	15.0
100	63	0.47	5.0	11.0	18.0	15.0
VDC	VAC	Capacitance Value (μF)	B (mm)	H (mm)	L (mm)	Lead Spacing

Instrumentation amplifier

Video

Using **source superposition**:



$$V_{o,A}|_{V_B=0} = \left(1 + \frac{R_{4b}}{R_{3b}}\right) V_a = \left(1 + \frac{R_{4b}}{R_{3b}}\right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}}\right) V_A$$

$$V_{o,B}|_{V_A=0} = -\frac{R_{4b}}{R_{3b}} V_B$$

$$V_o = V_{o,A}|_{V_B=0} + V_{o,B}|_{V_A=0} = \left(\frac{R_{3b} + R_{4b}}{R_{3b}}\right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}}\right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$

By matching resistance values, we can simplify the circuit equation.

$$V_o = \frac{R_{4a}}{R_{3a}} (V_A - V_B) \quad R_{3a} = R_{3b} \quad R_{4a} = R_{4b}$$

3 op-amp model of instrumentation amplifier

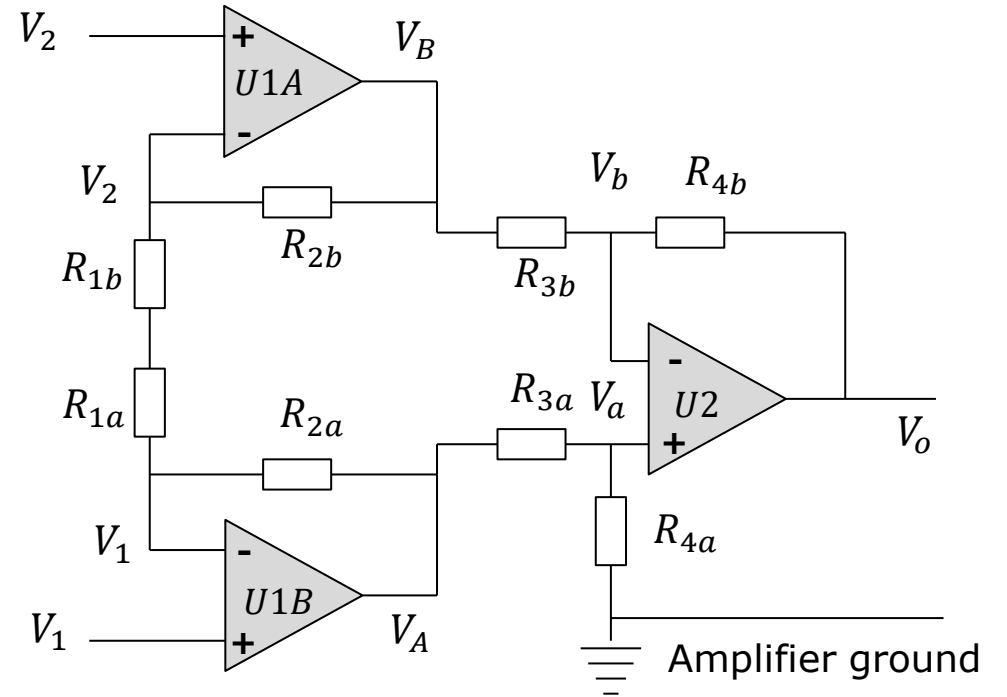
To increase the input impedance, we can place a **non-inverting op-amp** in front of nodes V_A and V_B .

Using **source superposition** we obtain:

$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}}\right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

$$V_B = \left(1 + \frac{R_{2b}}{R_{1a} + R_{1b}}\right) \cdot V_2 - \frac{R_{2b}}{R_{1a} + R_{1b}} \cdot V_1$$

$$V_o = \left(\frac{R_{3b} + R_{4b}}{R_{3b}}\right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}}\right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$



These expressions are greatly simplified if we **match resistors**.

$$R_{3a} = R_{3b} = R_{4a} = R_{4b}$$

$$R_{2a} = R_{2b}$$

$$R_{1a} = R_{1b} = \frac{R_G}{2}$$

Unmatched resistors

$$V_o = \left(\frac{R_{3b} + R_{4b}}{R_{3b}} \right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}} \right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$

$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

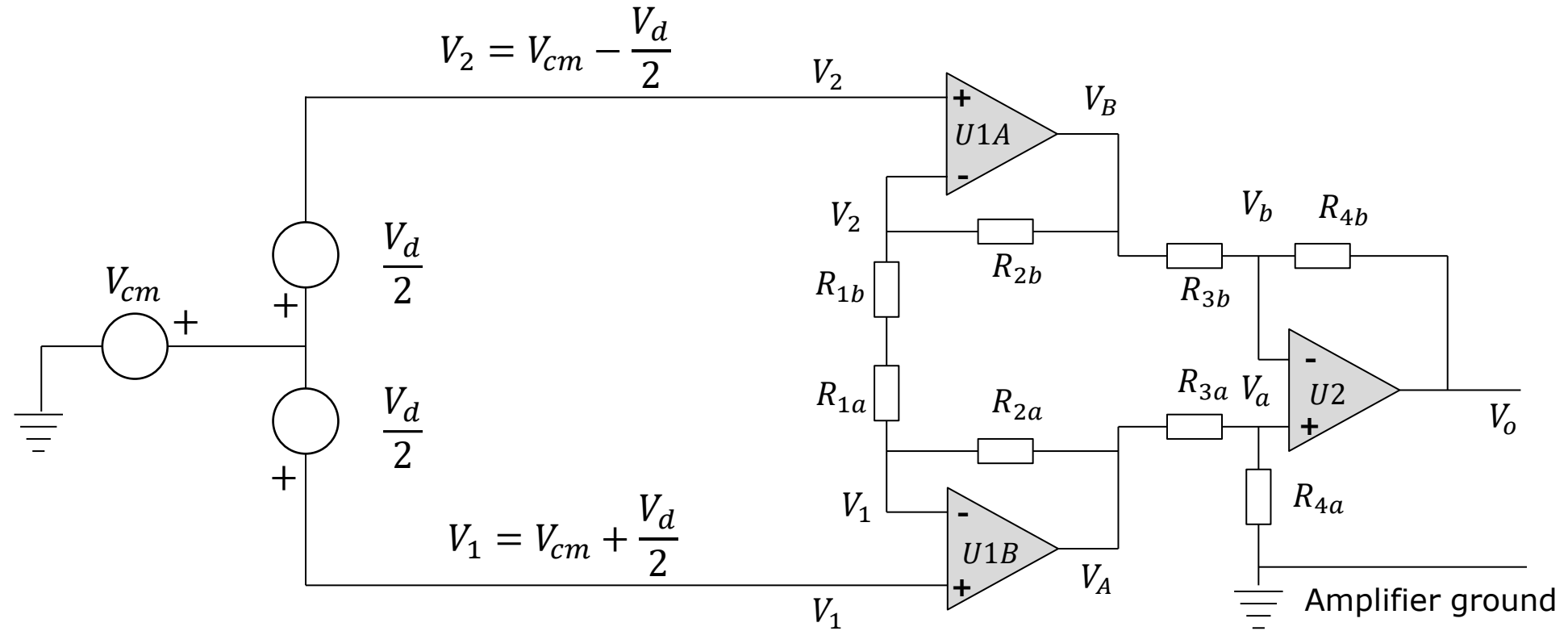
$$V_B = \left(1 + \frac{R_{2b}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{R_{2b}}{R_{1a} + R_{1b}} \cdot V_1$$

Matched resistors

$$V_o = V_A - V_B$$

$$V_A = \left(1 + \frac{R_2}{R_G} \right) \cdot V_1 - \frac{R_2}{R_G} \cdot V_2$$

$$V_B = \left(1 + \frac{R_2}{R_G} \right) \cdot V_2 - \frac{R_2}{R_G} \cdot V_1$$



We have two input signals V_1 and V_2 .

The difference between the signal is called the **differential signal**:

$$V_d \stackrel{\text{def}}{=} V_1 - V_2$$

The average is called the **common mode signal**:

$$V_{cm} \stackrel{\text{def}}{=} \frac{V_1 + V_2}{2}$$

An instrumentation amplifier processes differential and common mode signals very differently. To see this, we express the input signals in terms of V_d and V_{cm} .

$$V_d \stackrel{\text{def}}{=} V_1 - V_2 \quad V_1 = V_{cm} + \frac{V_d}{2}$$

$$V_{cm} \stackrel{\text{def}}{=} \frac{V_1 + V_2}{2} \quad V_2 = V_{cm} - \frac{V_d}{2}$$

$$V_A = \left(1 + \frac{R_2}{R_G}\right) \cdot V_1 - \frac{R_2}{R_G} \cdot V_2$$

$$V_A = \left(1 + \frac{R_2}{R_G}\right) \cdot \left(V_{cm} + \frac{V_d}{2}\right) - \frac{R_2}{R_G} \cdot \left(V_{cm} - \frac{V_d}{2}\right)$$

$$V_A = \left(V_{cm} + \frac{R_2}{R_G} V_{cm}\right) + \left(1 + \frac{R_2}{R_G}\right) \cdot \frac{V_d}{2} - \left(\frac{R_2}{R_G} V_{cm} - \frac{R_2}{R_G} \frac{V_d}{2}\right)$$

$$V_A = V_{cm} + \left(1 + \frac{2R_2}{R_G}\right) \cdot \frac{V_d}{2}$$

$$V_A = V_{cm} + G \cdot \frac{V_d}{2}$$

Observation:

V_{cm} is not amplified.

Similar derivation
for node B:

$$V_d \stackrel{\text{def}}{=} V_1 - V_2 \quad V_1 = V_{cm} + \frac{V_d}{2}$$

$$V_{cm} \stackrel{\text{def}}{=} \frac{V_1 + V_2}{2} \quad V_2 = V_{cm} - \frac{V_d}{2}$$

$$V_B = \left(1 + \frac{R_2}{R_G}\right) \cdot V_2 - \frac{R_2}{R_G} \cdot V_1$$

$$V_B = \left(1 + \frac{R_2}{R_G}\right) \cdot \left(V_{cm} - \frac{V_d}{2}\right) - \frac{R_2}{R_G} \cdot \left(V_{cm} + \frac{V_d}{2}\right)$$

$$V_B = \left(V_{cm} + \frac{R_2}{R_G} V_{cm}\right) - \left(1 + \frac{R_2}{R_G}\right) \cdot \frac{V_d}{2} - \left(\frac{R_2}{R_G} V_{cm} + \frac{R_2}{R_G} \frac{V_d}{2}\right)$$

$$V_B = V_{cm} - \left(1 + \frac{2R_2}{R_G}\right) \cdot \frac{V_d}{2}$$

$$V_B = V_{cm} - G \cdot \frac{V_d}{2}$$

Observation:

V_{cm} is not amplified.

To avoid signal saturation, we need to be concerned about the magnitude of the voltage in the internal nodes V_A and V_B .

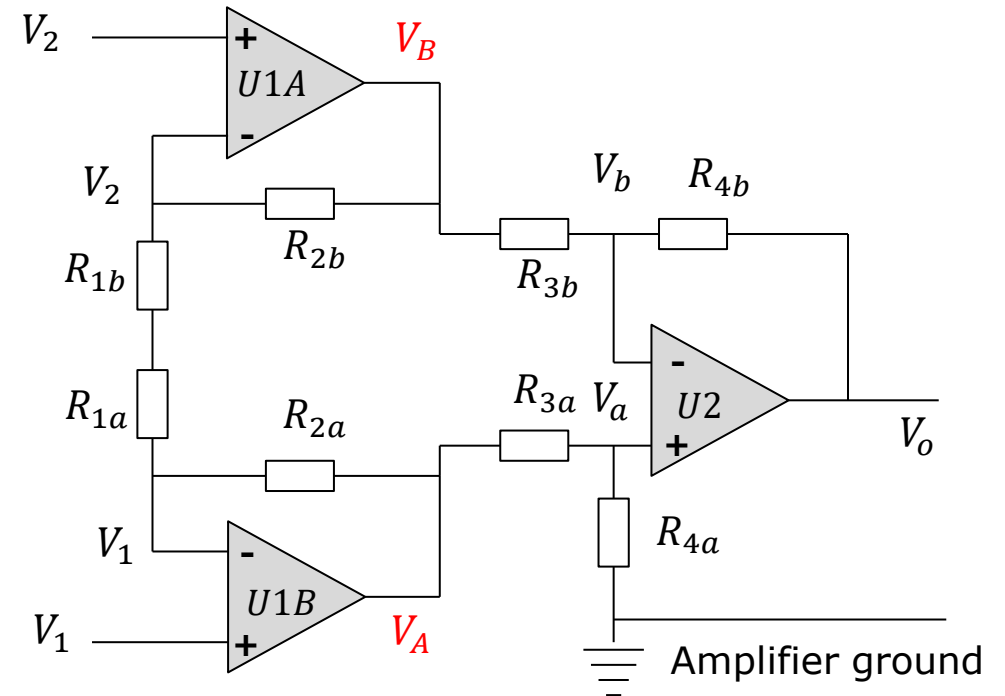
It is not uncommon that the common mode signal is significantly larger than the differential signal.

This would often be the case, if the common mode signal is due to electrical interference.

V_{cm} is not amplified in the gain stages U1A and U1B. This implies that a fairly large common mode signal may be tolerated without saturation.

$$V_2 = V_{cm} - \frac{V_d}{2}$$

$$V_B = V_{cm} - G \cdot \frac{V_d}{2}$$



$$V_1 = V_{cm} + \frac{V_d}{2}$$

$$V_A = V_{cm} + G \cdot \frac{V_d}{2}$$

Signal saturation threshold in an op-amp or in-amp depends on the voltage supply and if the in-amp or op-amp can operate with signal amplitudes equal to the voltage supply or if it has **rail margins**.

For a given component, the saturation voltage can be found in the data sheet.

If we know the amplitude of the common mode signal and the differential signal, we can calculate the maximum gain we can achieve without saturating internal nodes.

$$-V_{sat} < V_A = V_{cm} + G \cdot \frac{V_d}{2} < V_{sat}$$

$$-V_{sat} < V_B = V_{cm} - G \cdot \frac{V_d}{2} < V_{sat}$$

Example: LM358 op-amp.

$$|V_{sat}| < |V_{CC}| - |V_{rail-margin}|$$

$$V_{sat} < 5V - 1.5V = 3.5V$$

$$-3.5V < V_A, V_B < 3.5V$$

Output of non-saturated in-amp:

$$V_o = \frac{R_{4a}}{R_{3a}} (V_A - V_B)$$

$$V_A = V_{cm} + G \cdot \frac{V_d}{2} \qquad V_B = V_{cm} - G \cdot \frac{V_d}{2}$$

$$R_{3a} = R_{3b} = R_{4a} = R_{4b} \Rightarrow$$

$$V_o = (V_A - V_B)$$

Taking the difference, the common mode signal is subtracted.

$$V_o = V_{cm} + G \cdot \frac{V_d}{2} - V_{cm} + G \cdot \frac{V_d}{2}$$

$$V_o = G \cdot V_d$$

Observation:

An ideal instrumentation amplifier

- amplifies the differential signal
- cancels completely the common mode signal.

We derived earlier:

$$V_o = \left(\frac{R_{3b} + R_{4b}}{R_{3b}} \right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}} \right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$

$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

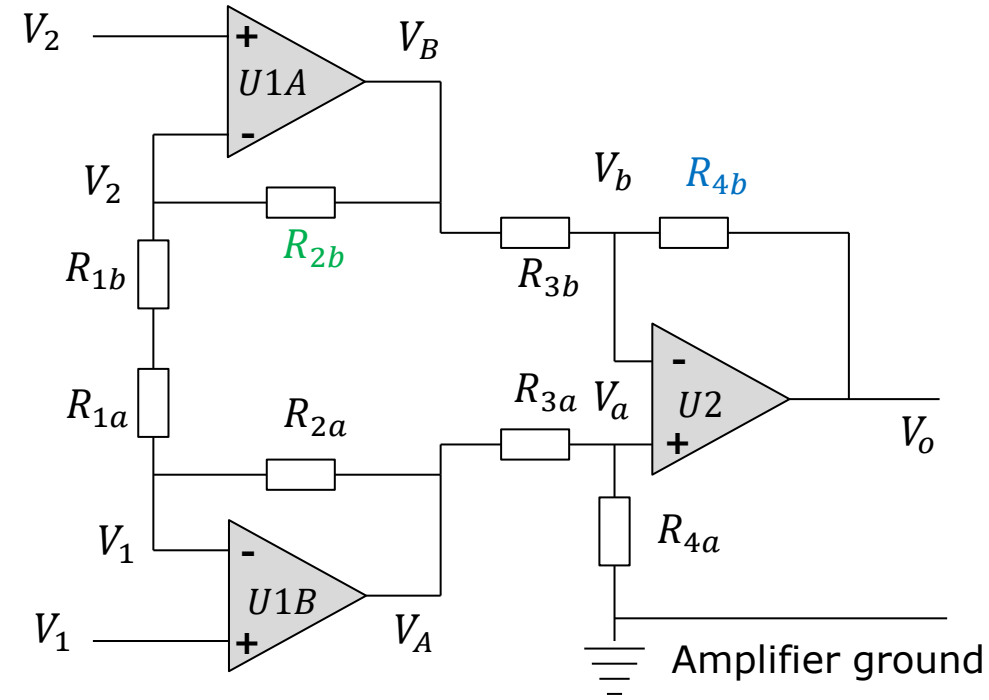
$$V_B = \left(1 + \frac{R_{2b}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{R_{2b}}{R_{1a} + R_{1b}} \cdot V_1$$

We now permit **imbalance** between certain internal resistors:

$$R_{3a} = R_{3b} = R_{4a}$$

$$R_{4b} = \alpha R_{4a}$$

$$R_{2b} = \beta R_{2a}$$



It is shown in an **appendix** that:

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

Output from
differential
signal:

$$V_1 = -V_2 = \frac{V_d}{2} \Rightarrow$$

$$V_{od} = \left(\frac{1 + 3\alpha}{4} + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) \right) V_d$$

Output from
common mode
signal:

$$V_1 = V_2 = V_{cm} \Rightarrow$$

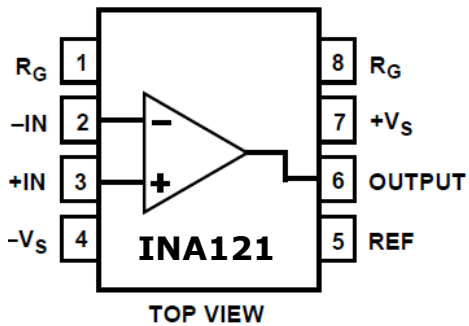
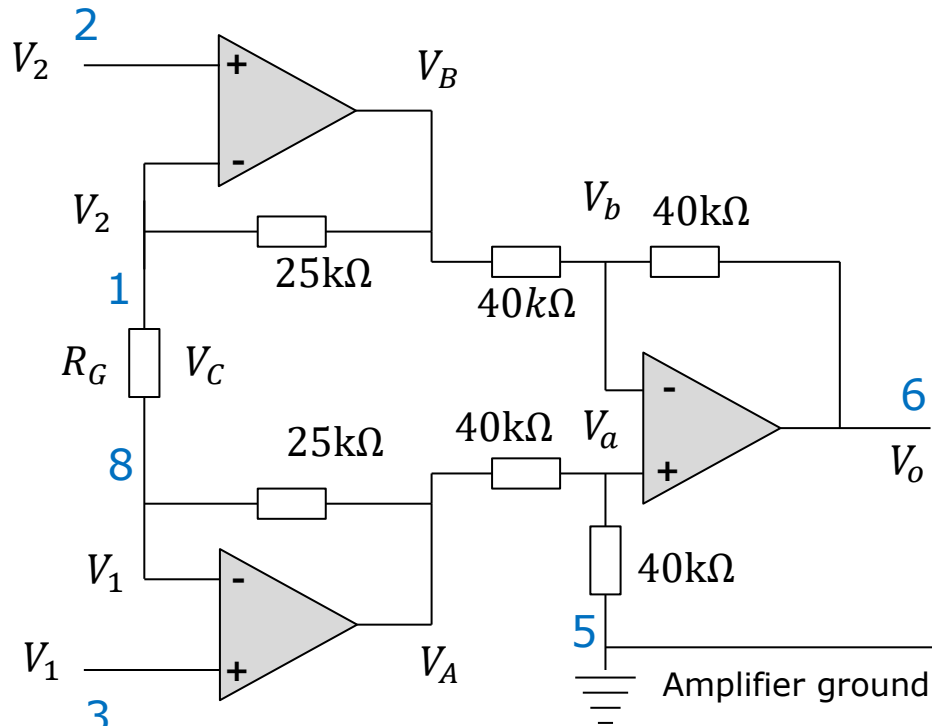
$$V_{oc} = \left(\frac{1 + \alpha}{2} \right) \cdot V_{cm} - \alpha \cdot V_{cm} + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_{cm} - V_{cm})$$

$$V_{oc} = \left(\frac{1 - \alpha}{2} \right) V_{cm} \quad G_c = \frac{V_{oc}}{V_{cm}} = \frac{1 - \alpha}{2}$$

Observation:

If we were to build an in-amp using three op-amps, we would be challenged to obtain perfect match of resistors. The result is a poor rejection of common mode signals. Although much pricier than three op-amps, a monolithic in-amp is a far better option.

The INA121 Instrumentation amplifier



$$V_o = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_G} \right) (V_1 - V_2)$$

$$V_o = \frac{40\text{k}\Omega}{40\text{k}\Omega} \left(1 + \frac{2 \cdot 25\text{k}\Omega}{R_G} \right) (V_1 - V_2)$$

$$V_o = \left(1 + \frac{50\text{k}\Omega}{R_G} \right) (V_1 - V_2)$$

Gain $G = 1 + \frac{50\text{k}\Omega}{R_G}$

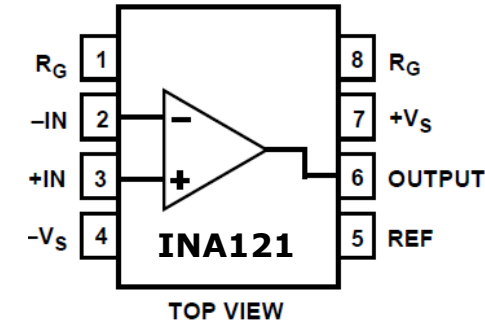
Gain resistor $R_G = \frac{50\text{k}\Omega}{G - 1}$

The INA121 Instrumentation amplifier

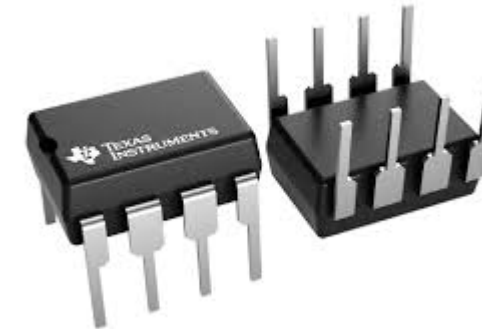
Many other instrumentation amplifier share the same pin layout, and it is therefore straight forward to replace an in-amp of one type with one of another type.

The expression for gain will vary from one in-amp to another.

Do not mistake an in-amp for an op-amp. They look the same.



Gain resistor $R_G = \frac{50k\Omega}{G - 1}$



Instrumentation amplifier

Common mode rejection ratio

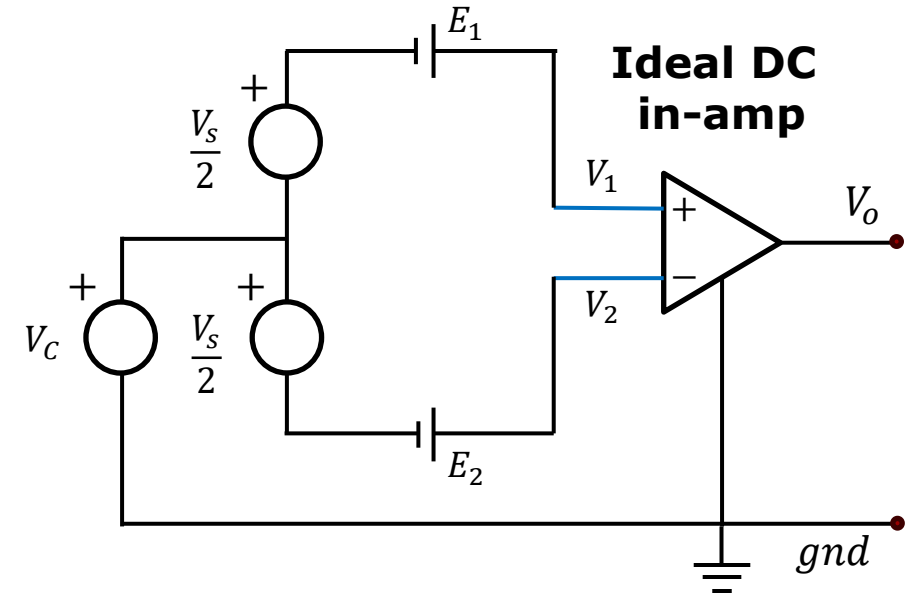
Video

To the right, a combination of sources are now feeding signals to an in-amp. We would like to set up equations to determine the output signal.

V_C is common mode signal. It can be either a DC or AC signal. A typical common mode signal is 50Hz noise from power lines.

The differential sources V_s and E_1, E_2 have been split into separate AC and DC sources.

If we assume an ideal in-amp with differential gain G_d , then output will be:



$$V_1 = V_C + \frac{V_s}{2} + E_1 \quad V_2 = V_C - \frac{V_s}{2} + E_2$$

$$V_d = V_1 - V_2 = V_s + E_1 - E_2$$

$$V_{cm} = \frac{V_1 + V_2}{2} = V_C + \frac{E_1 + E_2}{2}$$

$$V_o = G_d \cdot V_s + G_d \cdot (E_1 - E_2)$$

Now we will assume that the in-amp is **non-ideal**. This implies that some small common mode signal will be present at the output.

$$V_o = \underbrace{G_d \cdot V_d}_{V_{od}} + \underbrace{G_c \cdot V_{cm}}_{V_{oc}}$$

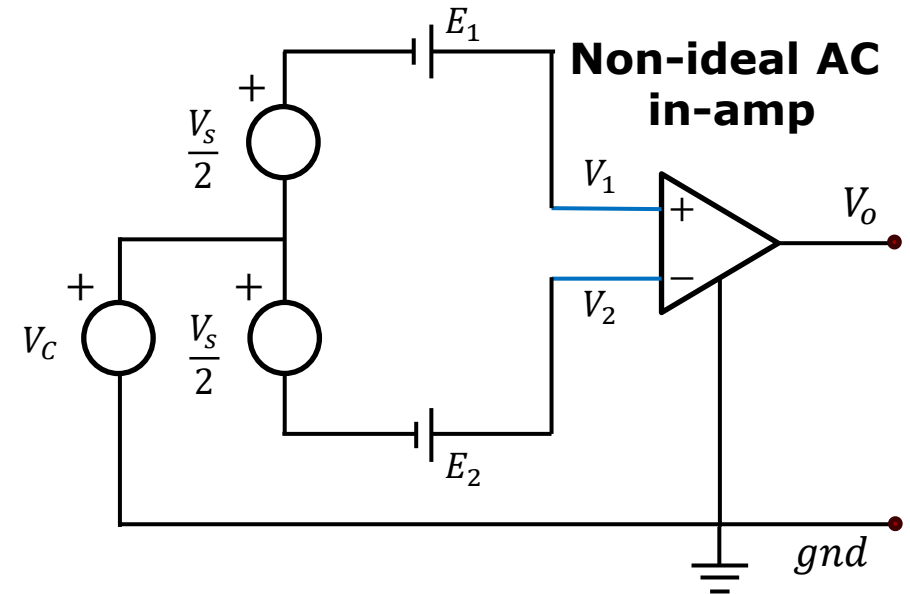
Differential mode gain $G_d \stackrel{\text{def}}{=} \frac{V_{od}}{V_d}$

Common mode gain $G_c \stackrel{\text{def}}{=} \frac{V_{oc}}{V_{cm}}$

G_c is not designed, it is a consequence of a design.

We can only measure it.

For a high-performance in-amp, G_d will be large ($G_d \gg 1$), and G_c will be very small ($G_c \ll 1$).



$$V_1 = V_C + \frac{V_s}{2} + E_1 \quad V_2 = V_C - \frac{V_s}{2} + E_2$$

$$V_d = V_1 - V_2 = V_s + E_1 - E_2$$

$$V_{cm} = \frac{V_1 + V_2}{2} = V_C + \frac{E_1 + E_2}{2}$$

$$V_o = \underbrace{G_d \cdot V_{id}}_{V_{od}} + \underbrace{G_c \cdot V_{cm}}_{V_{oc}}$$

A numerical measure of how well an in-amp rejects common mode signal is referred to as the **common mode rejection ratio**, CMRR.

$$V_o = \underbrace{G_d \cdot V_d}_{V_{od}} + \underbrace{G_c \cdot V_{cm}}_{V_{oc}}$$

$$G_d \stackrel{\text{def}}{=} \frac{V_{od}}{V_d} \quad G_c \stackrel{\text{def}}{=} \frac{V_{oc}}{V_{cm}}$$

$$CMRR \stackrel{\text{def}}{=} \frac{G_d}{G_c}$$

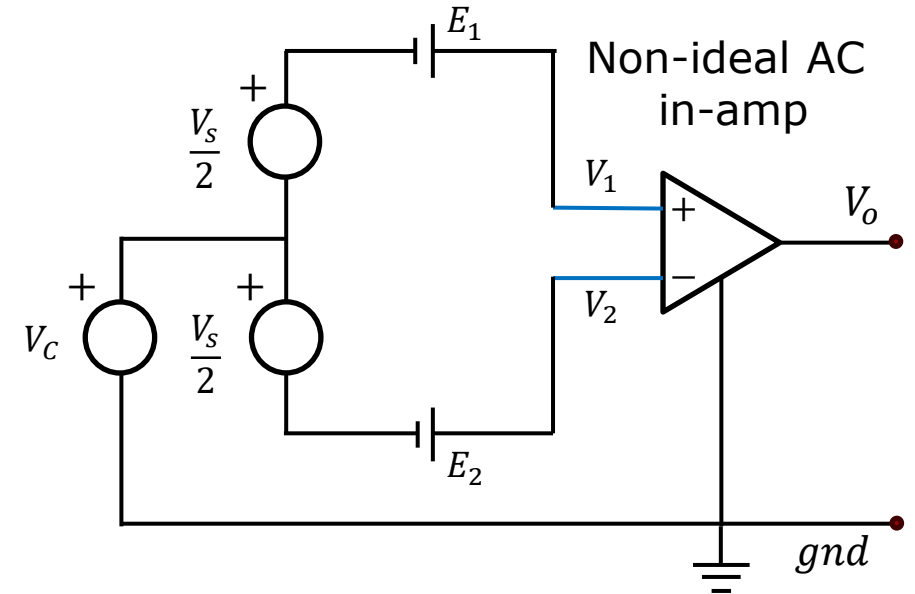
It is often presented in decibel:

$$CMRR_{dB} \stackrel{\text{def}}{=} 20 \log \frac{G_d}{G_c}$$

The common mode input signal can be a summation of many sources. For example:

E_1, E_2 : sensor offset voltages

V_C : Electrical interference radiating into sensor cables.



$$V_o = \underbrace{G_d \cdot V_d}_{V_{od}} + \underbrace{\frac{G_d}{CMRR} \cdot V_{cm}}_{V_{oc}}$$

$$V_{cm} = \frac{V_1 + V_2}{2} = V_C + \frac{E_1 + E_2}{2}$$

Example:

$$V_o = \underbrace{G_d \cdot V_d}_{V_{od}} + \underbrace{G_c \cdot V_{cm}}_{V_{oc}}$$

Calculate $CMRR$

$$CMRR \stackrel{\text{def}}{=} \frac{G_d}{G_c}$$

$$V_d = 5\text{mV}$$

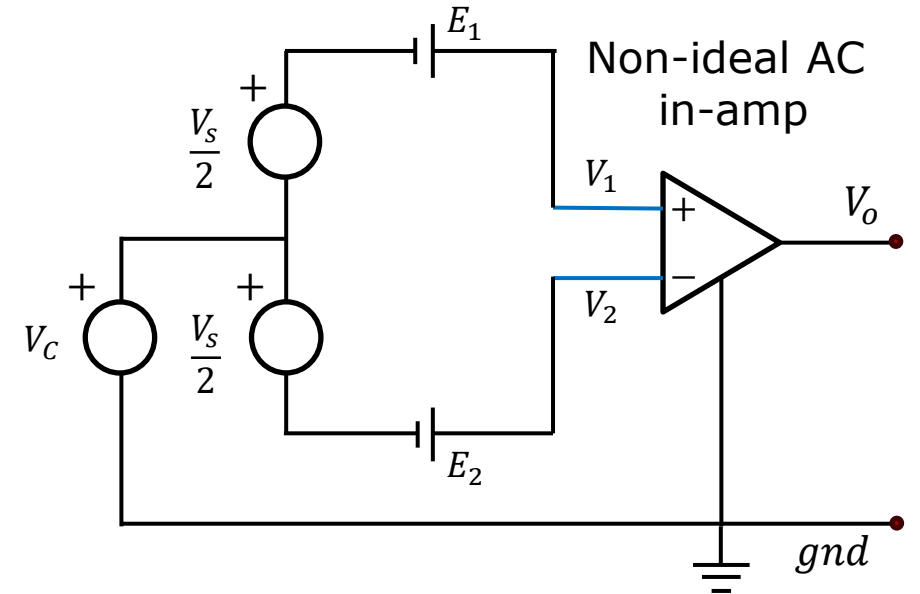
$$V_{od} = 5\text{V}$$

$$V_{cm} = 500\text{mV}$$

$$V_{oc} = 50\text{mV}$$

The $CMRR$ is typically determined by first determining G_d and G_c experimentally.

From this we get: $CMRR = G_d/G_c$



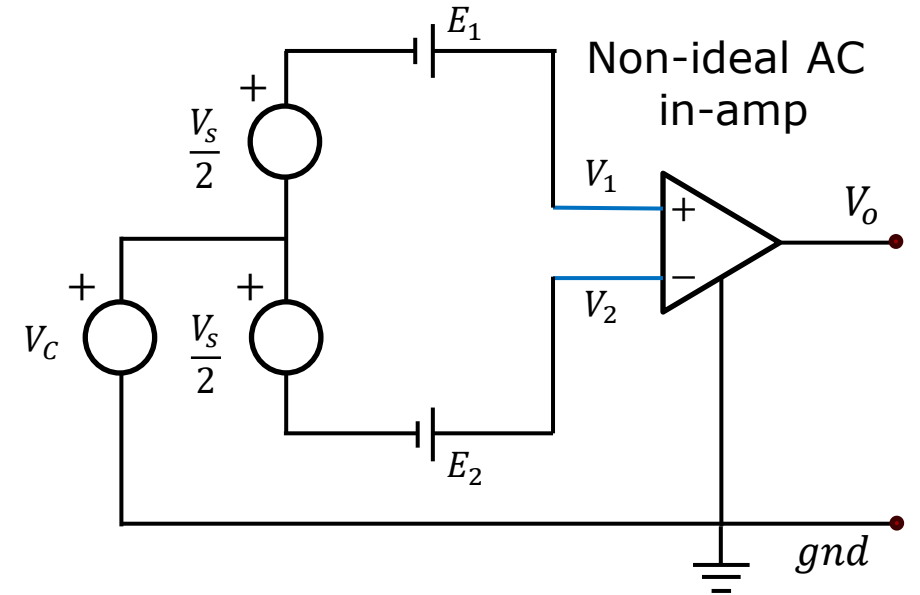
$$G_d = \frac{V_{od}}{V_d} = \frac{5\text{V}}{5\text{mV}} = 1000$$

$$G_c = \frac{V_{oc}}{V_{cm}} = \frac{50\text{mV}}{500\text{mV}} = 0.1$$

$$CMRR = \frac{G_d}{G_c} = \frac{1000}{0.1} = 10^4 = 80\text{dB}$$

What is the improvement in SNR?

$$V_{cm} = 500\text{mV} \quad V_{oc} = 50\text{mV}$$



Input SNR $\frac{V_d}{V_{cm}} = \frac{5\text{mV}}{500\text{mV}} = 10^{-2} \sim -40\text{dB}$

Output SNR $\frac{V_{od}}{V_{oc}} = \frac{5V}{0.05V} = 10^2 \sim 40\text{dB}$

The CMRR is the improvement in SNR from input to output.

$$\left(\frac{V_{od}}{V_{oc}}\right)_{dB} = \left(\frac{V_d}{V_{cm}}\right)_{dB} + CMRR_{dB}$$

This circuit model includes a source model and an AC-coupled in-amp.

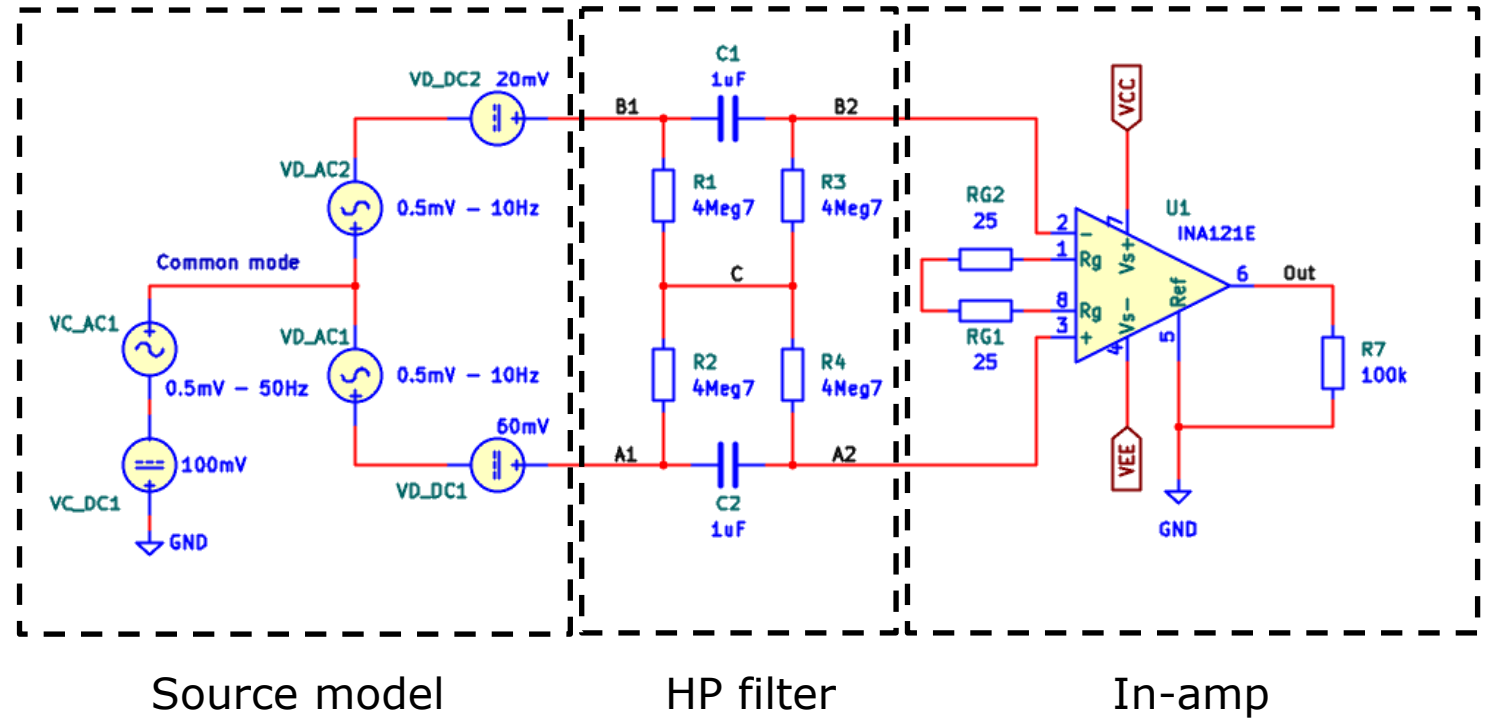
The source model has the following sources:

V_{C-AC1} : AC common mode

V_{C-DC1} : DC common mode

$V_{D-AC(1,2)}$: AC differential sources

$V_{D-DC(1,2)}$: DC differential sources



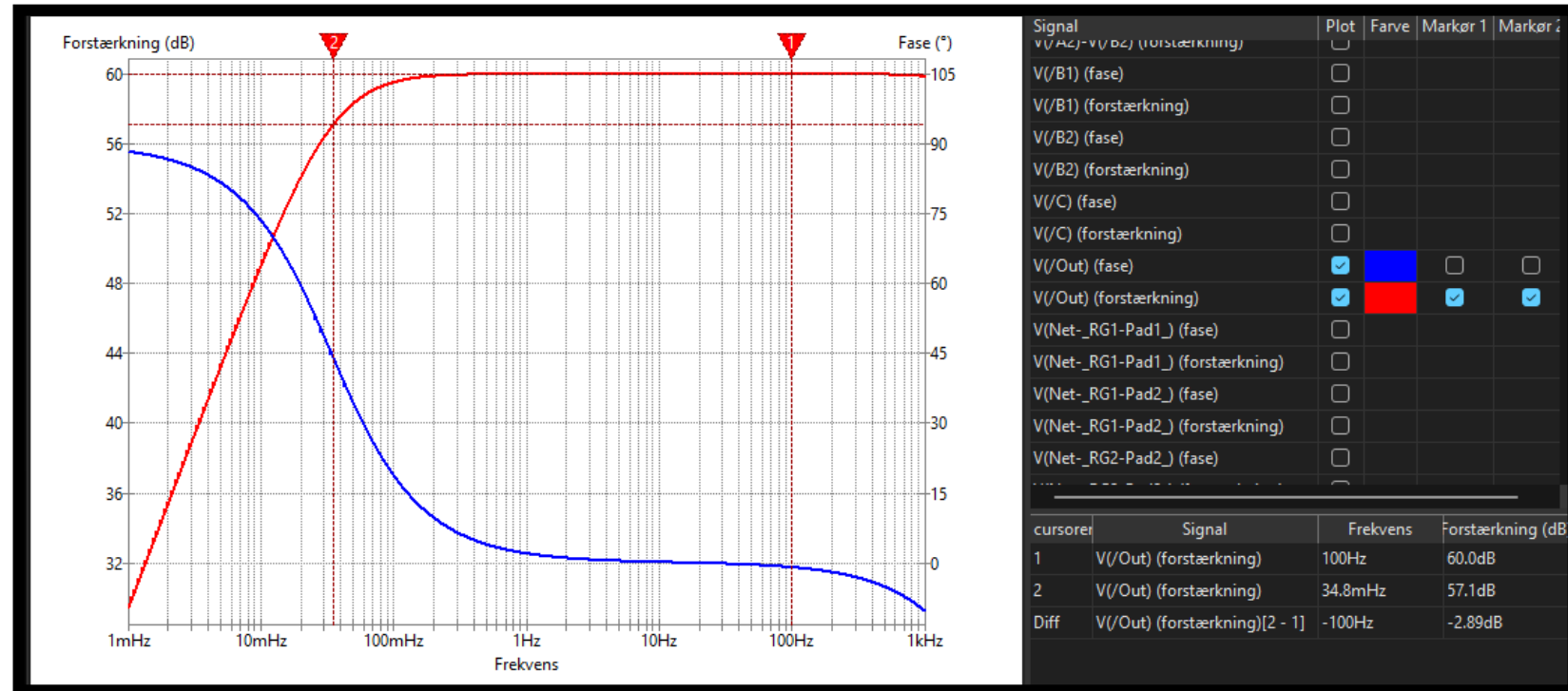
Differential high-pass filter is described in an appendix.

AC sweep of differential signal

In this AC sweep the two AC differential sources had an amplitude of 0.5V each. The DC differential sources and the common mode sources had an amplitude of 0V.

The differential gain
 $G_d \sim 1000$.

The red curve is the output from the in-amp.



Observations:

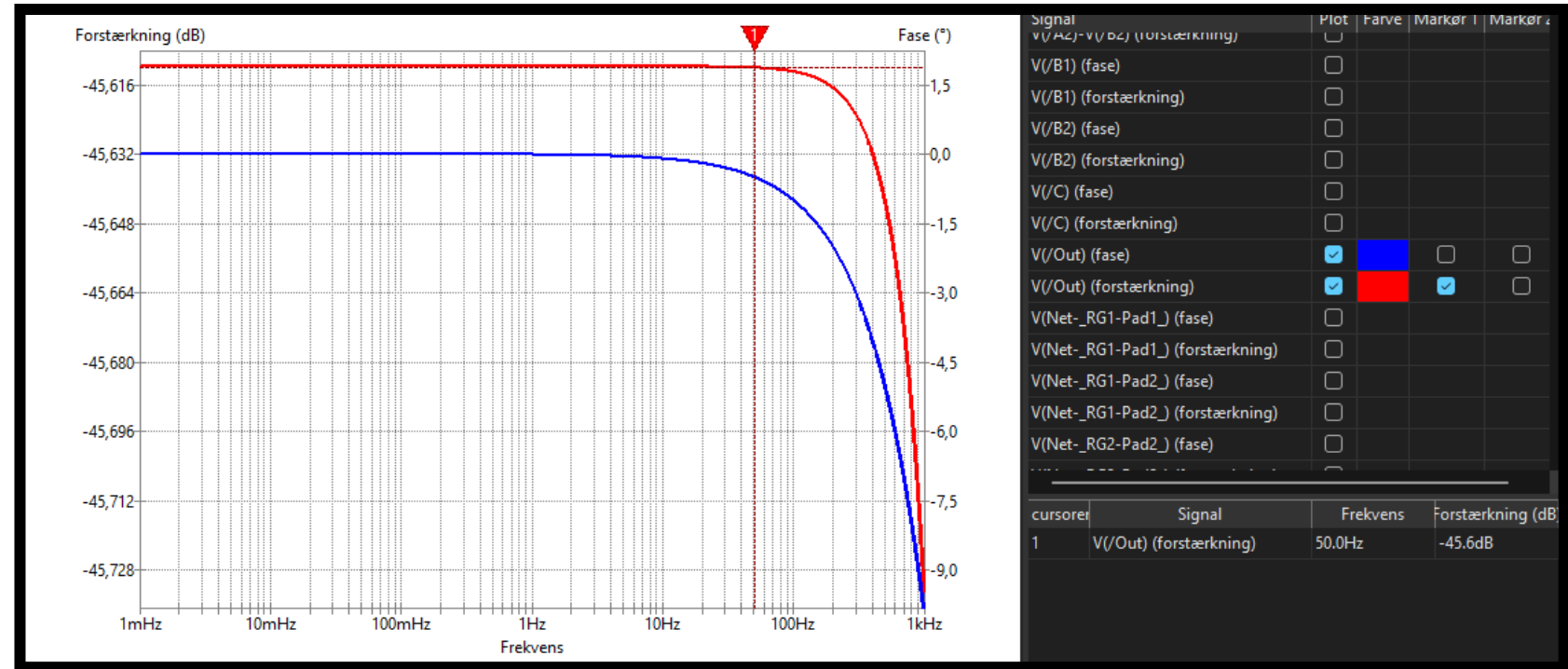
The output signal is 60dB (1000) larger than the input signal.

Here is shown the output when **only a 1V common mode signal** is present on the input. At 50Hz, the common mode signal is attenuated at least 45.6dB.

$$G_c = 0.00525$$

The differential gain
 $G_d = 1000$.

The CMRR is:

$$CMRR = \frac{G_d}{G_c} = \frac{1000}{0.00525} = 1.90 \times 10^5 = 105.6\text{dB} = 60\text{dB} - (-45.6\text{dB})$$


Problems

Problem 1 – EMG high-pass filter

Design a Butterworth high-pass filter with the following specifications:

$$G_{\infty} = 0\text{dB}$$

$$\omega_c = \omega_p = 2\pi \cdot 2\text{Hz}$$

$$G_p = -3\text{dB}$$

$$\omega_s = 2\pi \cdot 0.02\text{Hz}$$

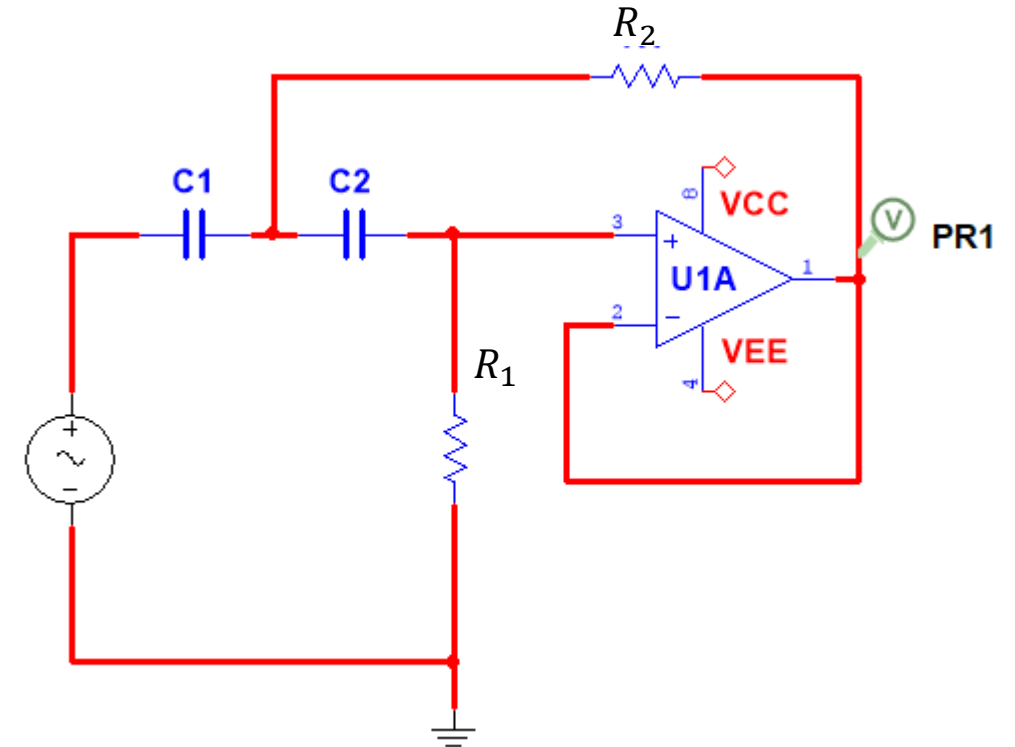
$$G_s = -60\text{dB}$$

Design constraints:

- Use a Sallen-Key circuit of lowest possible order.
- The filter should be very insensitive to component drift.
- Use resistors from the E12 series and capacitors from the E6 series as much as possible, but not at all costs.

Documentation:

- Draw the amplitude and phase spectra in Maple/Python for the frequency-normalized filter as well as for the final design.



Hint: Do not start from scratch. Start with the frequency-normalized HP filter designed in the lecture and scale it to the design-specifications.

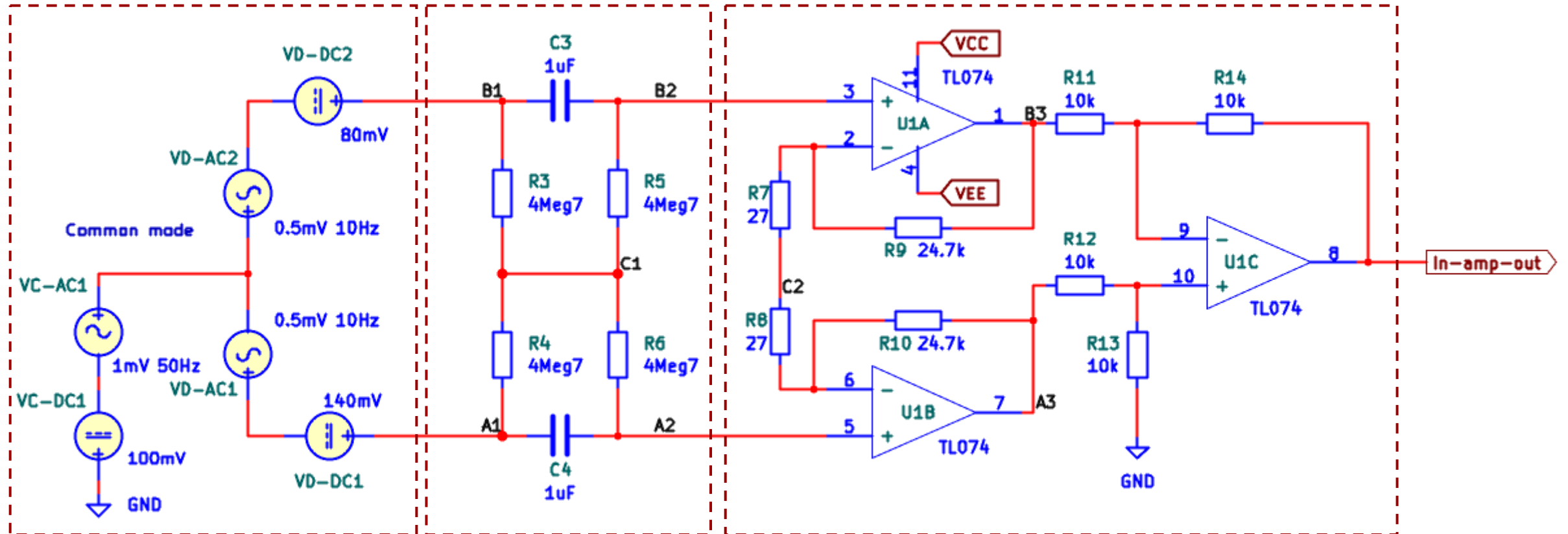
$$\hat{R}_1 \approx 0.68\Omega$$

$$\hat{R}_2 \approx 0.33\Omega$$

$$\hat{C} \approx 2.2F$$

Problem 2 Balanced In-amp

In this problem you will explore the AC and DC signals in an **AC-coupled** in-amp.
Sources are split in AC/DC and differential/common mode.



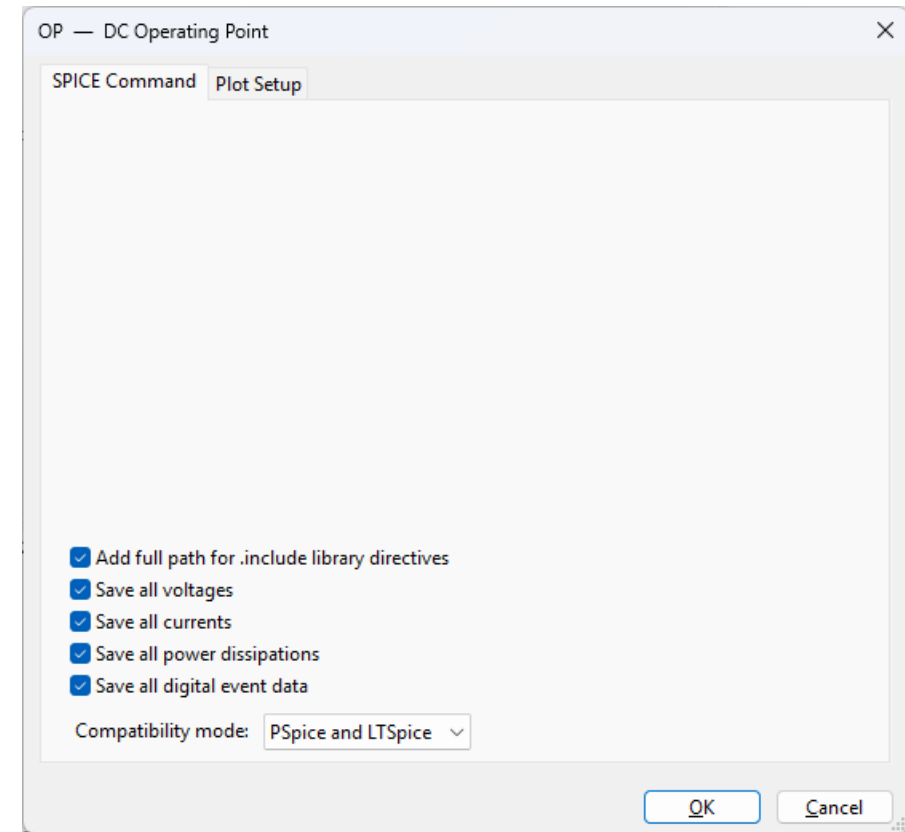
Source model

HP filter

3-op-amp model of in-amp

Part 1

Do an operating point simulation to obtain DC voltages in different nodal points.



Problem 2 Balanced In-amp

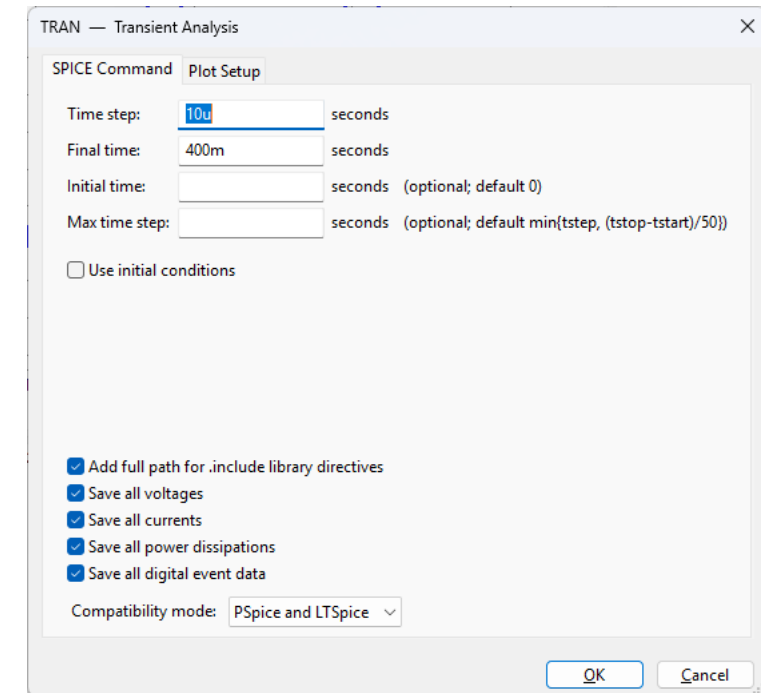
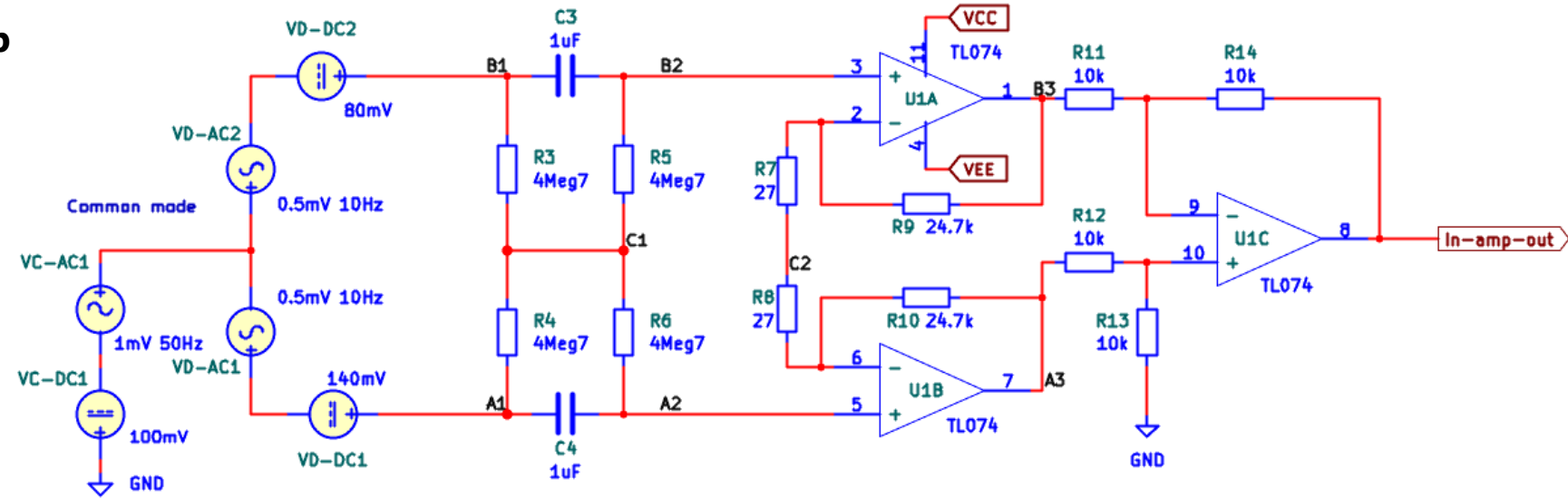
Part 2

Run a transient simulation.

Plot signals in nodes:

A1, B1, A2, B2.

Explain the signals making use of the DC operating voltages found earlier.



Problem 2 Balanced In-amp

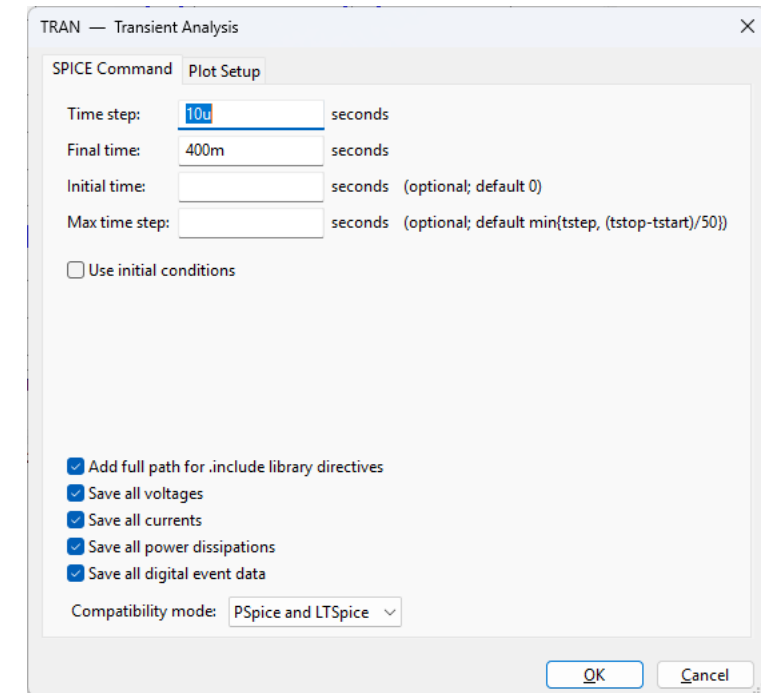
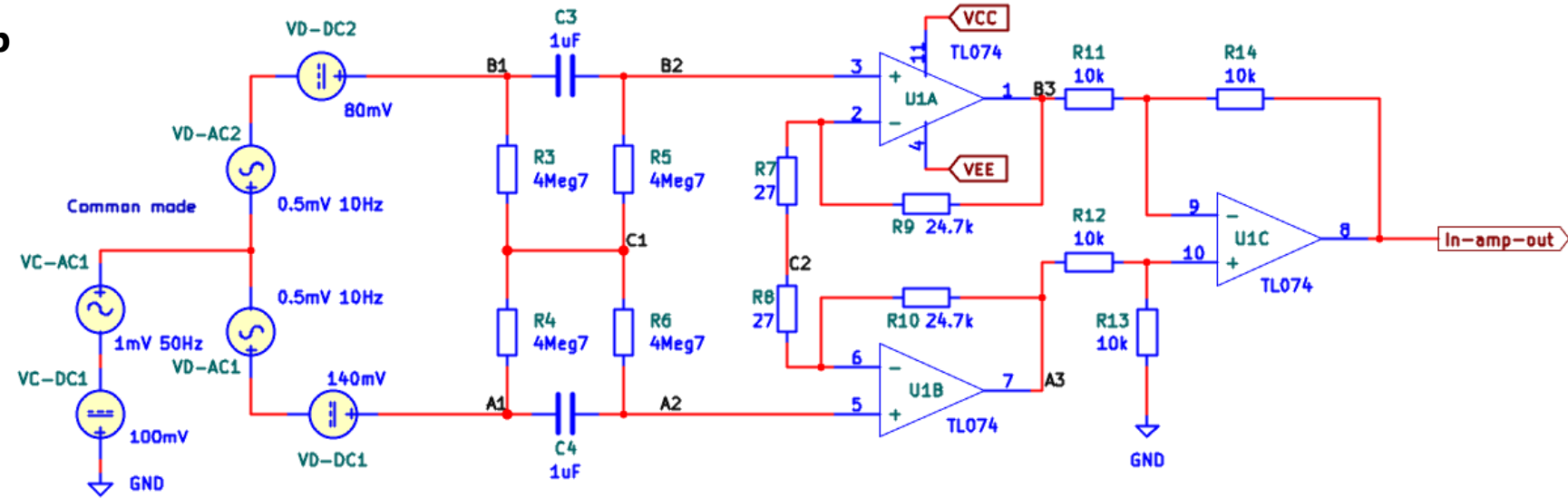
Part 3

Run a transient simulation.

Plot signals in nodes:

C1.

Explain the signal making use of the DC operating voltages found earlier.



Problem 2 Balanced In-amp

Part 4

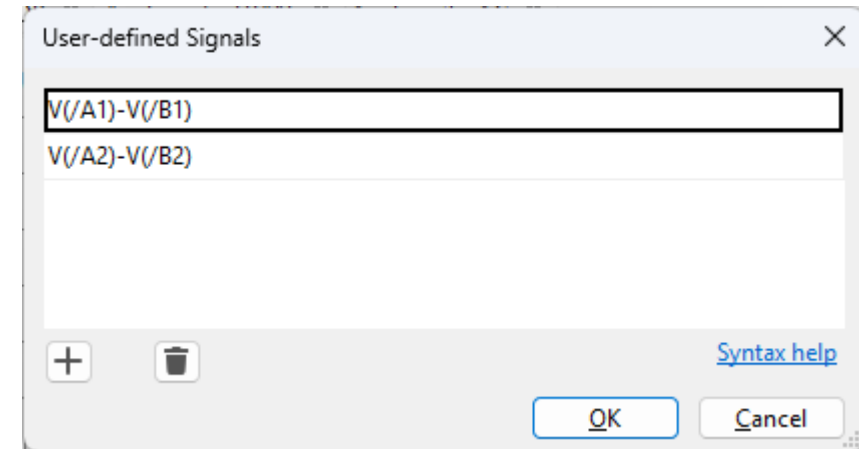
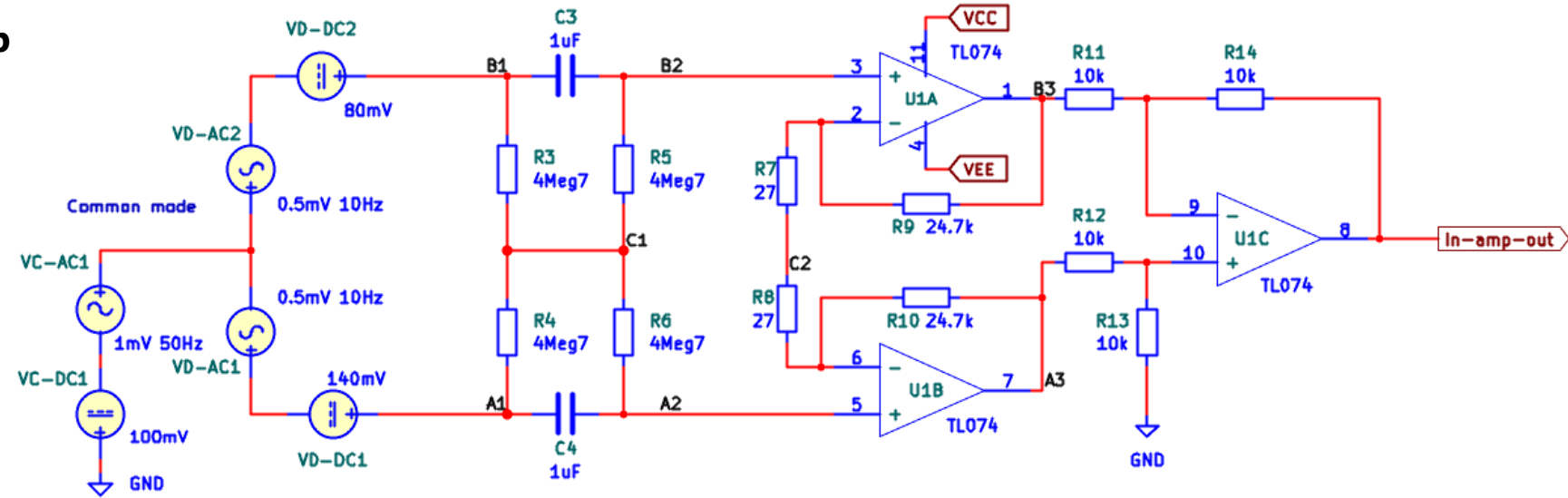
Run a transient simulation.

Create and plot two new differential signals:

A1 – B1

A2 – B2

Explain the signals making use of the DC operating voltages found earlier.

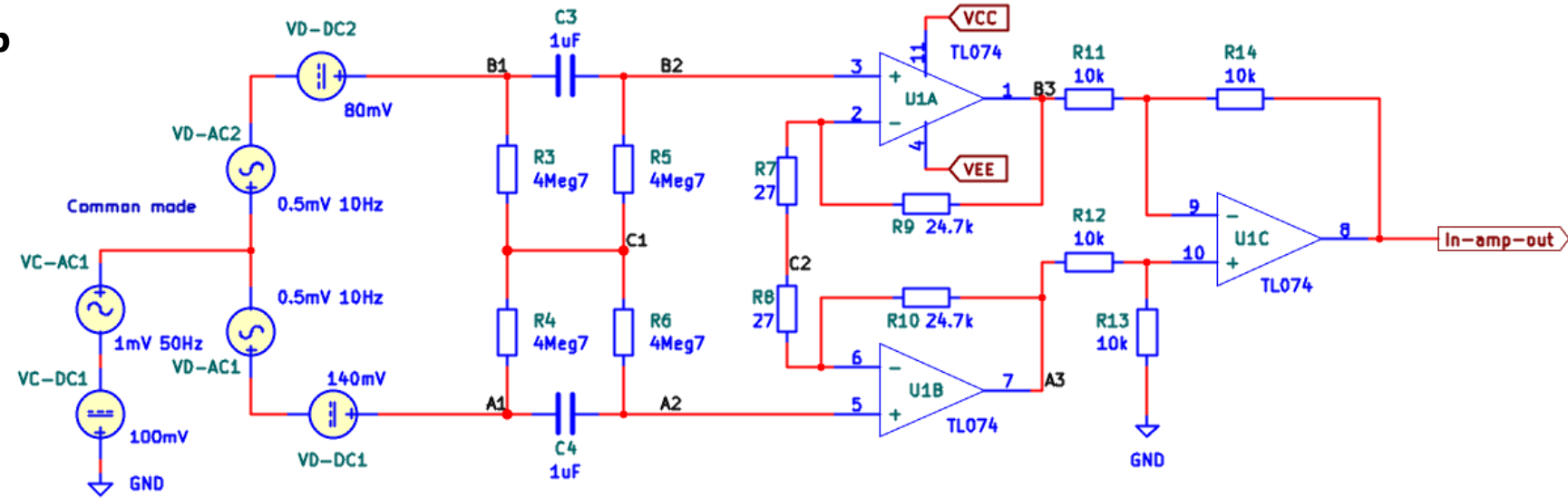


Problem 2 Balanced In-amp

Part 5

Run a transient simulation.

Plot the output signal of the in-amp.



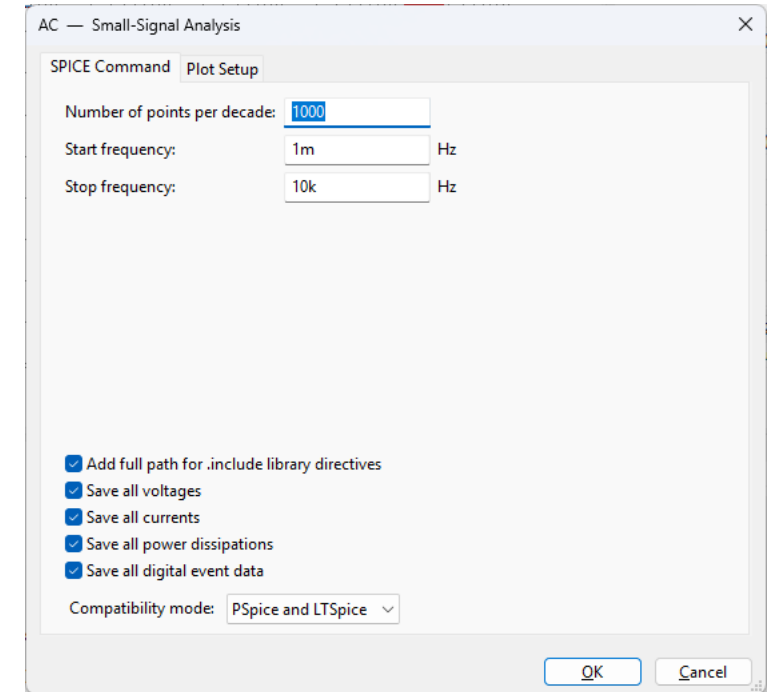
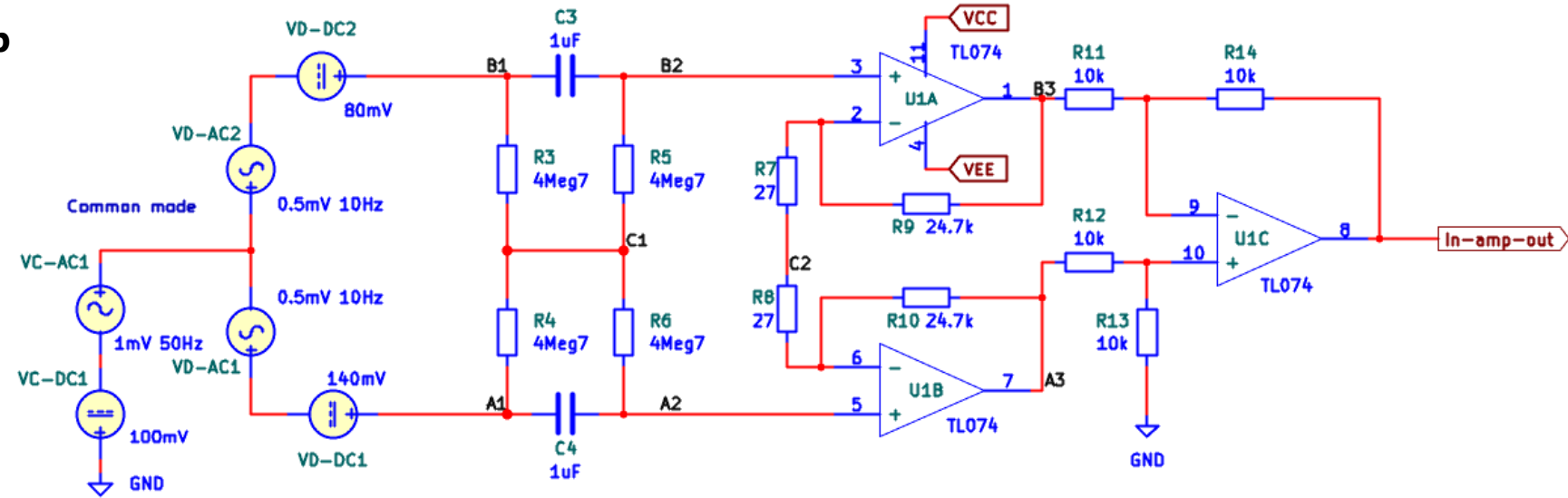
Explain the signals making use of the DC operating voltages found earlier.

Problem 2 Balanced In-amp

Part 6

Run an AC sweep simulation using VD-AC1 and VD-AC2.

Plot the amplitude spectra of the output signals of the in-amp output and filter.

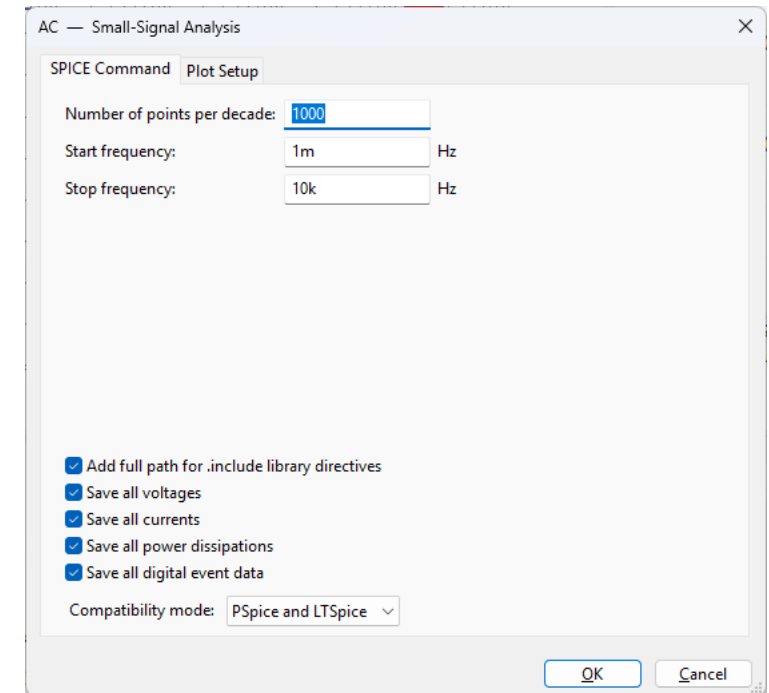
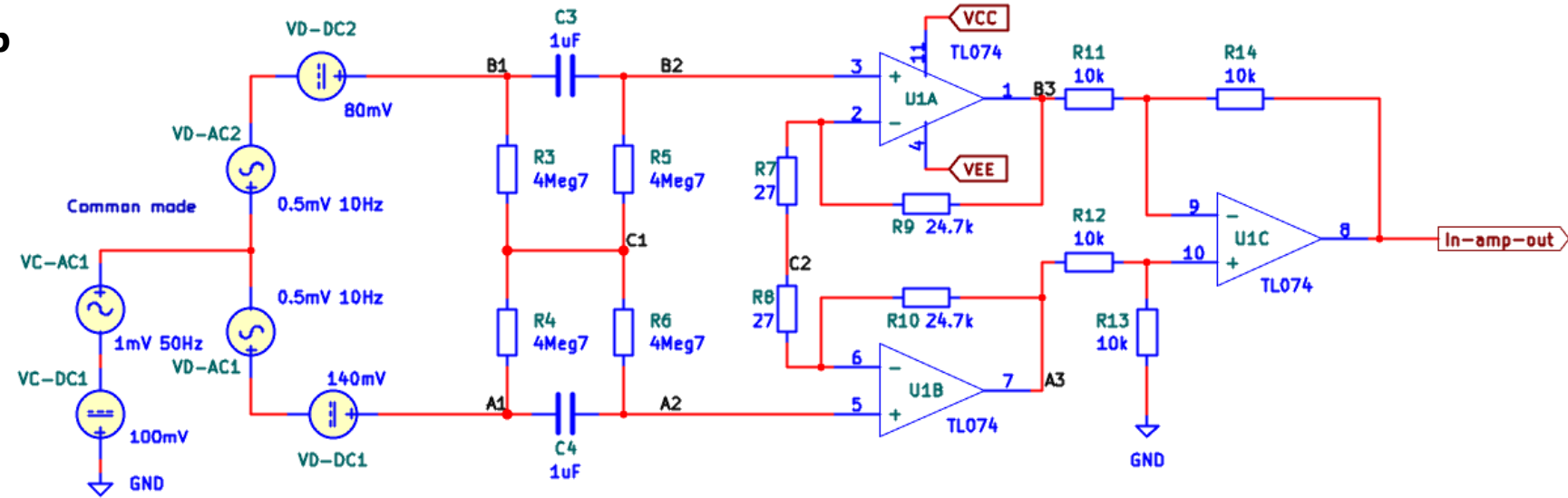


Problem 2 Balanced In-amp

Part 7

Run a **common mode** AC sweep simulation using VC-AC1.

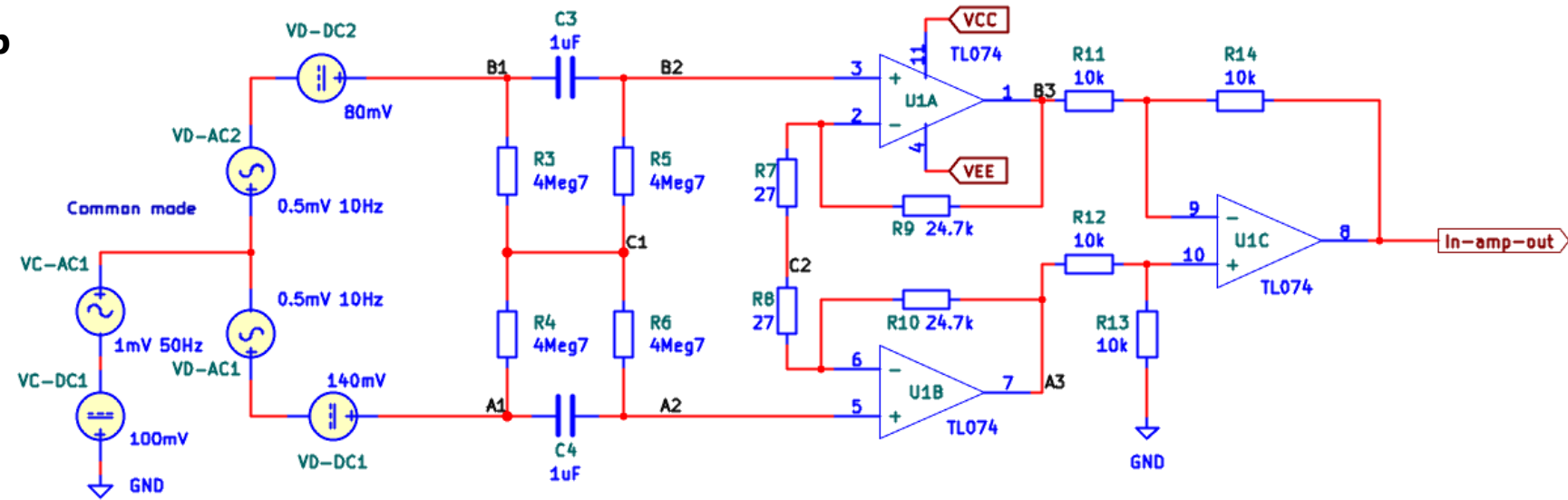
Plot the amplitude spectra of the output signals of the in-amp output and filter.



Problem 2 Balanced In-amp

Part 8

Using the differential and common mode AC sweeps, calculate the CMRR.

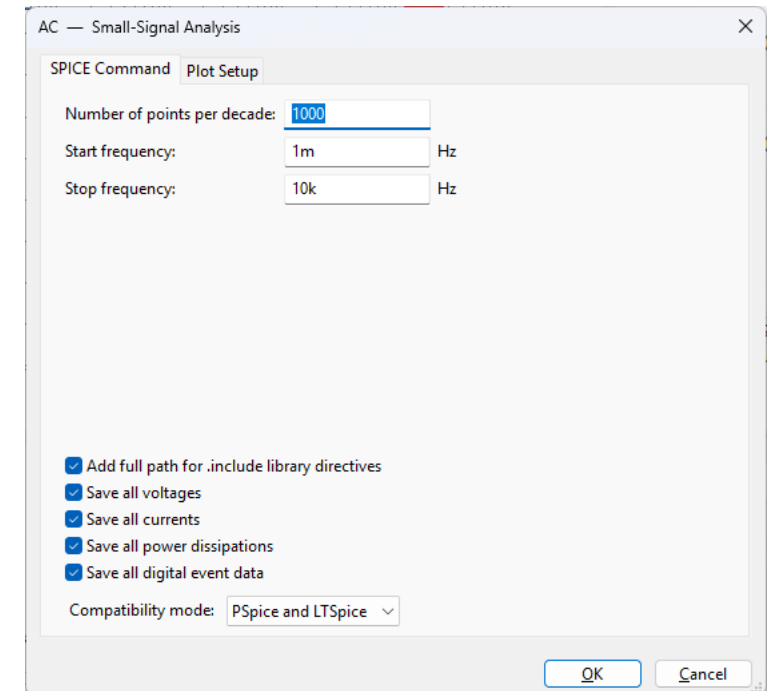
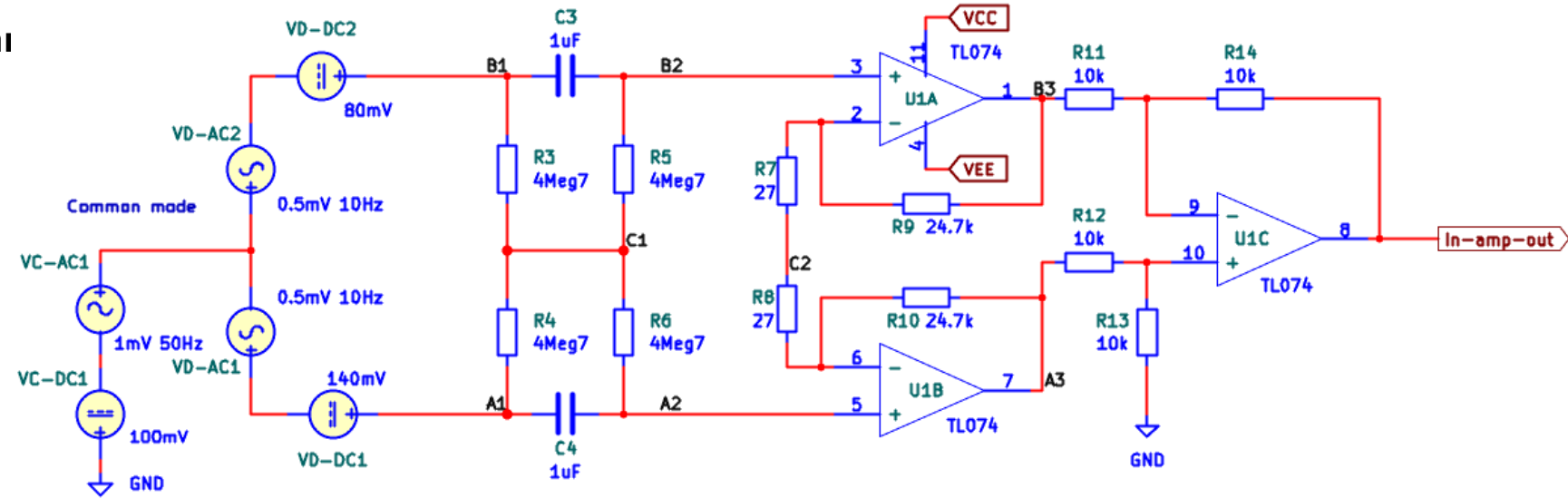


Part 9

Change R14 to 11k.

Run a new **common mode** AC sweep simulation using VC-AC1.

Plot the amplitude spectra of the output signals of the in-amp output and filter.



Solutions

Problem 1 – EMG high-pass filter

Design a Butterworth high-pass filter with the following specifications:

$$G_{\infty} = 0\text{dB}$$

$$\omega_c = \omega_p = 2\pi \cdot 2\text{Hz}$$

$$G_p = -3\text{dB}$$

$$\omega_s = 2\pi \cdot 0.02\text{Hz}$$

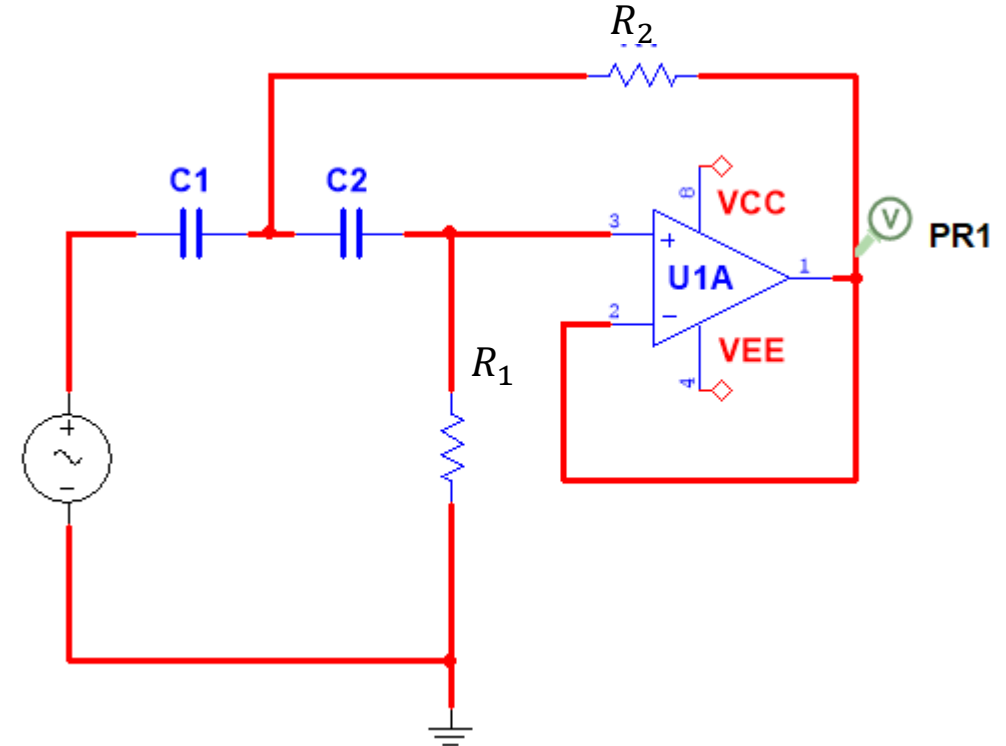
$$G_s = -60\text{dB}$$

Design constraints:

- Use a Sallen-Key circuit of lowest possible order.
- The filter should be very insensitive to component drift.
- Use resistors from the E12 series and capacitors from the E6 series as much as possible, but not at all costs.

Documentation:

- Draw the amplitude and phase spectra in Maple/Python for the frequency-normalized filter as well as for the final design.



Hint: Do not start from scratch. Start with the frequency-normalized HP filter designed in the lecture and scale it to the design-specifications.

$$\hat{R}_1 \approx 0.68\Omega$$

$$\hat{R}_2 \approx 0.33\Omega$$

$$\hat{C} \approx 2.2F$$

Problem 1 – EMG high-pass filter

Determination of filter order:

$$G_{\infty} = 0\text{dB}$$

$$\omega_c = \omega_p = 2\pi \cdot 2\text{Hz}$$

$$G_p = -3\text{dB}$$

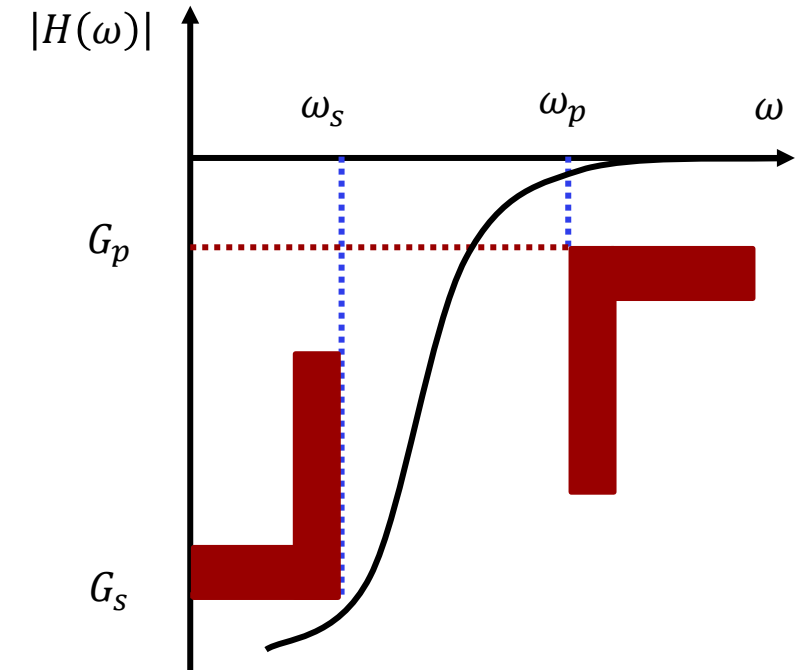
$$\omega_s = 2\pi \cdot 0.02\text{Hz}$$

$$G_s = -60\text{dB}$$

$$n_{HP} = \frac{1}{2\log_{10}(\omega_{p,HP}/\omega_{s,HP})} \log_{10} \left(\frac{10^{-\frac{G_{s,dB}}{10}} - 1}{10^{-\frac{G_{p,dB}}{10}} - 1} \right)$$

$$n_{HP} = \frac{1}{2\log_{10}(2/0.02)} \log_{10} \left(\frac{10^{-\frac{-60}{10}} - 1}{10^{-\frac{-3}{10}} - 1} \right) = 1.5$$

We need a 2nd order filter.



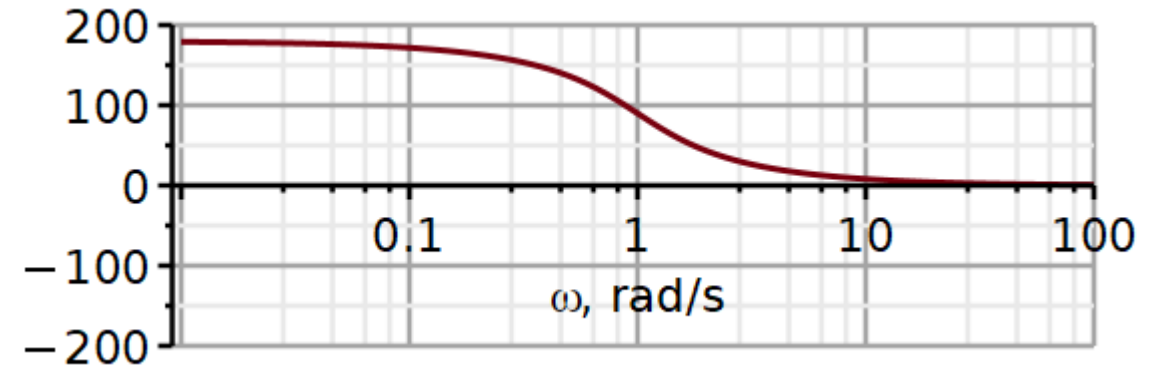
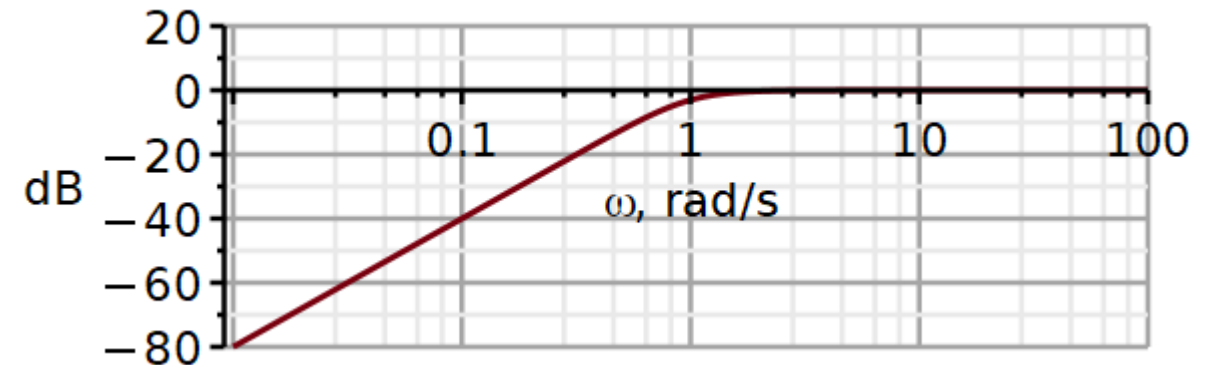
Problem 1 – EMG high-pass filter (sol)

For the frequency-normalized filter
Several pairs of component values were
considered in the lecture slides. Here I
have used:

$$\hat{R}_1 \approx 0.68\Omega$$

$$\hat{R}_2 \approx 0.33\Omega$$

$$\hat{C} \approx 2.2F$$



The low-frequency asymptote has a slope of +40dB/decade, the cut-off is at 1 rad/s and the phase is 90 degrees at the cut-off. So, it looks promising.

Frequency scaling

$$K_F = 2\pi \cdot 2 = 12.5663$$

$$C'_1 = \frac{\hat{C}_1}{K_F} = \frac{2.2\text{F}}{12.5663} = 175\text{mF}$$

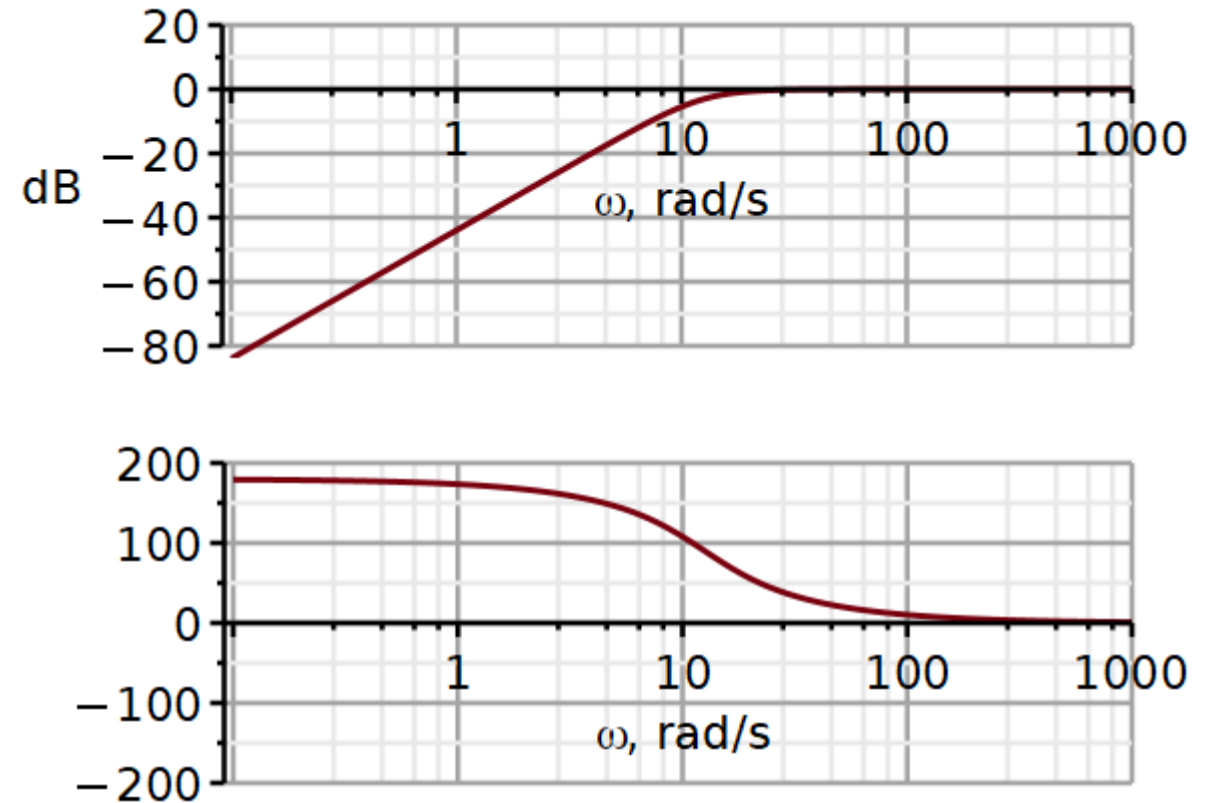
$$C'_2 = \frac{\hat{C}_1}{K_F} = 175\text{mF}$$

$$R'_1 = \hat{R}_1 = 0.68\Omega$$

$$R'_2 = \hat{R}_2 = 0.33\Omega$$

$$\tau_1 = R'_1 C'_1 = 0.119\text{s}$$

$$\tau_2 = R'_2 C'_2 = 0.05775\text{s}$$



Looking up the best component values in the time constant look-up table on the next slide, we find that a capacitor value of $1\mu\text{F}$ will work well.

A value of $1.2\mu\text{F}$ would also work well, but it belongs to the E12 series.

2nd order Sallen-Key high pass filter - Butterworth

$$\tau_1 = R_1 C = 0.119s$$

$$\tau_2 = R_2 C = 0.05775s$$



E6 E12 E6 E12

R1*C1	0,1	0,12	0,15	0,18	0,22	0,27	0,33	0,39	0,47	0,56	0,68	0,82	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10
0,1	0,010	0,012	0,015	0,018	0,022	0,027	0,033	0,039	0,047	0,056	0,068	0,082	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000
0,12	0,012	0,014	0,018	0,022	0,026	0,032	0,040	0,047	0,056	0,067	0,082	0,098	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200
0,15	0,015	0,018	0,023	0,027	0,033	0,041	0,050	0,059	0,071	0,084	0,102	0,123	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500
0,18	0,018	0,022	0,027	0,032	0,040	0,049	0,059	0,070	0,085	0,101	0,122	0,148	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800
0,22	0,022	0,026	0,033	0,040	0,048	0,059	0,073	0,086	0,103	0,123	0,150	0,180	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200
0,27	0,027	0,032	0,041	0,049	0,059	0,073	0,089	0,105	0,127	0,151	0,184	0,221	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700
0,33	0,033	0,040	0,050	0,059	0,073	0,089	0,109	0,129	0,155	0,185	0,224	0,271	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300
0,39	0,039	0,047	0,059	0,070	0,086	0,105	0,129	0,152	0,183	0,218	0,265	0,320	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900
0,47	0,047	0,056	0,071	0,085	0,103	0,127	0,155	0,183	0,221	0,263	0,320	0,385	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700
0,56	0,056	0,067	0,084	0,101	0,123	0,151	0,185	0,218	0,263	0,314	0,381	0,459	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600
0,68	0,068	0,082	0,102	0,122	0,150	0,184	0,224	0,265	0,320	0,381	0,462	0,558	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800
0,82	0,082	0,098	0,123	0,148	0,180	0,221	0,271	0,320	0,385	0,459	0,558	0,672	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200
1	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000
1,2	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200	1,440	1,800	2,160	2,640	3,240	3,960	4,680	5,640	6,720	8,160	9,840	12,000
1,5	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500	1,800	2,250	2,700	3,300	4,050	4,950	5,850	7,050	8,400	10,200	12,300	15,000
1,8	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800	2,160	2,700	3,240	3,960	4,860	5,940	7,020	8,460	10,080	12,240	14,760	18,000
2,2	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200	2,640	3,300	3,960	4,840	5,940	7,260	8,580	10,340	12,320	14,960	18,040	22,000
2,7	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700	3,240	4,050	4,860	5,940	7,290	8,910	10,530	12,690	15,120	18,360	22,140	27,000
3,3	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300	3,960	4,950	5,940	7,260	8,910	10,890	12,870	15,510	18,480	22,440	27,060	33,000
3,9	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900	4,680	5,850	7,020	8,580	10,530	12,870	15,210	18,330	21,840	26,520	31,980	39,000
4,7	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700	5,640	7,050	8,460	10,340	12,690	15,510	18,330	22,090	26,320	31,960	38,540	47,000
5,6	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600	6,720	8,400	10,080	12,320	15,120	18,480	21,840	26,320	31,360	38,080	45,920	56,000
6,8	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800	8,160	10,200	12,240	14,960	18,360	22,440	26,520	31,960	38,080	46,240	55,760	68,000
8,2	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200	9,840	12,300	14,760	18,040	22,140	27,060	31,980	38,540	45,920	55,760	67,240	82,000
10	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000	12,000	15,000	18,000	22,000	27,000	33,000	39,000	47,000	56,000	68,000	82,000	100,00

Problem 1 – EMG high-pass filter (sol)

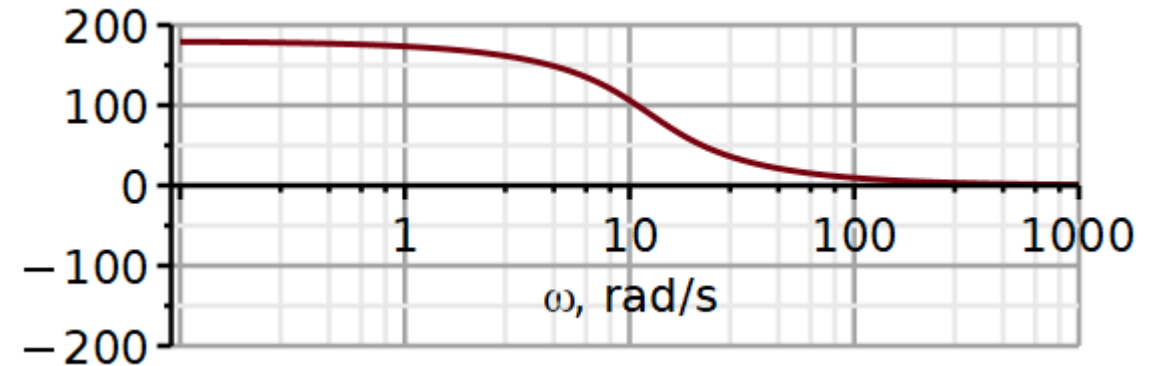
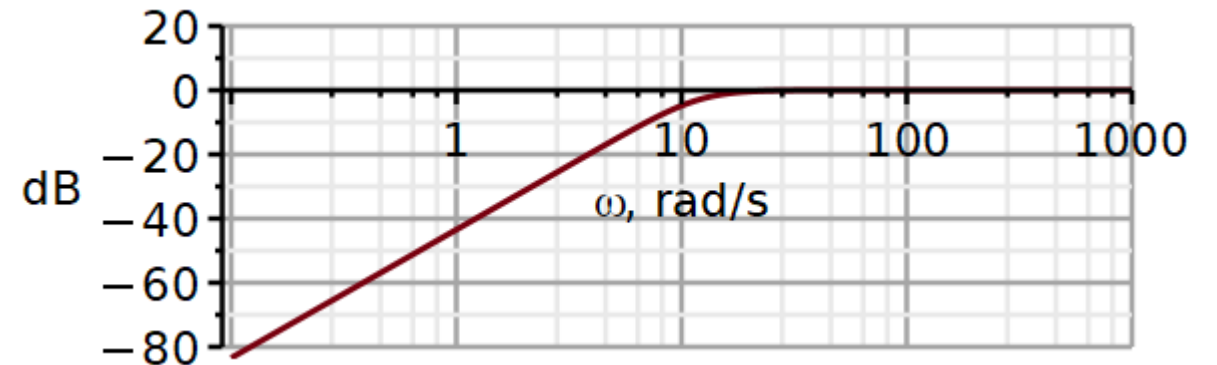
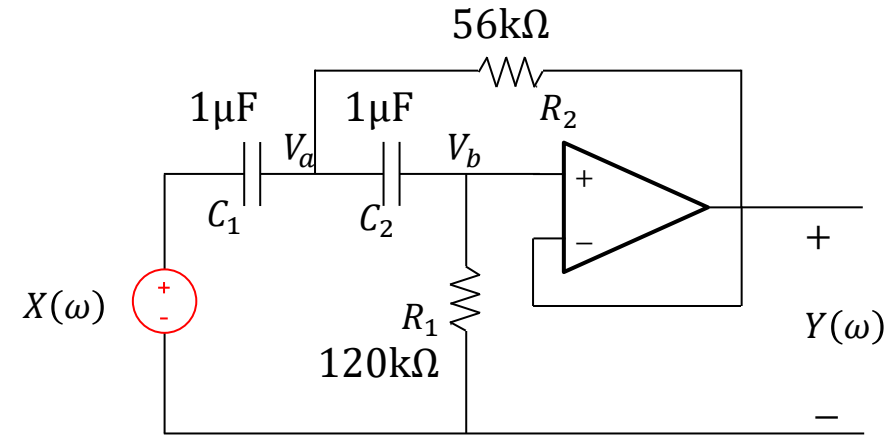
Impedance scaling:

$$C_1 = C_2 = 1\mu F \quad \leftarrow \text{chosen capacitor value}$$

$$K_z = \frac{C'_1}{1 \cdot 10^{-6} \text{F}} = \frac{175 \text{mF}}{1 \cdot 10^{-6} \text{F}} = 175000$$

$$R_1 = K_z \cdot 0.68\Omega = 119.0\text{k}\Omega \approx 120\text{k}\Omega$$

$$R_2 = K_z \cdot 0.33\Omega = 57.77\text{k}\Omega \approx 56\text{k}\Omega$$



Problem 1 – EMG high-pass filter (sol)

Design a Butterworth high-pass filter with the following specifications:

$$G_{\infty} = 0\text{dB}$$

$$\omega_c = \omega_p = 2\pi \cdot 2\text{Hz}$$

$$G_p = -3\text{dB}$$

$$\omega_s = 2\pi \cdot 0.02\text{Hz}$$

$$G_s = -60\text{dB}$$

Validation:

$$C1 := 1 \cdot 10^{-6} ; C2 := 1 \cdot 10^{-6} ; R1 := 120 \cdot 10^3 ; R2 := 56 \cdot 10^3 ;$$

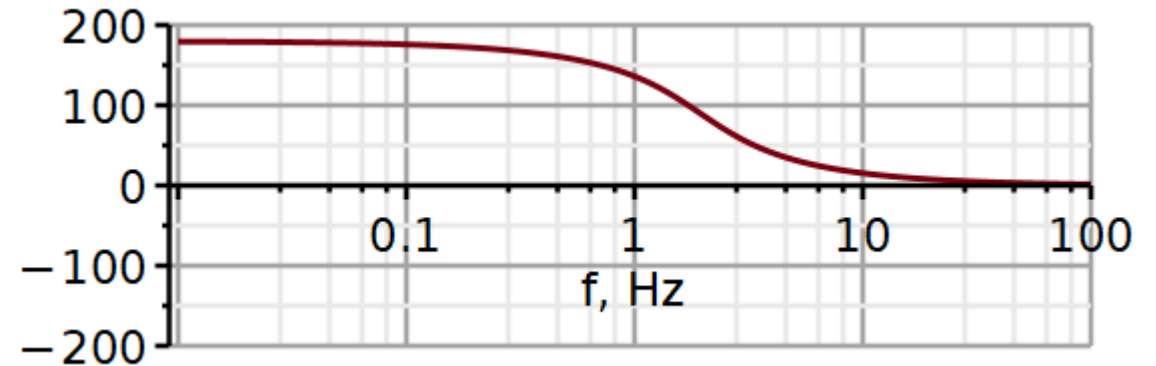
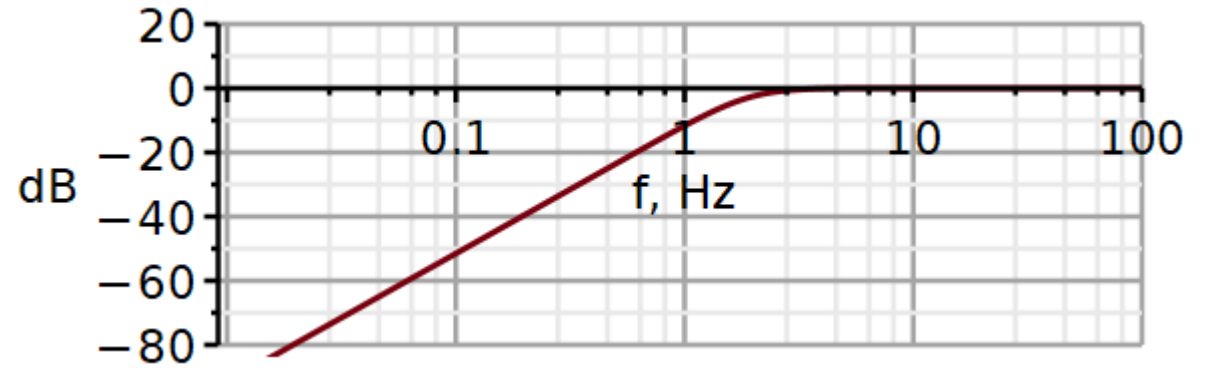
Test:

$$\text{evalf}(\text{dB}(H(2 \cdot \pi \cdot 2)))$$

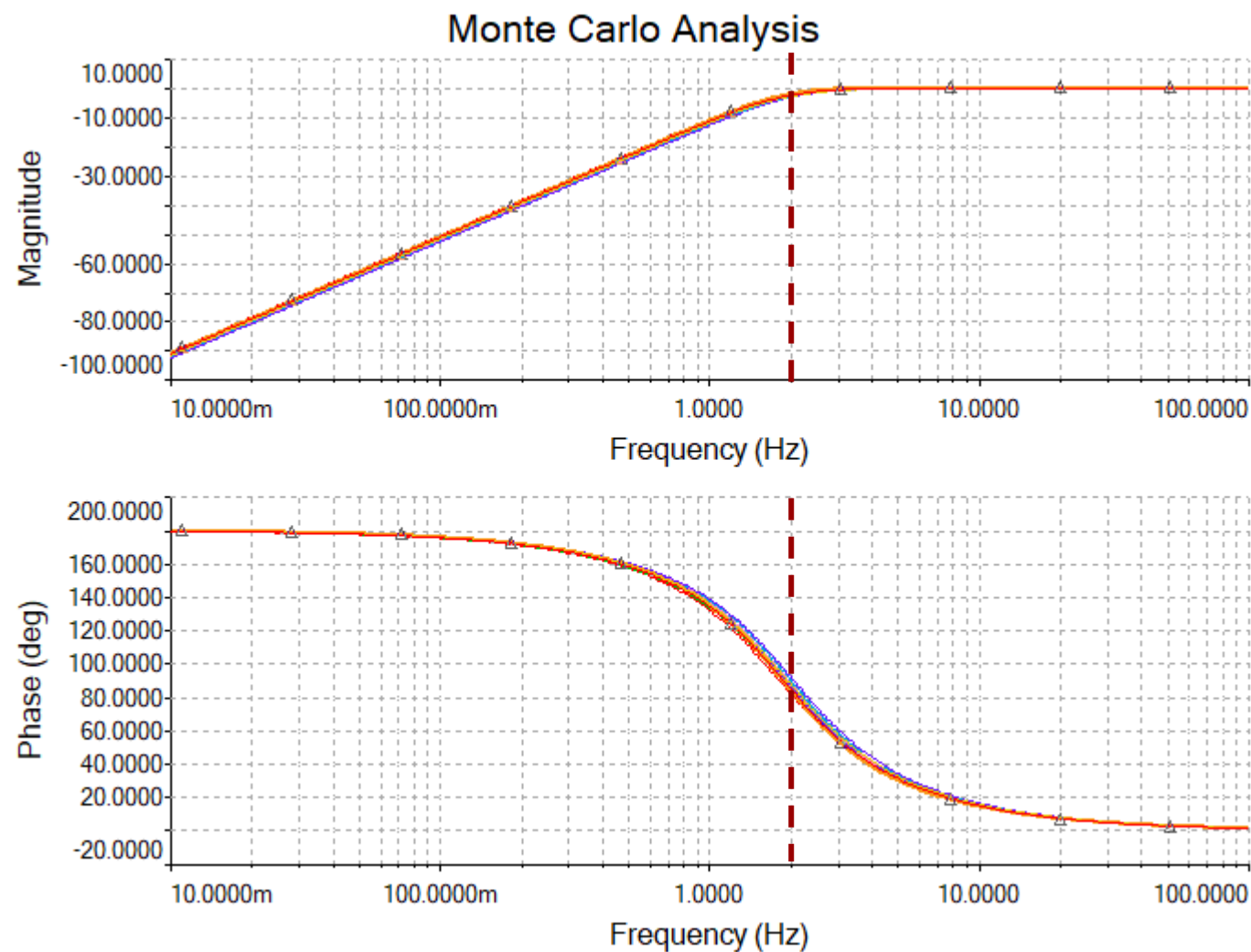
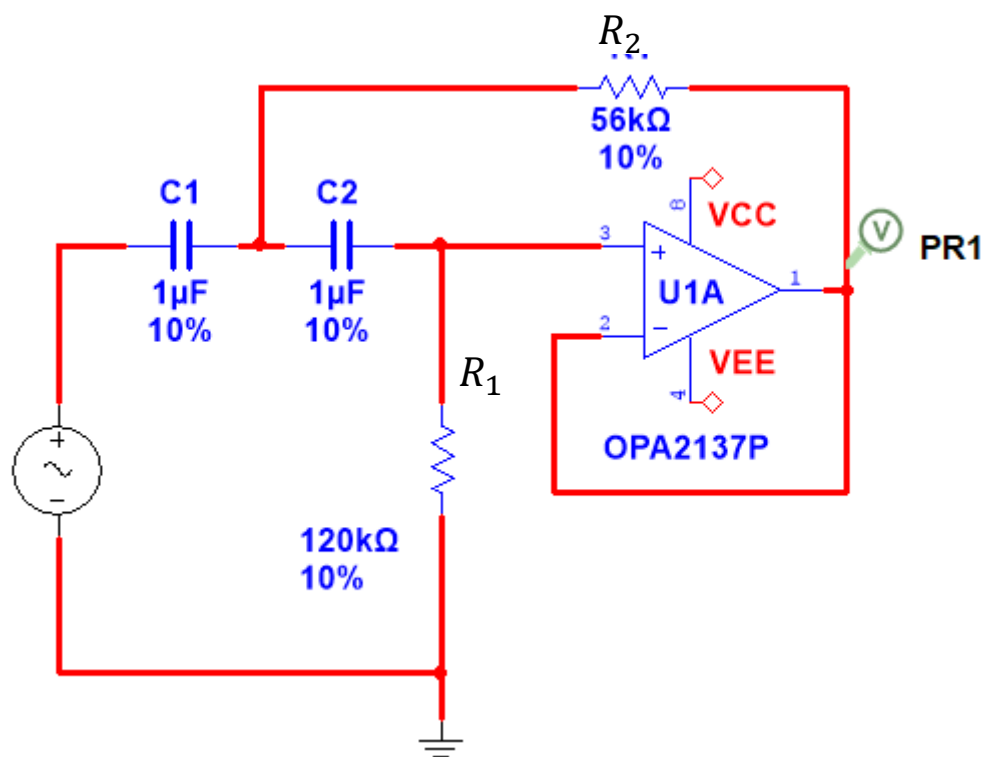
$$-2.460976232$$

$$\text{evalf}(\text{dB}(H(2 \cdot \pi \cdot 0.02)))$$

$$-79.48415858$$

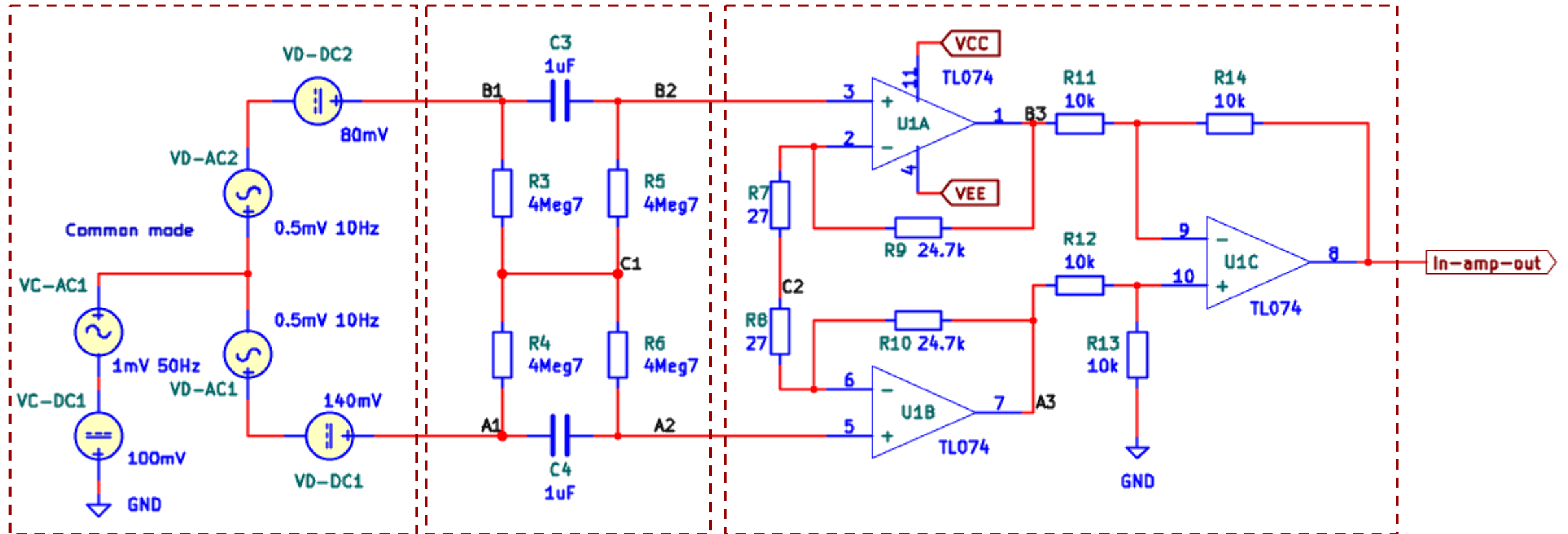


Problem 1 – EMG high-pass filter (sol)



Problem 2 Balanced In-amp

In this problem you will explore the AC and DC signals in an **AC-coupled** in-amp.
Sources are split in AC/DC and differential/common mode.



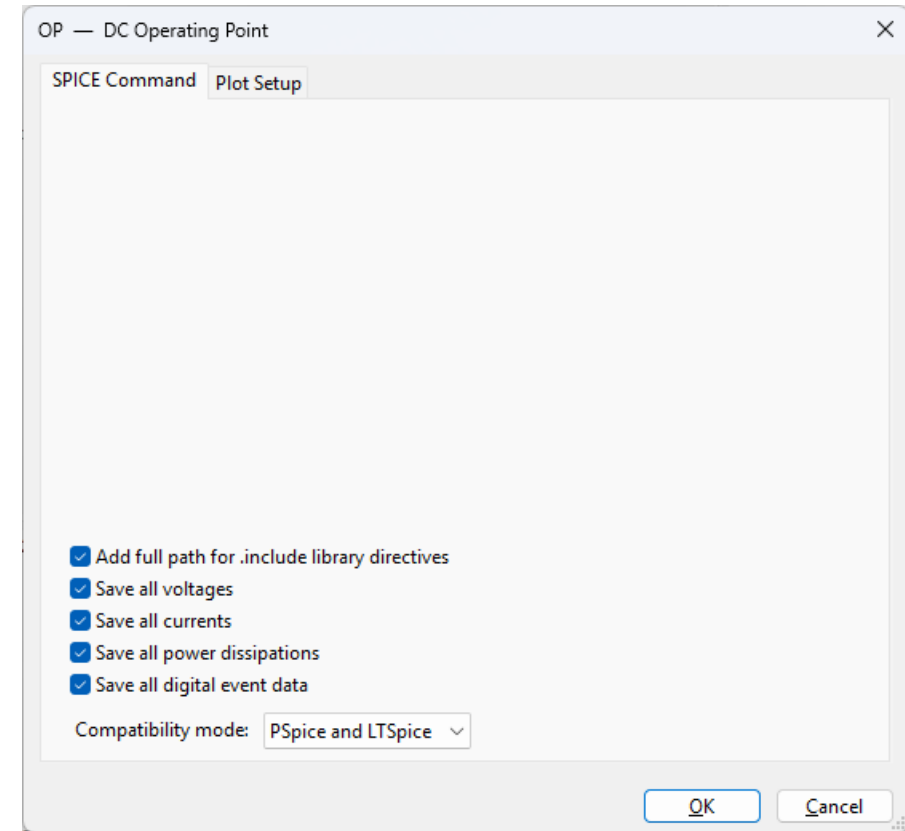
Source model

HP filter

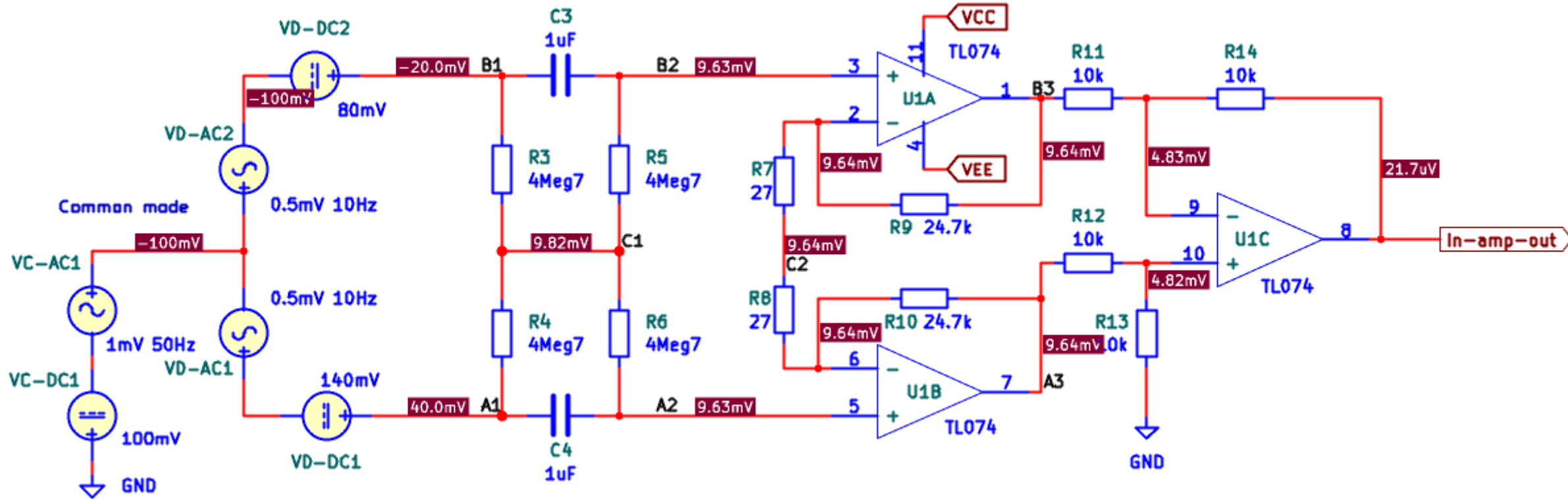
3-op-amp model of in-amp

Part 1

Do an operating point simulation to obtain DC voltages in different nodal points.



Problem 2(sol)



DC operating voltages.

V_{C_1} is the average of V_{A_1} and V_{B_1} . The highpass filter is a differential highpass filter. It blocks a differential DC signal. It does not block common mode DC signals.

Unless the in-amp is saturated, it will attenuate both AC and DC common mode signals.

The output DC signal is likely due to bias currents in op-amps.

Problem 2 Balanced In-amp

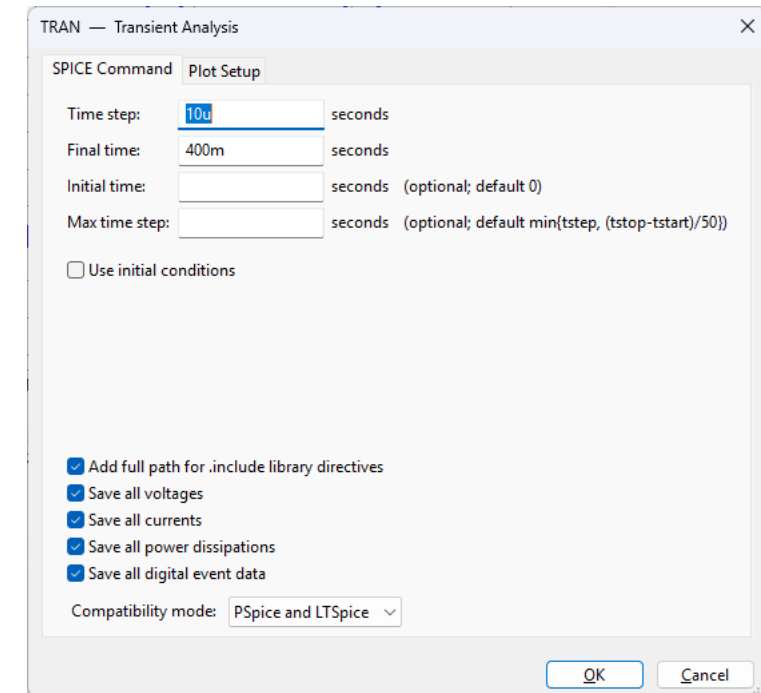
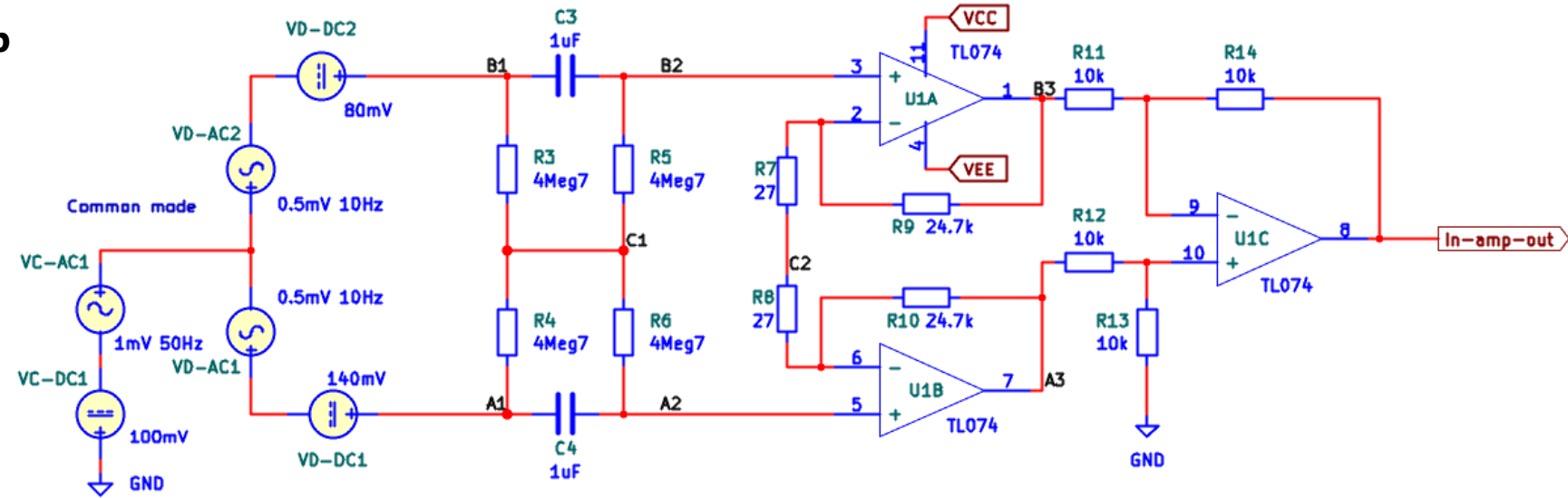
Part 2

Run a transient simulation.

Plot signals in nodes:

A1, B1, A2, B2.

Explain the signals making use of the DC operating voltages found earlier.



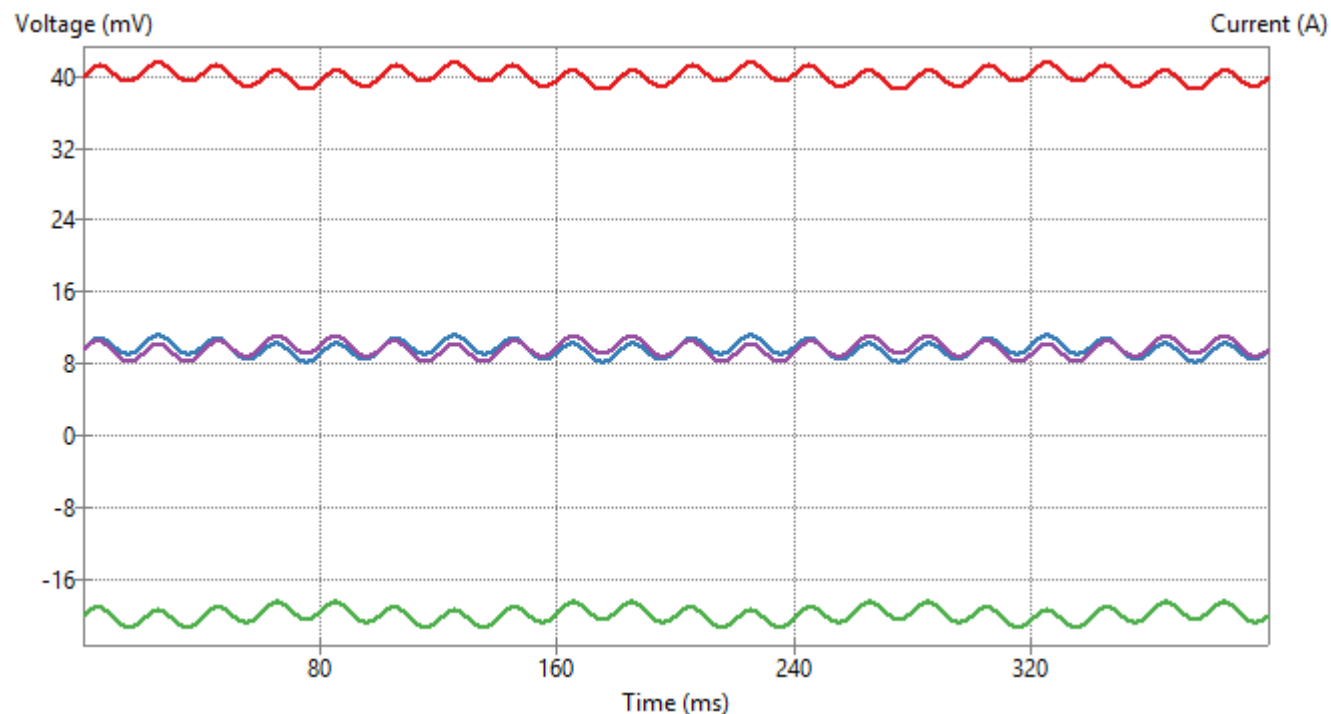
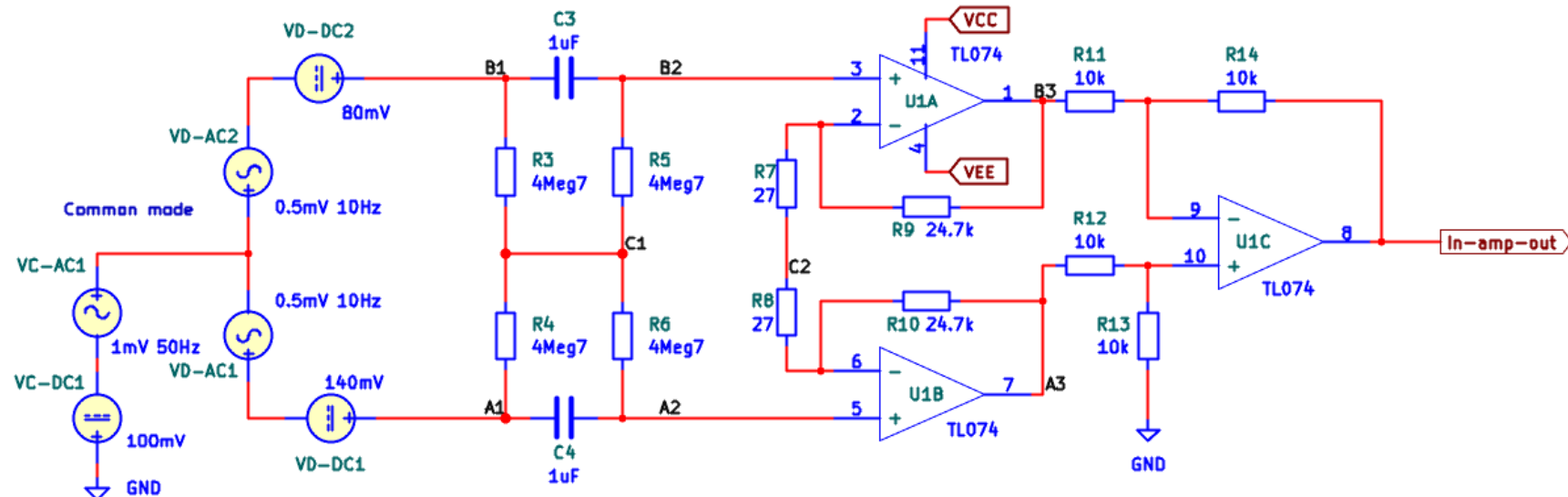
Problem 2 (sol)

$$V_{A_1} = 40\text{mV} + V_{C-AC1} + V_{D-AC1}$$

$$V_{B_1} = -20\text{mV} + V_{C-AC1} - V_{D-AC2}$$

$$V_{A_2} = 10\text{mV} + V_{C-AC1} + V_{D-AC1}$$

$$V_{B_2} = 10\text{mV} + V_{C-AC1} - V_{D-AC2}$$



Signal	Plot	Color	Cursor 1	Cursor 2
V(/A1)	☒	■	☐	☐
V(/A2)	☒	■	☐	☐
V(/A3)	☐			
V(/B1)	☒	■	☐	☐
V(/B2)	☒	■	☐	☐
V(/B3)	☐			
V(/C1)	☐			
V(/C2)	☐			
V(/OutLP)	☐			
V(In-amp-out)	☐			
V(Net-_C2-Pad1_)	☐			
V(Net-_U1A--_)	☐			
V(Net-_U1B--_)	☐			
V(Net-_U1C+_)	☐			

Cursor	Signal	Time	Value

Problem 2 Balanced In-amp

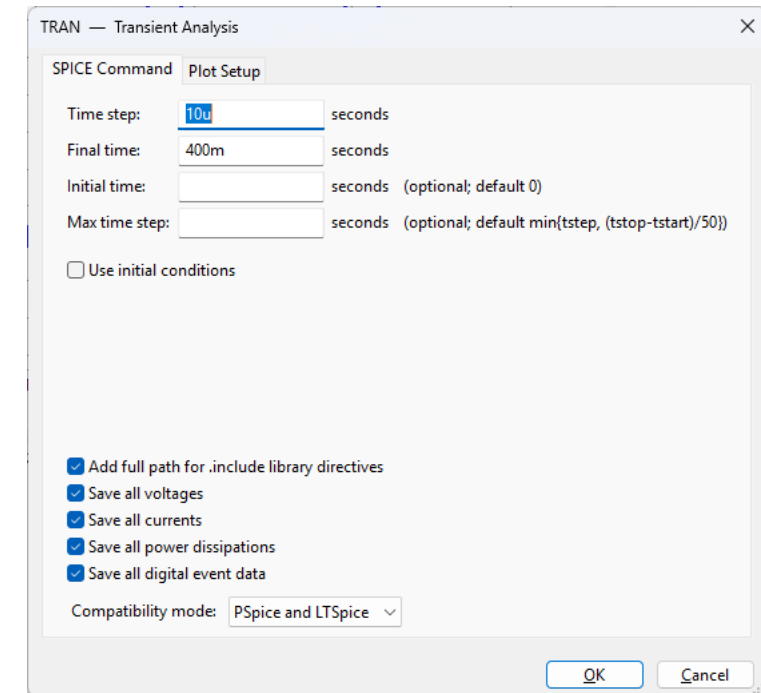
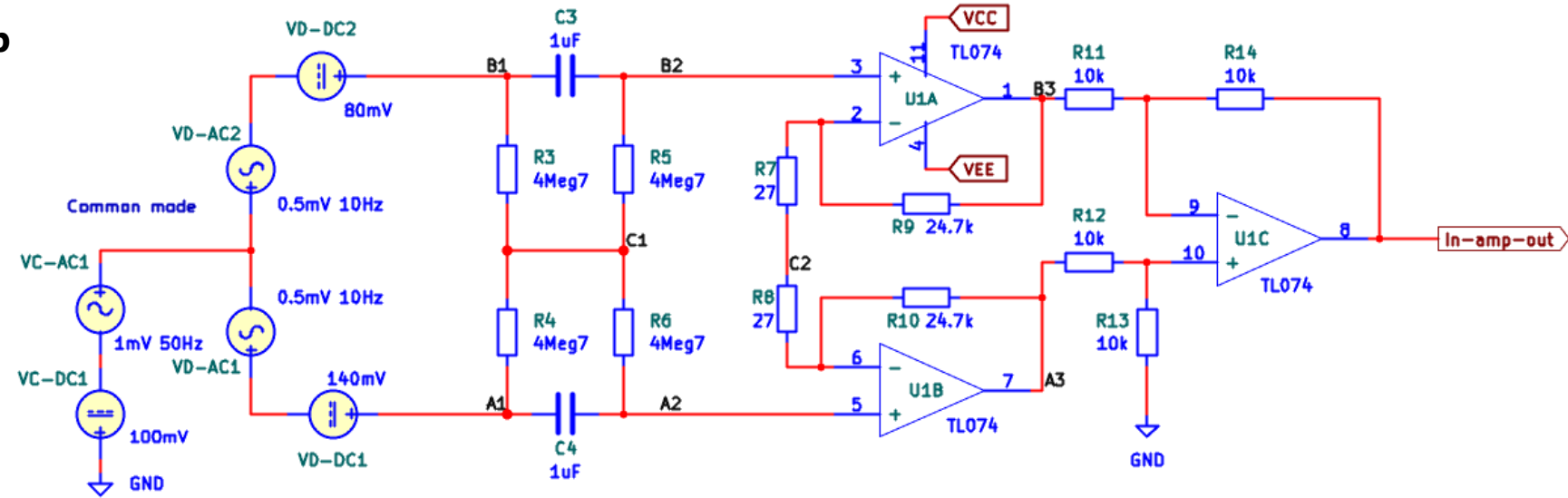
Part 3

Run a transient simulation.

Plot signals in nodes:

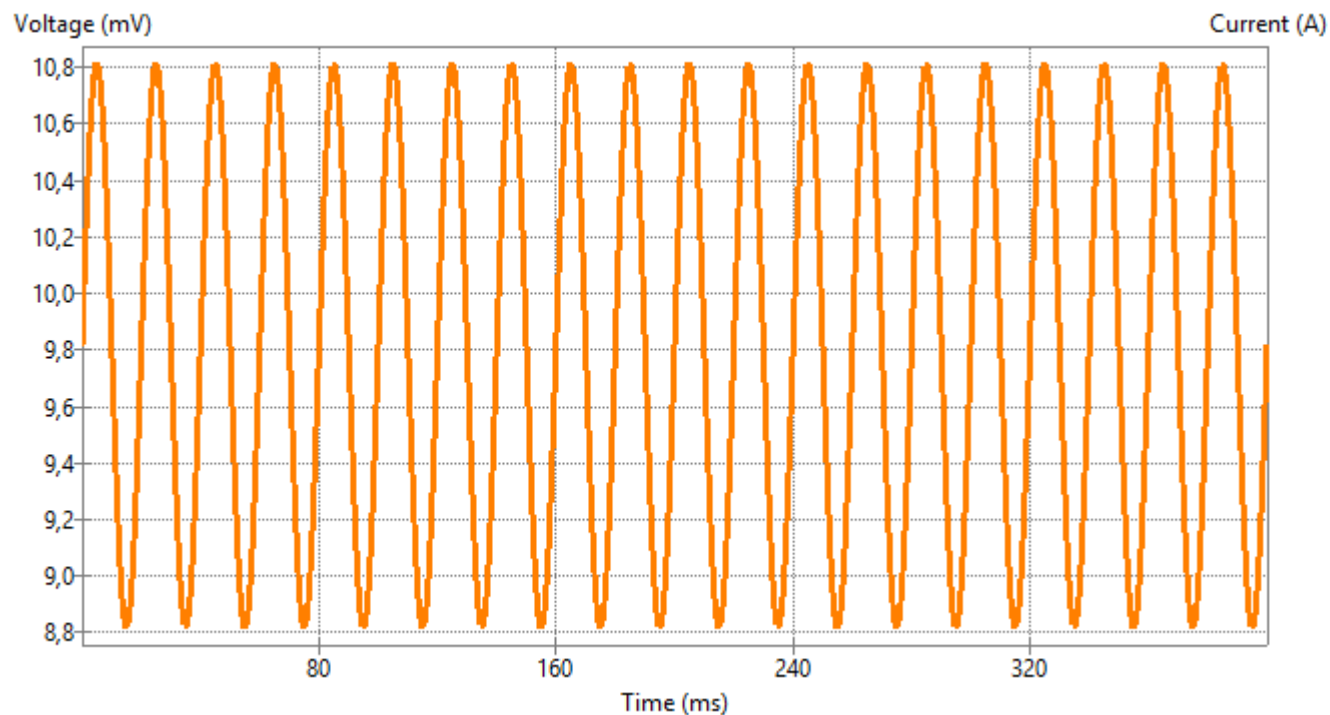
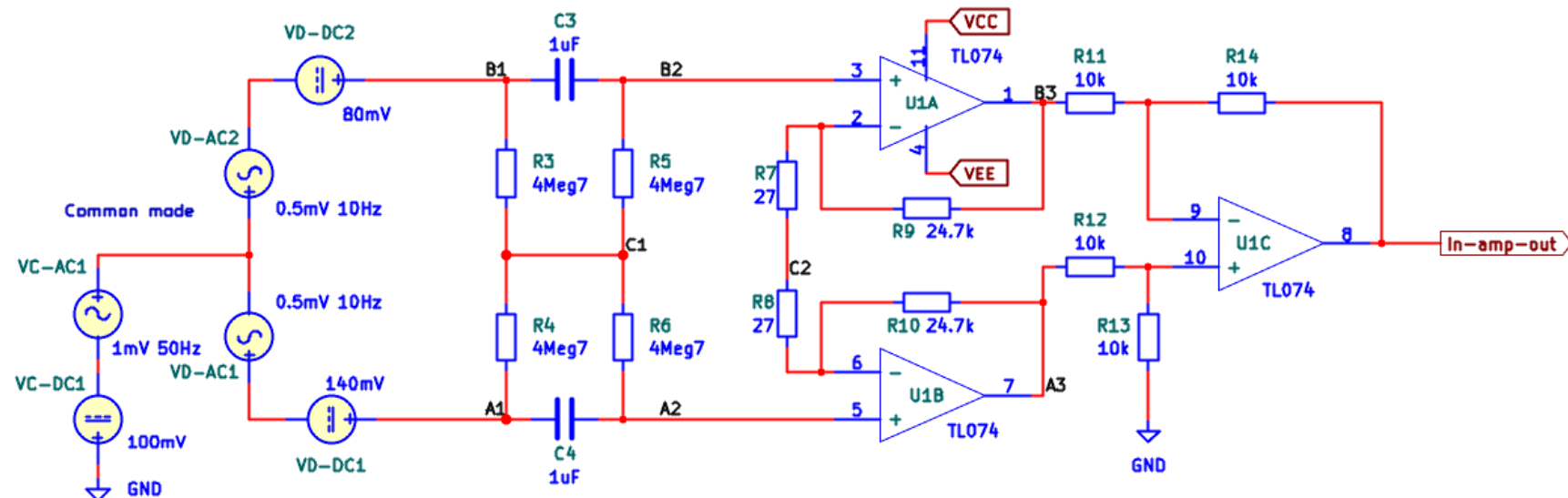
C1.

Explain the signal making use of the DC operating voltages found earlier.



Problem 2 (sol)

$$V_{C_1} = 10\text{mV} + V_{C-AC1}$$



Signal	Plot	Color	Cursor 1	Cursor 2
V(/A1)	☐			
V(/A2)	☐			
V(/A3)	☐			
V(/B1)	☐			
V(/B2)	☐			
V(/B3)	☐			
V(/C1)	☒		☐	☐
V(/C2)	☐			
V(/OutLP)	☐			
V(In-amp-out)	☐			
V(Net_C2-Pad1_)	☐			
V(Net_U1A--_)	☐			
V(Net_U1B--_)	☐			
V(Net_U1C+_)	☐			

Cursor	Signal	Time (ms)	Value

Problem 2 Balanced In-amp

Part 4

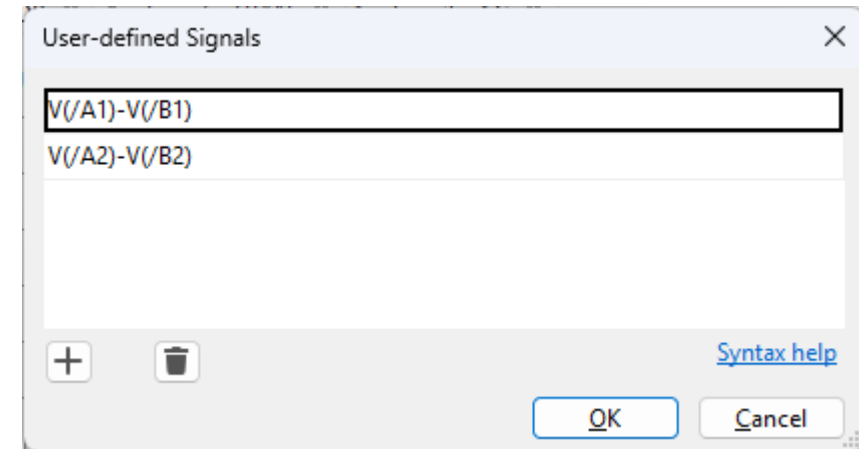
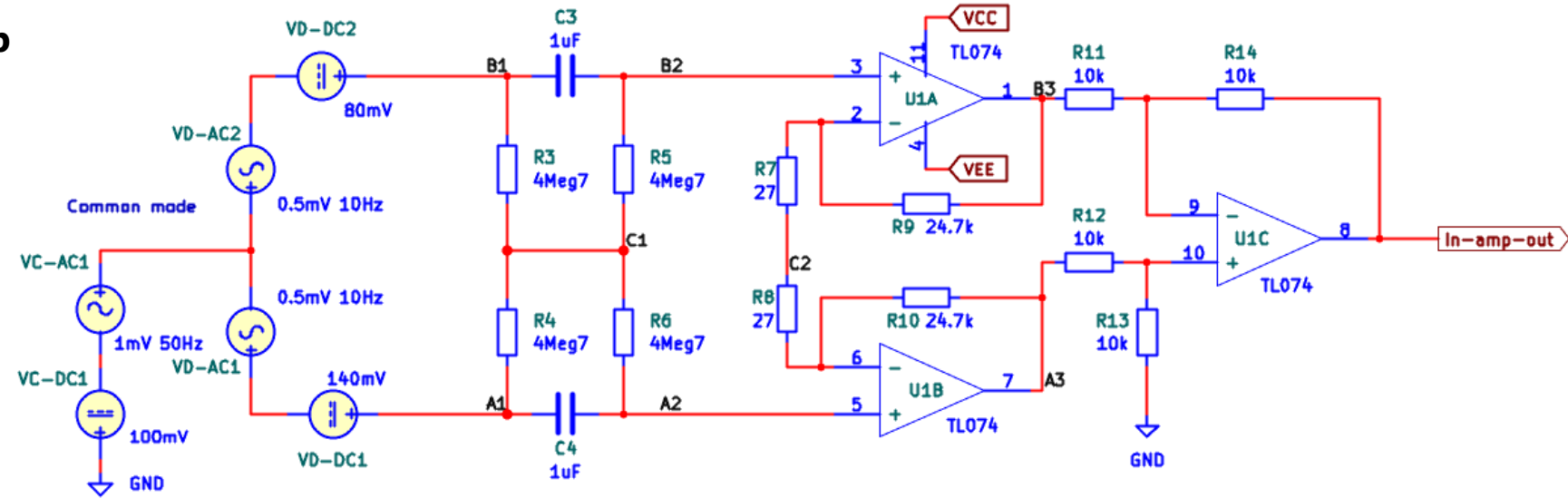
Run a transient simulation.

Create and plot two new differential signals:

A1 – B1

A2 – B2

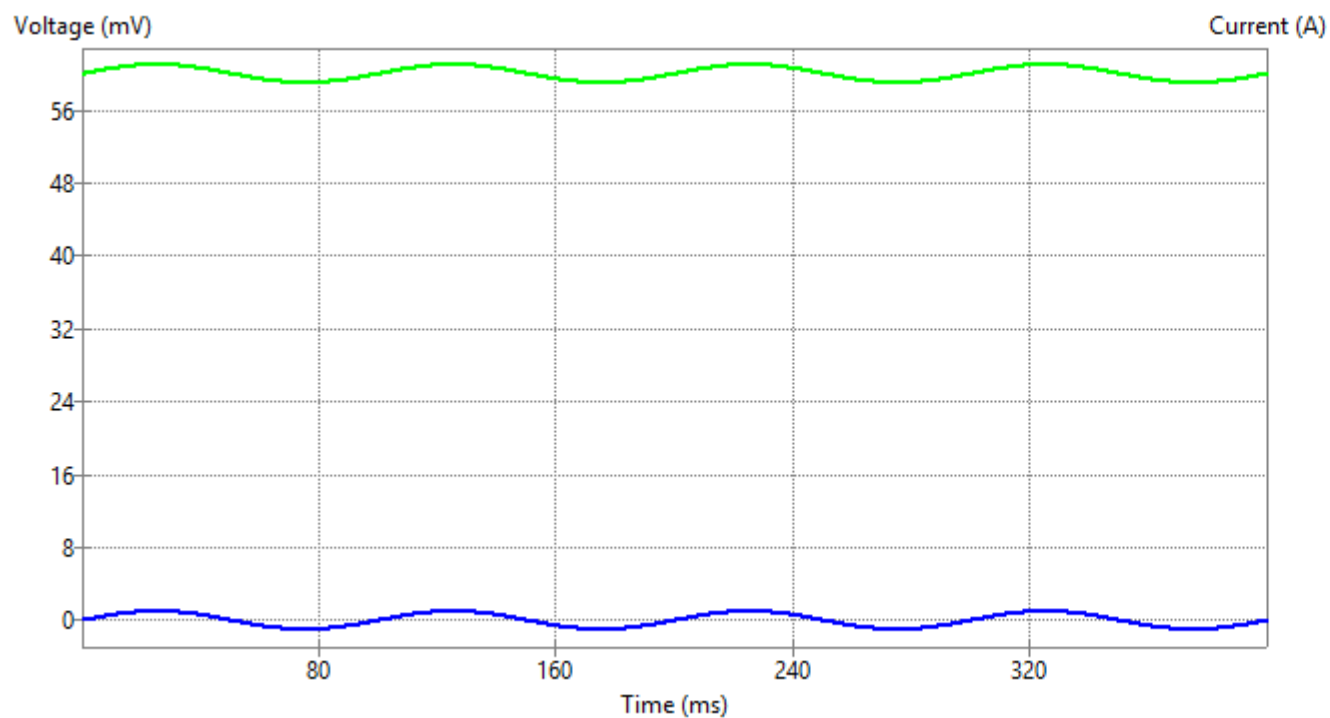
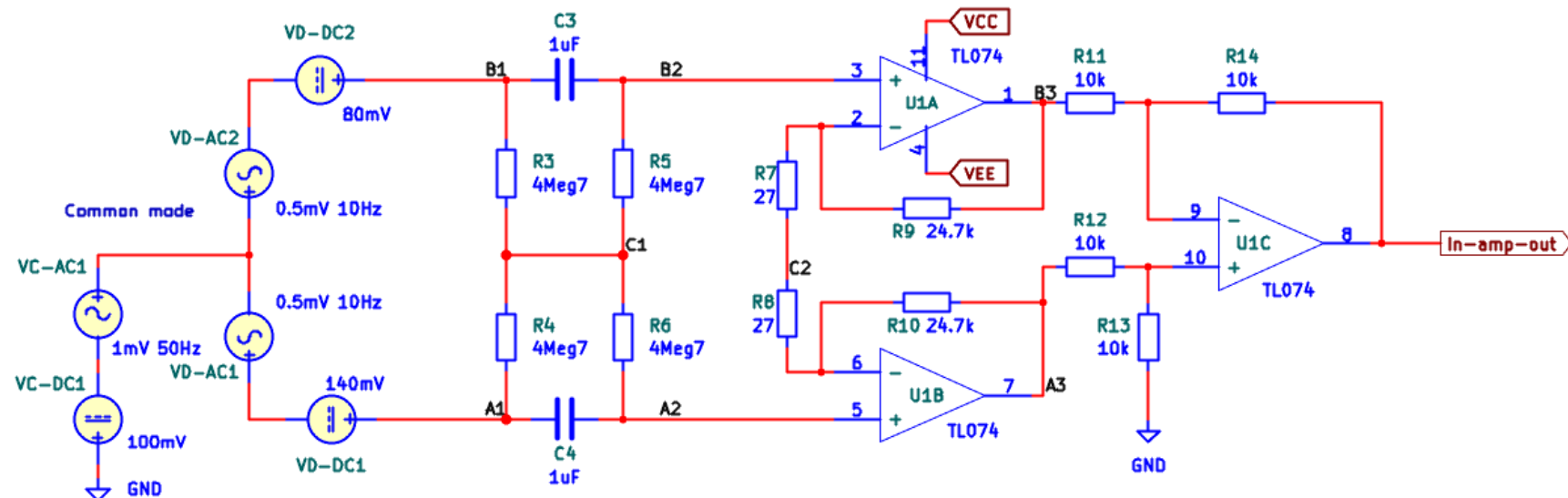
Explain the signals making use of the DC operating voltages found earlier.



Problem 2 (sol)

$$V_{A_1} - V_{B_1} = 60\text{mV} + V_{D-AC1} + V_{D-AC2}$$

$$V_{A_2} - V_{B_2} = V_{D-AC1} + V_{D-AC2}$$

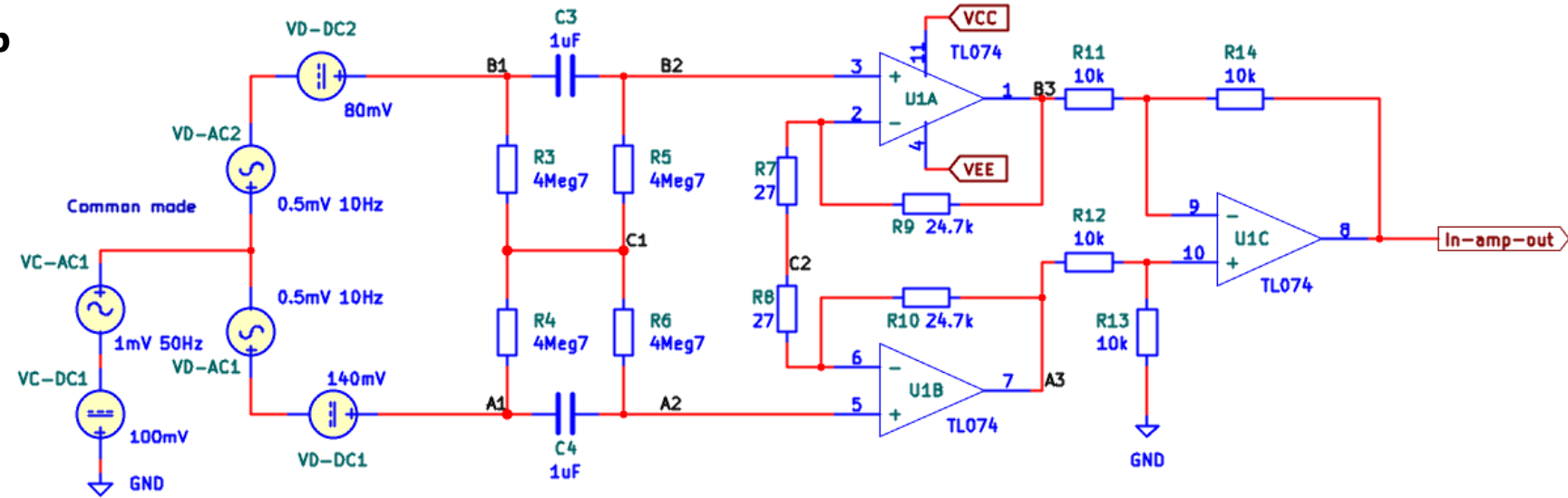


Signal	Plot	Color	Cursor 1	Cursor 2
V(/A1)	☐		☐	☐
V(/A1)-V(/B1)	☒	Green	☐	☐
V(/A2)	☐		☐	☐
V(/A2)-V(/B2)	☒	Blue	☐	☐
V(/A3)	☐		☐	☐
V(/B1)	☐		☐	☐
V(/B2)	☐		☐	☐
V(/B3)	☐		☐	☐
V(/C1)	☐		☐	☐
V(/C2)	☐		☐	☐
V(/OutLP)	☐		☐	☐
V(In-amp-out)	☐		☐	☐
V(Net-_C2-Pad1_)	☐		☐	☐
V(Net-_U1A--_)	☐		☐	☐

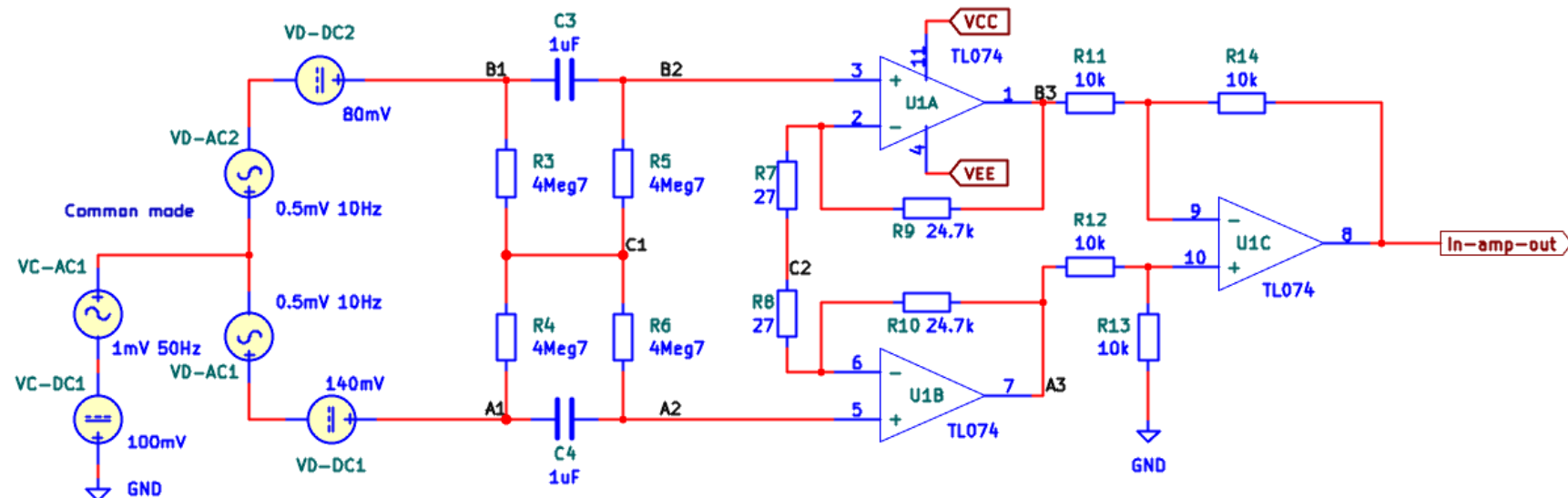
Cursor	Signal	Time	Value

Run a transient simulation.

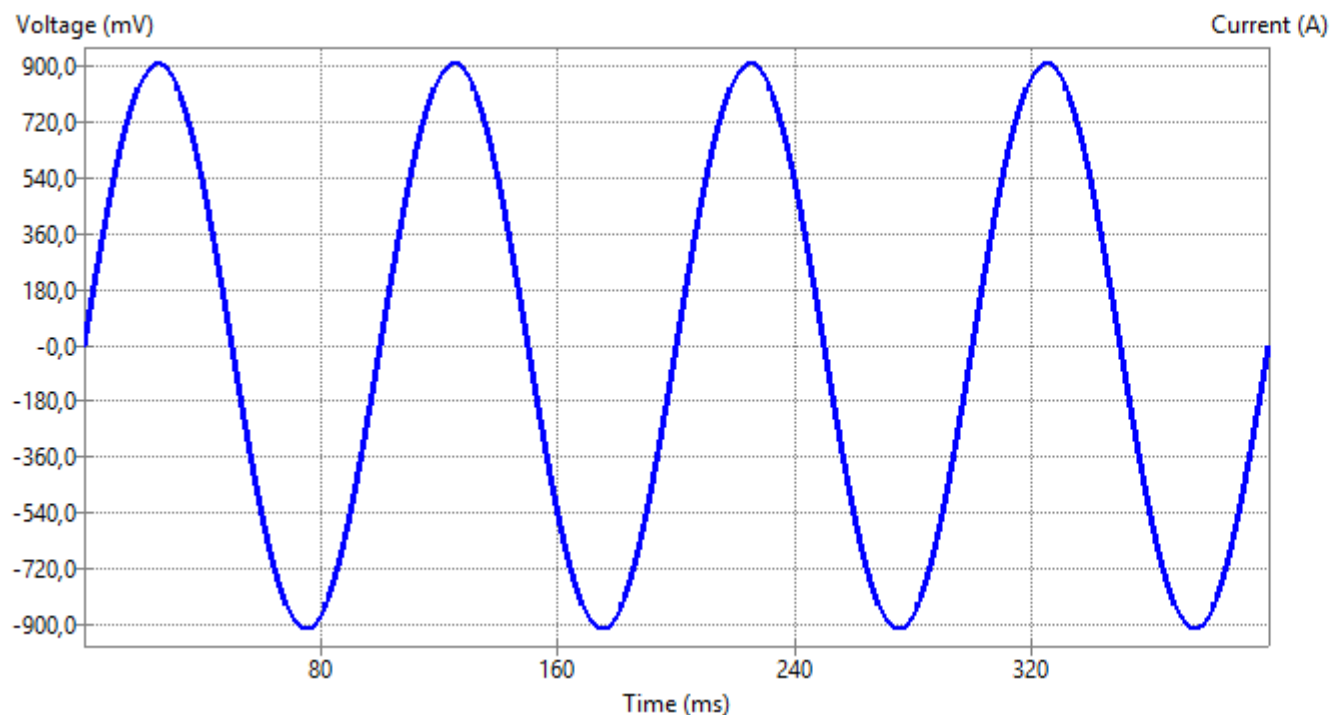
Explain the signals making use of the DC operating voltages found earlier.



Problem 2 (sol)



$$V_{in-amp-out} = G_d(V_{D-AC1} + V_{D-AC2})$$



Signal	Plot	Color	Cursor 1	Cursor 2
V(/A1)	☐			
V(/A1)-V(/B1)	☐			
V(/A2)	☐			
V(/A2)-V(/B2)	☐			
V(/A3)	☐			
V(/B1)	☐			
V(/B2)	☐			
V(/B3)	☐			
V(/C1)	☐			
V(/C2)	☐			
V(/OutLP)	☐			
V(In-amp-out)	☒	Blue	☐	☐
V(Net-_C2-Pad1_)	☐			
V(Net-_U1A--_)	☐			

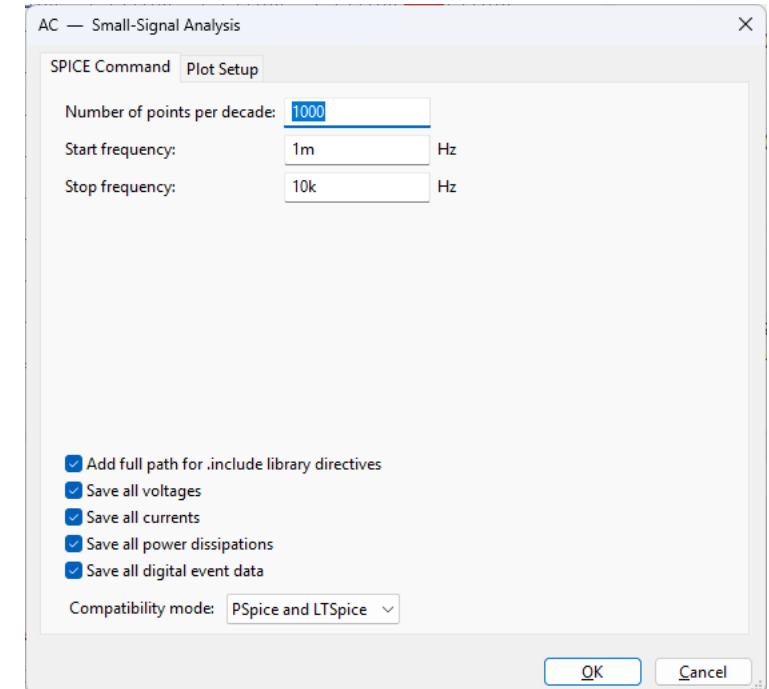
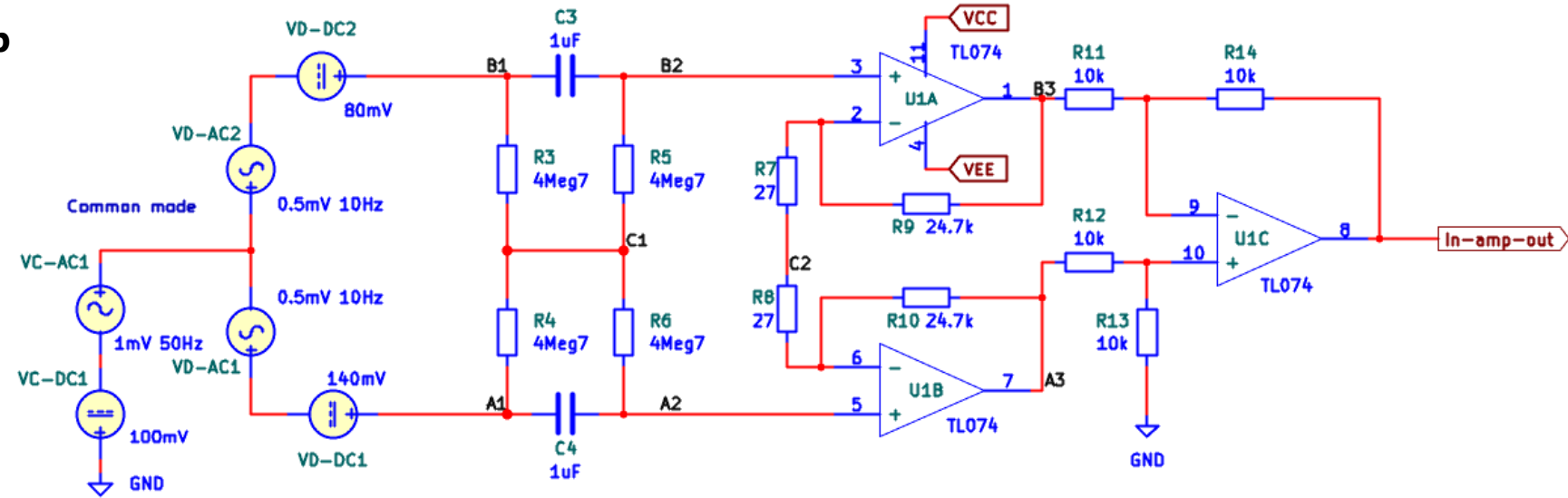
Cursor	Signal	Time	Value

Problem 2 Balanced In-amp

Part 6

Run an AC sweep simulation using VD-AC1 and VD-AC2.

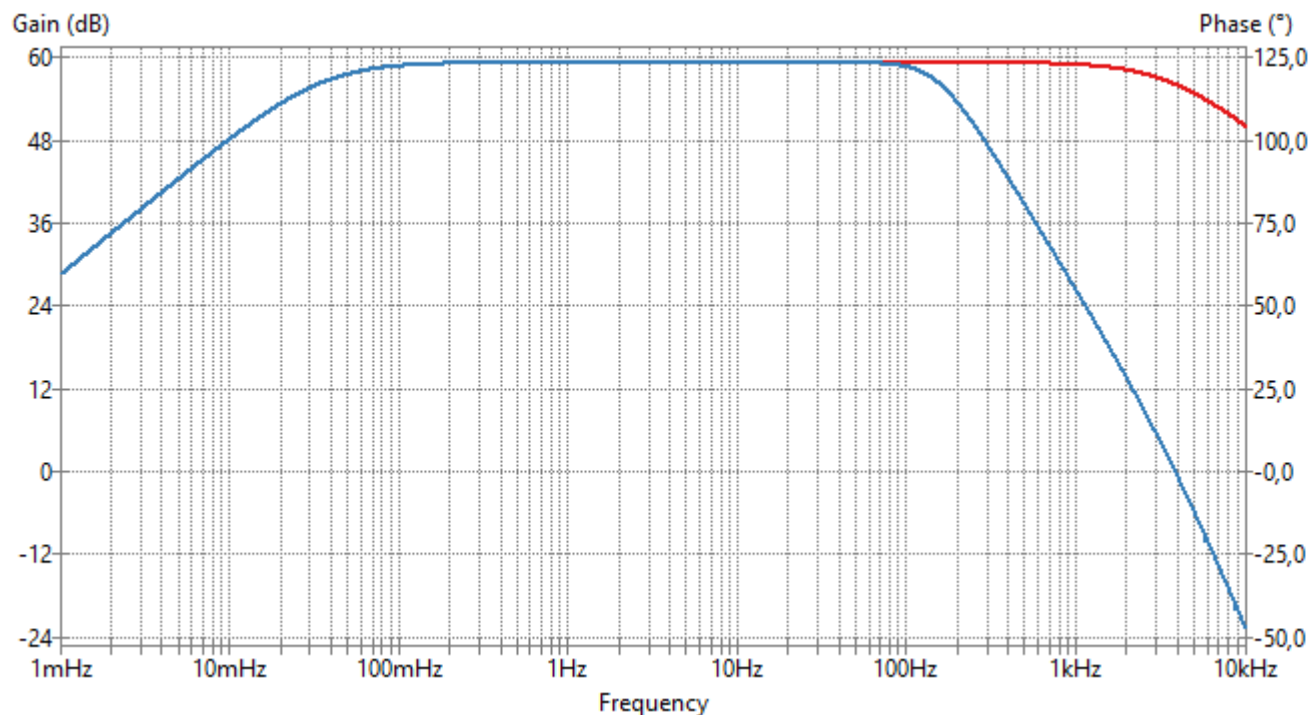
Plot the amplitude spectra of the output signals of the in-amp output and filter.



Problem 2 (sol)

AC sweep

Simulate a **differential signal** AC sweep from 1mHz – 10kHz. The differential gain should be $1000 = 60\text{dB}$.



Signal	Plot	Color	Cursor 1	Cursor 2
V(/B3) (gain)	☐			
V(/B3) (phase)	☐			
V(/C1) (gain)	☐			
V(/C1) (phase)	☐			
V(/C2) (gain)	☐			
V(/C2) (phase)	☐			
V(/OutLP) (gain)	☒	Blue	☐	☐
V(/OutLP) (phase)	☐			
V(In-amp-out) (gain)	☒	Red	☐	☐
V(In-amp-out) (phase)	☐			
V(Net-_C2-Pad1_) (gain)	☐			
V(Net-_C2-Pad1_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			

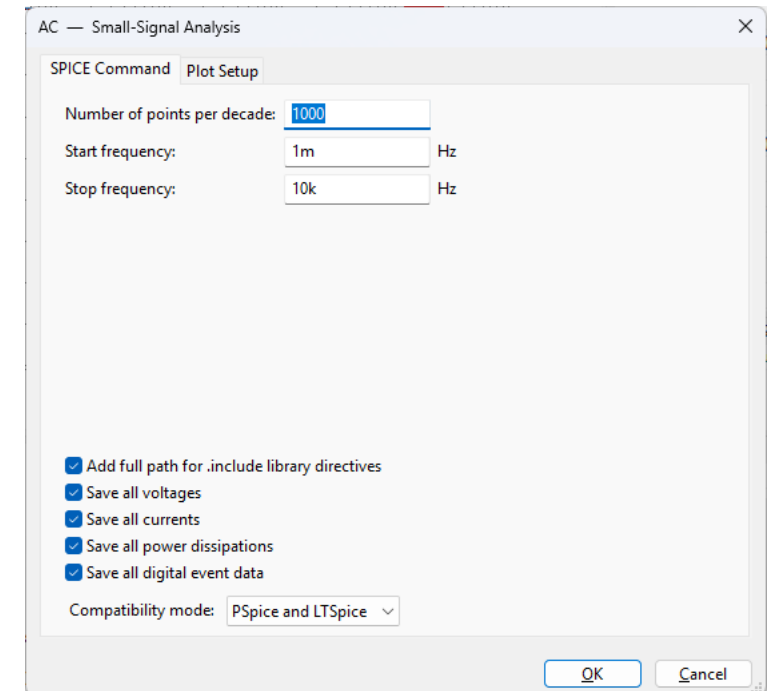
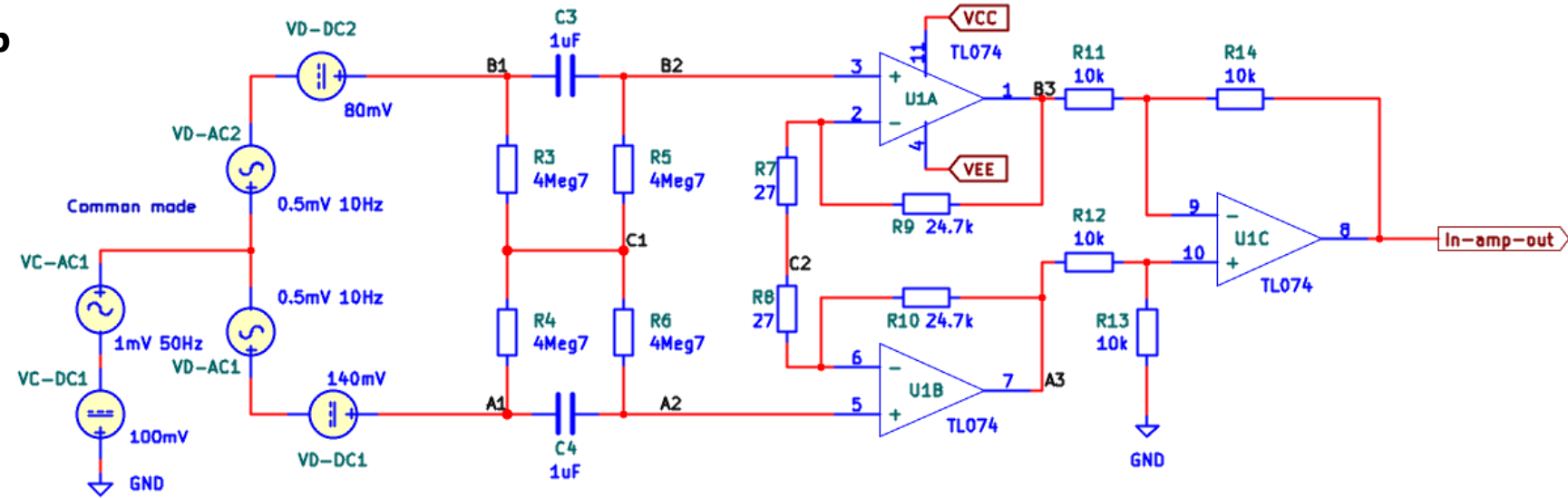
Cursor	Signal	Frequency	Value

Problem 2 Balanced In-amp

Part 7

Run a **common mode** AC sweep simulation using VC-AC1.

Plot the amplitude spectra of the output signals of the in-amp output and filter.

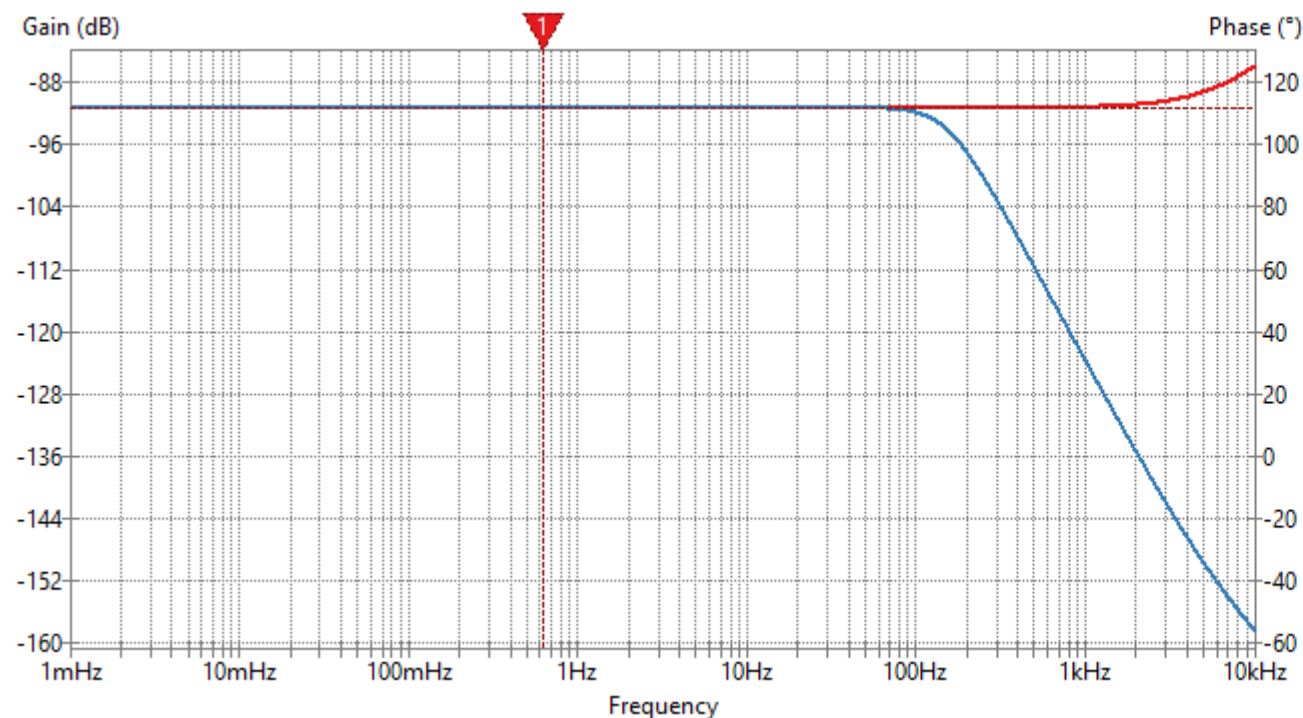


Problem 2 Balanced In-amp (sol)

AC sweep

Simulate a **common mode** AC sweep from 1mHz – 10kHz.

The attenuation is about -91dB.

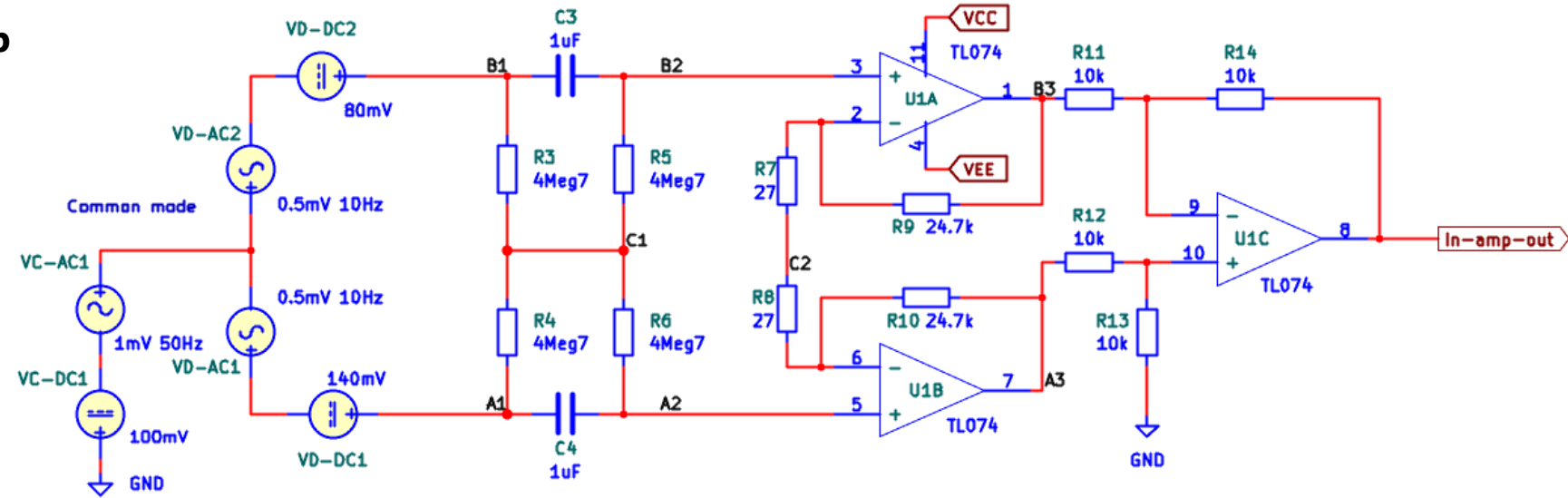


Signal	Plot	Color	Cursor 1	Cursor 2
V(/B3) (gain)	☐			
V(/B3) (phase)	☐			
V(/C1) (gain)	☐			
V(/C1) (phase)	☐			
V(/C2) (gain)	☐			
V(/C2) (phase)	☐			
V(/OutLP) (gain)	☒		☐	☐
V(/OutLP) (phase)	☐			
V(In-amp-out) (gain)	☒		☒	☐
V(In-amp-out) (phase)	☐			
V(Net-_C2-Pad1_) (gain)	☐			
V(Net-_C2-Pad1_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			
V(Net-_U1A--_) (phase)	☐			
V(Net-_U1A--_) (gain)	☐			

Problem 2 Balanced In-amp

Part 8

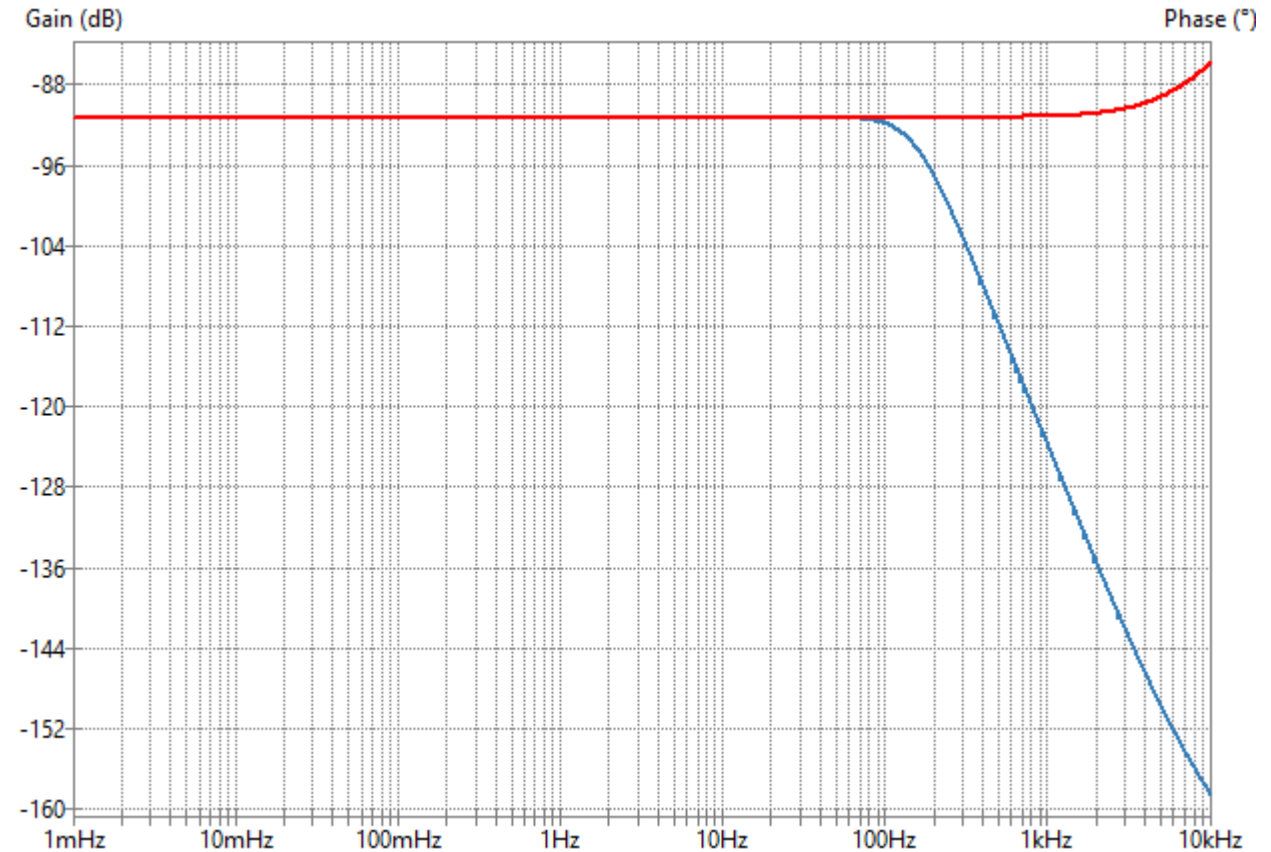
Using the differential and common mode AC sweeps, calculate the CMRR.



Problem 2 Balanced In-amp (sol)

The solution shows a common mode gain of -91.3dB within the passband.

$$\text{CMRR} = 60\text{dB} - (-91.3\text{dB}) = 151.3\text{dB}$$



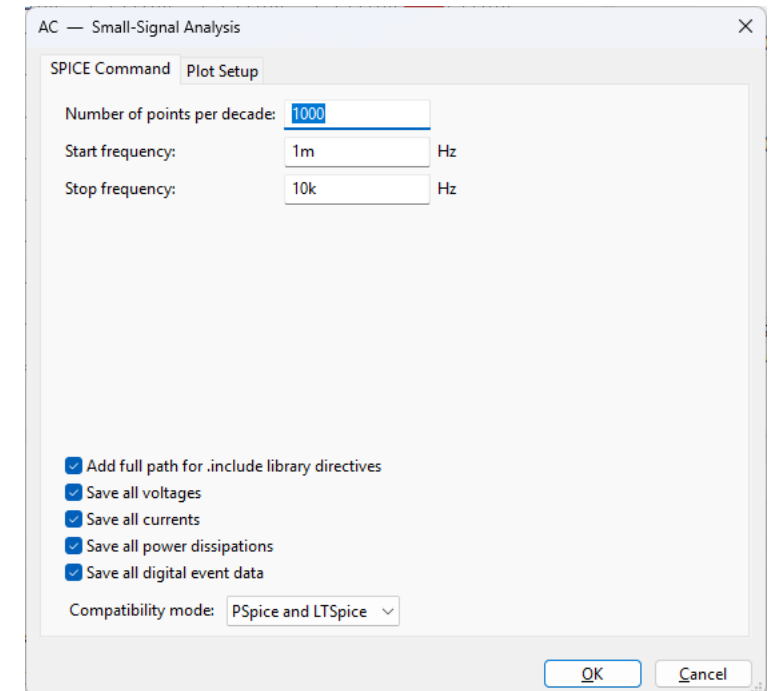
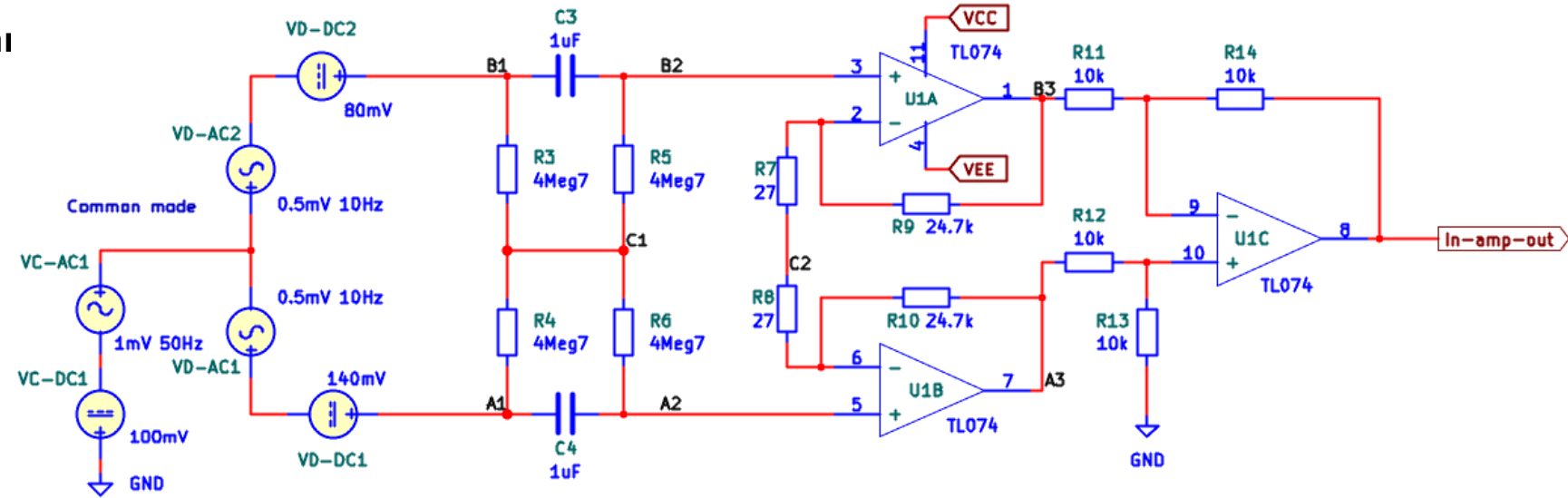
Problem 2 Imbalanced In-a

Part 9

Change R14 to 11k.

Run a new **common mode** AC sweep simulation using VC-AC1.

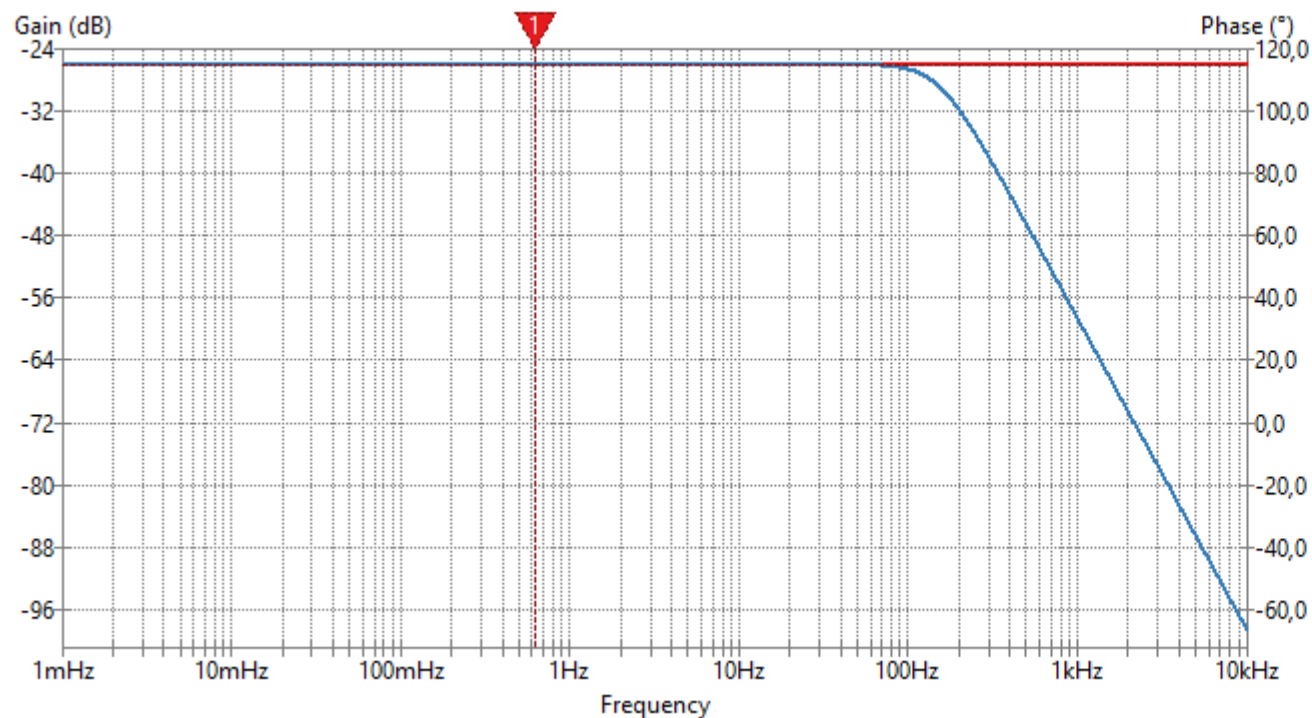
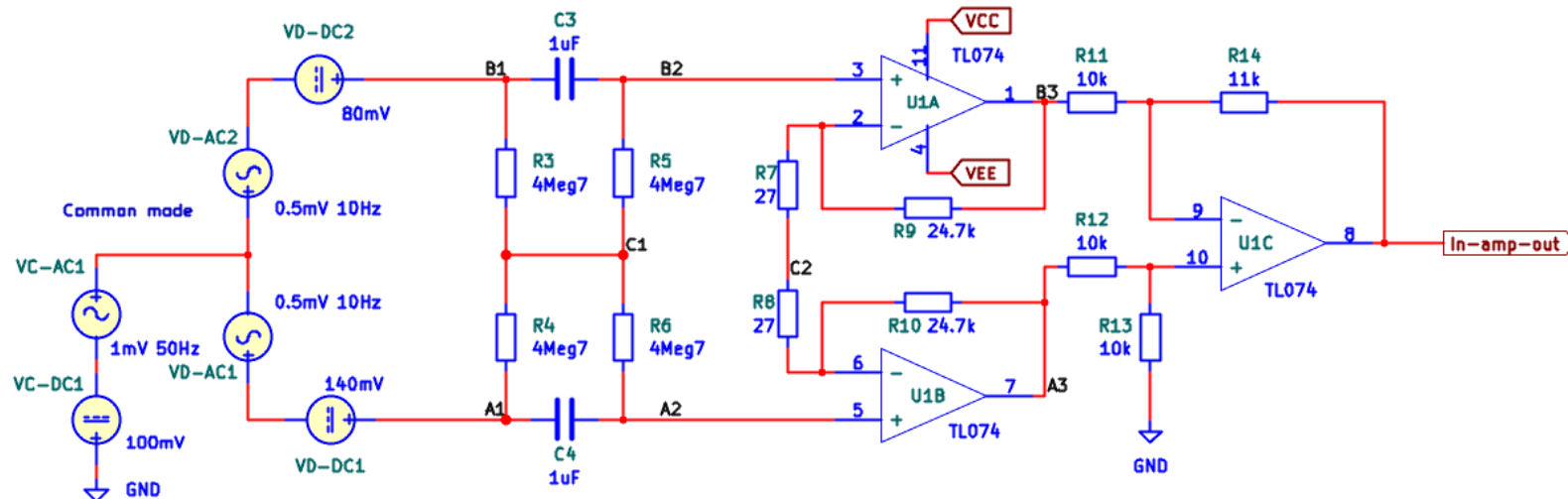
Plot the amplitude spectra of the output signals of the in-amp and filter.



Problem 2 (sol)

The solution shows a common mode gain of -26dB within the passband.

$$\text{CMRR} = 60\text{dB} - (-26\text{dB}) = 86\text{dB}$$



Signal	Plot	Color	Cursor 1	Cursor 2
V(/A1) (gain)	☐			
V(/A1) (phase)	☐			
V(/A1)-V(/B1) (gain)	☐			
V(/A1)-V(/B1) (phase)	☐			
V(/A2) (gain)	☐			
V(/A2) (phase)	☐			
V(/A2)-V(/B2) (gain)	☐			
V(/A2)-V(/B2) (phase)	☐			
V(/A3) (gain)	☐			
V(/A3) (phase)	☐			
V(/B1) (gain)	☐			
V(/B1) (phase)	☐			
V(/B2) (gain)	☐			
V(/B2) (phase)	☐			

Cursor	Signal	Frequency	Gain (dB)
1	V(In-amp-out) (gain)	634mHz	-26.0dB

Appendix

Resistor imbalance inside in-amp

We derived earlier:

$$V_o = \left(\frac{R_{3b} + R_{4b}}{R_{3b}} \right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}} \right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$

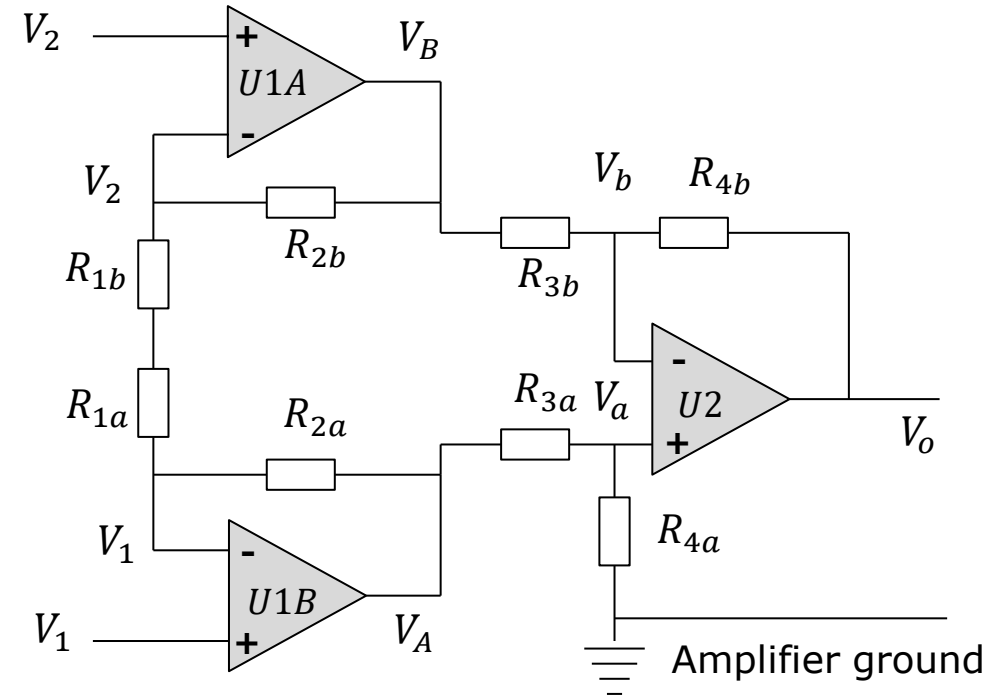
$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

$$V_B = \left(1 + \frac{R_{2b}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{R_{2b}}{R_{1a} + R_{1b}} \cdot V_1$$

We then assumed that internal resistors were balanced:

$$R_{3a} = R_{3b} = R_{4a} = R_{4b}$$

$$R_{2a} = R_{2b}$$



$$V_o = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{R_G} \right) (V_1 - V_2)$$

We now permit **imbalance** between certain internal resistors:

$$R_{3a} = R_{3b} = R_{4a}$$

$$R_{4b} = \alpha R_{4a}$$

$$R_{2b} = \beta R_{2a}$$

$$V_o = \left(\frac{R_{3b} + R_{4b}}{R_{3b}} \right) \left(\frac{R_{4a}}{R_{3a} + R_{4a}} \right) V_A - \frac{R_{4b}}{R_{3b}} V_B$$

$$V_o = \left(\frac{R_{3a} + \alpha R_{4a}}{R_{3a} + R_{4a}} \right) \left(\frac{R_{4a}}{R_{3a}} \right) V_A - \frac{\alpha R_{4a}}{R_{3a}} V_B$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) V_A - \alpha V_B$$

$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

$$V_B = \left(1 + \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \cdot V_1$$

Inserting V_A and V_B in V_o :

$$V_o = \left(\frac{1 + \alpha}{2} \right) V_A - \alpha V_B$$

$$V_A = \left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

$$V_B = \left(1 + \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \cdot V_1$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \left(\left(1 + \frac{R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_1 - \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2 \right) - \alpha \left(\left(1 + \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \right) \cdot V_2 - \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \cdot V_1 \right)$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \left(\frac{1 + \alpha}{2} \right) \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_1 - \alpha \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \cdot V_2 + \alpha \frac{\beta R_{2a}}{R_{1a} + R_{1b}} \cdot V_1 - \left(\frac{1 + \alpha}{2} \right) \frac{R_{2a}}{R_{1a} + R_{1b}} \cdot V_2$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_{1a} + R_{1b}} \left(\left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \beta \cdot V_2 + \alpha \beta \cdot V_1 - \left(\frac{1 + \alpha}{2} \right) \cdot V_2 \right)$$

Further reorganisation of terms:

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_{1a} + R_{1b}} \left(\left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha\beta \cdot V_2 + \alpha\beta \cdot V_1 - \left(\frac{1 + \alpha}{2} \right) \cdot V_2 \right)$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_{1a} + R_{1b}} \left(\left(\frac{1 + \alpha}{2} + \alpha\beta \right) \cdot V_1 - \left(\frac{1 + \alpha}{2} + \alpha\beta \right) \cdot V_2 \right)$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_{1a} + R_{1b}} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

$$R_G \stackrel{\text{def}}{=} R_{1a} + R_{1b}$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

Verifying result by reinserting balanced resistors:

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

$$\beta = 1$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + 3\alpha}{2} \right) (V_1 - V_2)$$

$$\alpha = 1$$

$$V_o = \left(1 + \frac{(1 + \beta)R_{2a}}{R_G} \right) (V_1 - V_2)$$

$$\alpha = \beta = 1$$

$$V_o = \left(1 + \frac{2R_{2a}}{R_G} \right) (V_1 - V_2)$$

This final reduced expression is identical to the expression derived for the instrumentation amplifier with balanced resistors.

Let us now derive the output from an imbalanced instrumentation amplifier with **only a common mode input**.

$$V_1 = V_2 = V_{cm} \Rightarrow$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_{cm} - \alpha \cdot V_{cm} + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_{cm} - V_{cm})$$

$$V_o = \left(\frac{1 - \alpha}{2} \right) V_{cm}$$

From this result we can define a **common mode gain** G_c :

$$G_c = \frac{V_o}{V_{cm}} = \frac{1 - \alpha}{2}$$

Let us now derive the output from an imbalanced instrumentation amplifier with **only a differential mode input**.

$$V_1 = -V_2 = \frac{V_d}{2}$$

$$G_{d,bal} \stackrel{\text{def}}{=} 1 + \frac{2R_{2a}}{R_G}$$

$$\frac{R_{2a}}{R_G} = \frac{G_{d,bal} - 1}{2}$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot V_1 - \alpha \cdot V_2 + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) (V_1 - V_2)$$

$$V_o = \left(\frac{1 + \alpha}{2} \right) \cdot \frac{V_d}{2} + \alpha \cdot \frac{V_d}{2} + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) V_d$$

$$V_o = \left(\frac{1 + 3\alpha}{4} \right) \cdot V_d + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) V_d$$

$$V_o = \left(\frac{1 + 3\alpha}{4} + \frac{R_{2a}}{R_G} \left(\frac{1 + \alpha}{2} + \alpha\beta \right) \right) V_d$$

$$V_o = \left(\frac{1 + 3\alpha}{4} + \left(\frac{G_{d,bal} - 1}{2} \right) \left(\frac{1 + \alpha}{2} + \alpha\beta \right) \right) V_d$$

We continue manipulating:

$$V_o = \left(\frac{1 + 3\alpha}{4} + \left(\frac{G_{d,bal} - 1}{2} \right) \left(\frac{1 + \alpha}{2} + \alpha\beta \right) \right) V_d$$

We can now define the differential gain for the imbalanced instrumentation amplifier:

$$G_{d,imbal} \stackrel{\text{def}}{=} \frac{1 + 3\alpha}{4} + \left(\frac{G_{d,bal} - 1}{2} \right) \left(\frac{1 + \alpha}{2} + \alpha\beta \right)$$

$$G_{d,imbal} = \frac{1 + 3\alpha}{4} + (G_{d,bal} - 1) \left(\frac{1 + \alpha}{4} + \frac{\alpha\beta}{2} \right)$$

$$G_{d,imbal} = \frac{1 + 3\alpha}{4} - \frac{1 + \alpha}{4} - \frac{\alpha\beta}{2} + G_{d,bal} \left(\frac{1 + \alpha}{4} + \frac{\alpha\beta}{2} \right)$$

$$G_{d,imbal} = \frac{\alpha}{2} (1 - \beta) + G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{4} \right)$$

Numerical example:

Assume that $\alpha = \beta = 0.9$

The first term is negligible.

$$G_{d,imbal} = \frac{\alpha}{2}(1 - \beta) + G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{4} \right)$$

$$G_{d,imbal} = \frac{0.9}{2}(1 - 0.9) + G_{d,bal} \left(\frac{1 + 0.9 + 2 \cdot 0.9 \cdot 0.9}{4} \right)$$

$$G_{d,imbal} = \cancel{0.045} + 0.88G_{d,bal}$$

$$\alpha = \beta = 0.9 \Rightarrow$$

$$G_{d,imbal} = 0.88G_{d,bal}$$

$$\alpha = \beta = 1.1 \Rightarrow$$

$$G_{d,imbal} = 1.13G_{d,bal}$$

The purpose of a differential amplifier is to amplify differential signals and block common mode signals.

In a real-life instrumentation amplifier, some level of imbalance is to be expected in the internal resistors. The result is that a **common mode input signal will not be blocked completely**.

We can express the ability to suppress common mode signals as the **ratio of differential gain to common mode gain**. Here the term in red is small (see previous slide) and has been ignored.

$$G_{d,imbal} = \frac{V_o}{V_d} = \frac{\alpha}{2}(1 - \beta) + G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{4} \right)$$

$$G_c = \frac{V_o}{V_{cm}} = \frac{1 - \alpha}{2}$$

$$\frac{G_{d,imbal}}{G_c} = \frac{\frac{\alpha}{2}(1 - \beta) + G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{4} \right)}{\frac{1 - \alpha}{2}}$$

$$\frac{G_{d,imbal}}{G_c} = G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{2(1 - \alpha)} \right)$$

The most influential factor is α .

Let us assume $\beta = 1$:

$$\frac{G_{d,imbal}}{G_c} = G_{d,bal} \left(\frac{1 + \alpha + 2\alpha\beta}{2(1 - \alpha)} \right)$$

$$\frac{G_{d,imbal}}{G_c} = G_{d,bal} \left(\frac{1 + 3\alpha}{2(1 - \alpha)} \right)$$

$$1\% \quad \alpha = 0.99$$

$$\frac{G_{d,imbal}}{G_c} = 198.5 G_{d,bal}$$

$$10\% \quad \alpha = 0.9$$

$$\frac{G_{d,imbal}}{G_c} = 18.5 G_{d,bal}$$

The gain ratio G_d/G_c is called the **common mode rejection ratio**:

$$CMRR = \frac{G_d}{G_c}$$

CMRR is usually given in decibel:

$$CMRR_{dB} = 20 \log_{10} \frac{G_d}{G_c}$$

Assume as an example $G_{d,bal} = 10$.

$$1\% \quad \alpha = 0.99$$

$$CMRR_{dB} = 20 \log_{10} 198.5 \cdot 10 = 66\text{dB}$$

$$10\% \quad \alpha = 0.9$$

$$CMRR_{dB} = 20 \log_{10} 18.5 \cdot 10 = 45\text{dB}$$

Assume as an example $G_{d,bal} = 100$.

$$1\% \quad \alpha = 0.99$$

$$CMRR_{dB} = 20 \log_{10} 198.5 \cdot 100 = 86\text{dB}$$

$$10\% \quad \alpha = 0.9$$

$$CMRR_{dB} = 20 \log_{10} 18.5 \cdot 100 = 65\text{dB}$$

Observation:

A large differential gain improves the CMRR.

Appendix

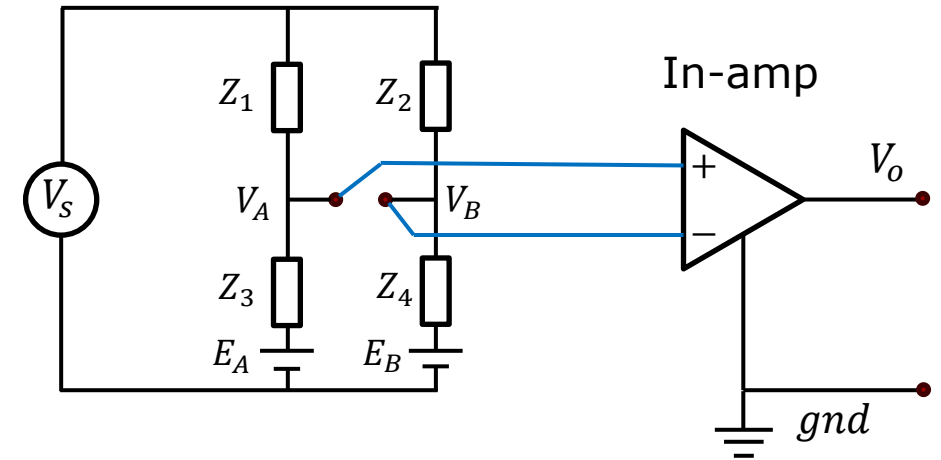
AC coupled instrumentation amplifier

Video

Now we will add a DC source in series with Z_3 and Z_4 . This may challenge the amplifier.

We need to derive an expression for $V_A - V_B$.

Using the technique of current superposition:



Node A: $\frac{V_A}{Z_A} = \frac{V_S}{Z_1} + \frac{E_A}{Z_3}$ $Z_A \stackrel{\text{def}}{=} \frac{Z_1 Z_3}{Z_1 + Z_3}$

$$V_A = \frac{Z_A}{Z_1} V_S + \frac{Z_A}{Z_3} E_A = \frac{Z_3}{Z_1 + Z_3} V_S + \frac{Z_1}{Z_1 + Z_3} E_A$$

Node B: $\frac{V_B}{Z_B} = \frac{V_S}{Z_2} + \frac{E_B}{Z_4}$ $Z_B \stackrel{\text{def}}{=} \frac{Z_2 Z_4}{Z_2 + Z_4}$

$$V_B = \frac{Z_B}{Z_2} V_S + \frac{Z_B}{Z_4} E_B = \frac{Z_4}{Z_2 + Z_4} V_S + \frac{Z_2}{Z_2 + Z_4} E_B$$

As expected, the amplifier now must amplify both an AC and a DC signal:

$$V_A - V_B = \underbrace{\left(\frac{Z_3}{Z_1 + Z_3} - \frac{Z_4}{Z_2 + Z_4} \right) V_S}_{AC \text{ signal}} + \underbrace{\left(\frac{Z_1}{Z_1 + Z_3} E_A - \frac{Z_2}{Z_2 + Z_4} E_B \right)}_{DC \text{ signal}}$$

We have not been specific about the context, so we do not know anything about the magnitudes of the impedances, nor the magnitude of the AC and DC sources.

We will assume that all impedances have very similar values, but that there is a small detectable difference ΔZ between Z_3 and Z_4 .

$$Z_1 = Z_2$$

$$Z_3 = Z_4 + \Delta Z$$

$$\Delta Z \ll \min(Z_1, Z_2, Z_3, Z_4)$$

$$V_A - V_B = \underbrace{\left(\frac{Z_3}{Z_1 + Z_3} - \frac{Z_4}{Z_2 + Z_4} \right) V_s}_{AC \text{ signal}} + \underbrace{\left(\frac{Z_1}{Z_1 + Z_3} E_A - \frac{Z_2}{Z_2 + Z_4} E_B \right)}_{DC \text{ signal}}$$

$$V_{AC} = \left(\frac{Z_3}{Z_1 + Z_3} - \frac{Z_4}{Z_2 + Z_4} \right) V_s = \left(\frac{Z_4 + \Delta Z}{Z_1 + Z_4 + \Delta Z} - \frac{Z_4}{Z_1 + Z_4} \right) V_s$$

$$\approx \left(\frac{Z_4 + \Delta Z}{Z_1 + Z_4} - \frac{Z_4}{Z_1 + Z_4} \right) V_s \approx \frac{\Delta Z}{Z_1 + Z_4} V_s$$

$$V_{DC} = \frac{Z_1}{Z_1 + Z_3} E_A - \frac{Z_2}{Z_2 + Z_4} E_B = \frac{Z_1}{Z_1 + Z_4 + \Delta Z} E_A - \frac{Z_1}{Z_1 + Z_4} E_B$$

$$\approx \frac{Z_1}{Z_1 + Z_4} (E_A - E_B)$$

$$V_A - V_B = V_{AC} + V_{DC}$$

For the sake of illustration, we will assume that the AC signal is in the range of $\pm 2.5\text{mV}$:

We will assume that the DC sources are half-cell voltages originating from within the sensors Z_3 and Z_4 . These DC sources can be hundreds of millivolts thus their difference can also be hundreds of millivolts. Here we will assume a DC voltage of 100mV .

We don't care about the DC signal. However, we want to amplify the AC signal to the full range of a 5V ADC.

To amplify the AC signal adequately, we need a gain of:

This gain will saturate the amplifier because the DC voltage on the output would have to be 100V .

If we are to obtain a gain of 1000 for the AC signal, we must remove the DC signal with a high-pass filter.

$$V_{AC} = \frac{\Delta Z}{\underbrace{Z_1 + Z_4}_{\ll 1}} V_s \approx \pm 2.5\text{mV}$$

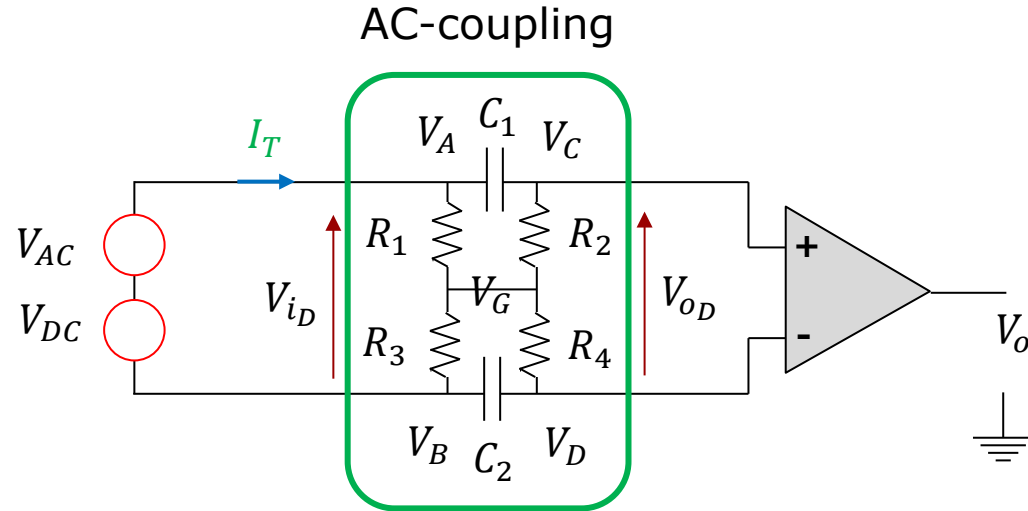
$$V_{DC} = \frac{Z_1}{\underbrace{Z_1 + Z_4}_{\approx 1/2}} (E_A - E_B) \approx 100\text{mV}$$

$$V_o = G_d \cdot V_{AC} + G_d \cdot V_{DC}$$

$$G_d = \frac{V_o}{V_{AC}} = \frac{5\text{V}}{5\text{mV}} = 1000$$

$$V_{o,DC} = 1000 \cdot 100\text{mV} = 100\text{V}$$

We need to block the DC potential before the instrumentation amplifier. Hence, we need an AC-coupled instrumentation amplifier.



V_{iD} : Differential input signal

V_{oD} : Differential output signal

Starting at the filter output:

$$V_{oD} = R_2 I_{R_2} + R_4 I_{R_4}$$

Current dividers:

$$I_{R_2} = I_T \cdot \frac{R_1}{R_1 + R_2 + Z_{C_1}}$$

$$I_{R_4} = I_T \cdot \frac{R_3}{R_3 + R_4 + Z_{C_2}}$$

$$I_T = \frac{V_{iD}}{Z_T}$$

$$Z_T = R_1 || (R_2 + Z_{C_1}) + R_3 || (R_4 + Z_{C_2})$$

From Maple:

$$\frac{V_{oD}}{V_{iD}} = \frac{\left(\left((R_2 + R_4)R_3 + R_2(R_4 + Z_{C_2}) \right) R_1 + R_3 R_4 (R_2 + Z_{C_1}) \right)}{\left((R_2 + R_4 + Z_{C_1} + Z_{C_2})R_3 + (R_4 + Z_{C_2})(R_2 + Z_{C_1}) \right) R_1 + R_3 (R_4 + Z_{C_2})(R_2 + Z_{C_1})}$$

To obtain symmetry we set:

$$R_3 = R_1 \quad R_4 = R_2 \quad C_2 = C_1$$

Letting Maple simplify again:

$$\frac{V_{oD}}{V_{iD}} = \frac{R_2}{R_2 + Z_{C_1}}$$

If all components are matched perfectly:

$$R_1 = R_2 = R_3 = R_4 \quad C_2 = C_1$$

$$\frac{V_{oD}(j\omega)}{V_{iD}(j\omega)} = \frac{(j\omega) R_1 C_1}{(j\omega) R_1 C_1 + 1}$$

We obtain a classic first order high-pass filter with cut-off frequency:

$$f_c = \frac{1}{2\pi R_1 C_1} \quad C_1 = \frac{1}{2\pi R_1 f_c}$$

We need a high input impedance, hence R_1 and R_2 must be chosen in the $M\Omega$ range.

$$R_1 = R_2 = 4.7M\Omega$$

$$f_c = 0.03 \text{ Hz} \Rightarrow C_1 = C_2 = \frac{1}{2\pi \cdot 4.7M\Omega \cdot 0.03\text{Hz}} = 1.12\mu\text{F} \approx 1\mu\text{F}$$

$$f_c = 20 \text{ Hz} \Rightarrow C_1 = C_2 = \frac{1}{2\pi \cdot 4.7M\Omega \cdot 20\text{Hz}} = 1.7\text{nF} \approx 1.5\text{nF}$$

